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Effects of Percutaneous Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients in the Stenting Era: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Effects of Percutaneous Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients in the Stenting Era: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Abstract

Objective

Whether the use of percutaneous mechanical circulatory support (pMCS) can reduce mortality in acute myocardial infarction (AMI) patients is still controversial. We sought to investigate the impact on clinical outcomes of pMCS in AMI patients who undergo stent implantation.

Design

Systematic review and meta-analysis.

Data sources

The Embase, Cochrane Library, Medline, PubMed, Web of Science, ClinicalTrials.gov and Clinicaltrialsregister.eu databases were searched from database inception to February 2020.

Eligibility criteria

Randomized clinical trials (RCTs) on pMCS use in AMI patients undergoing stent implantation were included.

Data extraction and synthesis

Data from the included studies were extracted and summarized independently by two reviewers. RR and 95% CI were calculated for clinical outcomes according to random effects model.

Results

Eleven studies were included, 8 studies including 1340 patients compared intra-aortic balloon pump (IABP) with non-IABP, and 3 studies including 115 patients compared percutaneous ventricular assist devices (pVADs) versus IABP. Compared with non-IABP, IABP was not associated with short-term (RR=0.92; 95% CI, 0.57-1.47) and long-term (RR=0.87; 95% CI, 0.60-1.27) mortality reductions but decreased the risk of heart failure (RR=0.56; 95% CI, 0.31-1.00). In subgroup analyses, IABP was associated with a long-term mortality reduction in patients without CS

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(RR=0.49; 95% CI, 0.27-0.90) but was not associated with short-term (RR=1.14; 95% CI, 0.67-1.93) or long-term (RR=1.03; 95% CI, 0.88-1.20) mortality reductions in patients with CS. For pVAD compared with IABP, all patients complicating CS, pVADs did not decrease all-cause mortality (RR=0.95; 95% CI, 0.64-1.41).

Conclusions

In all AMI patients undergoing stent implantation, the use of IABP does not reduce all-cause mortality but decreases the risk of heart failure. Patients without CS can benefit from IABP with a long-term mortality reduction, while patients with CS seem not. More convincing randomized data are required to verify the effect of pVADs in patients with CS.

Keywords: acute myocardial infarction (AMI), percutaneous coronary intervention (PCI), stent implantation, intra-aortic balloon pump (IABP), percutaneous ventricular assist devices (pVADs)

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Article Summary

Strengths and limitations of this study

- We conducted a comprehensive systematic search of literature from various databases to find out the application of pMCS in patients with AMI.
- To our knowledge, this meta-analysis is the first to explore the effect of pMCS in AMI patients undergoing stent implantation by strict inclusion criteria.
- Few RCTs and small sample size comparing pVADs with IABP, and different types of pVAD, the conclusion is cautious.
- Due to the limited number of included RCTs and participants, subgroup analyses of secondary endpoints could not be performed.

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Introduction

Cardiovascular diseases (CVDs) are the leading cause of morbidity and mortality, among which acute myocardial infarction (AMI) is the most common cause of hospitalization in elderly patients [1,2].

Percutaneous mechanical circulatory support (pMCS), including intra-aortic balloon pump (IABP) and percutaneous ventricular assist devices (pVADs), is an important haemodynamic support strategy. In early guidelines, IABP is a reasonable and class I recommendation for severe ischaemia in UA/NSTEMI patients [3,4]. However, IABP and left ventricular assist devices (LVADs) were recommended to class IIa and class IIb, even class III, in AMI patients with cardiogenic shock (CS) [5,6], because no reduction in mortality or other adverse events was shown in a large-scale randomized controlled trial (RCT) [7]. However, recent meta-analyses have shown different effects of IABP on mortality in AMI patients with or without CS, with no detailed distinction between the methods of revascularization [8,9]. Percutaneous coronary intervention (PCI) with stenting is now the standard treatment strategy in AMI patients [5,10]. It is still unknown whether AMI patients can benefit from the use of pMCS in the current stenting era. It is necessary to reassess the effects of pMCS in AMI patients undergoing stent implantation.

Therefore, we conducted a systematic review and meta-analysis using rigorous inclusion criteria to explore the effect of pMCS application in the stenting era and make full use of the evidence to provide a more comprehensive evaluation with more precise results. Moreover, we also performed subgroup analyses of patients with or without CS to explore the effects among different populations to investigate heterogeneity and identify modifying factors.

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Methods

The current meta-analysis was conducted in accordance with the PRISMA guidelines [11].

Search Strategy

We conducted a systematic search of the literature in February 2020 without restrictions on region, publication type, or language. The Embase, Cochrane Library, Medline, PubMed, and Web of Science databases were included. ClinicalTrials.gov and Clinicaltrialsregister.eu were searched as well. The MeSH search terms included the following: myocardial infarction, acute coronary syndrome, heart-assist devices, intra-aortic balloon pumping and extracorporeal membrane oxygenation. We excluded nonhuman studies. Details of the search strategy were presented in online supplementary file 2.

Study Selection

In the current meta-analysis, only RCTs in which stent implantation was the dominant form of revascularization and pMCS was used during the perioperative period in patients with AMI were included. We excluded nonrandomized trials, cohort studies, cross-sectional studies, ongoing trials, trials terminated early and other publication types, including review, case reports or case series, editorials, letters, and meeting abstracts. Studies in which patients underwent coronary artery bypass grafting (CABG), percutaneous transluminal coronary angioplasty (PTCA) or systemic thrombolysis as the main revascularization method were also excluded.

Endpoints

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The primary outcome was all-cause mortality, including short-term mortality and long-term mortality. Short-term mortality was defined as mortality occurring within 30 days. Long-term mortality occurred at least 6 months of follow-up. The secondary endpoints included reinfarction, repeat revascularization, heart failure (also included necessity of any additional ventricular assist device), stroke-TIA (transient ischaemic attack), bleeding (moderate to major bleeding, blood transfusion or surgery to control the bleeding), arrhythmias and vascular complications (peripheral ischaemic vascular complications, major dissection, pseudoaneurysm or arteriovenous fistula) occurring within 30 days. The endpoint definitions as applied in each study were presented in online supplementary file 3.

Data Extraction and Quality Assessment

Data from the included studies were extracted and summarized independently by two reviewers (Shi and Wang). All extracted data from the included studies were independently collected in duplicate using a standardized data extraction form. Any disagreements were resolved by a third reviewer (Sun). The quality of the included studies was assessed using the Cochrane Collaborations tool [12].

Statistical Analysis

We applied risk ratios (RRs) and 95% confidence intervals (CIs) to summarize the statistics. Statistical heterogeneity was assessed using Cochran's Q test and quantified using the I^2 statistic; $p < 0.10$ and $I^2 > 50\%$ indicated statistically significant heterogeneity among studies. Considering the clinical differences among the included studies, a random-effects model was applied for statistical analyses. Sensitivity analyses were performed for primary outcomes with the following: (1) fixed

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versus random effects models, (2) exclusion of research with a high risk of bias, and (3) the one-study-out method. Funnel plots were used to evaluate publication bias. Statistical analyses were performed using STATA/SE 12.0 (Stata Corp LP, College Station, Texas, USA). P<0.05 was considered statistically significant.

Results

Overall, 11877 citations were identified through the search criteria. The full texts of 81 potentially eligible articles were scrutinized in detail. Among these, 14 RCTs used PCI as the main revascularization therapy, 3 RCTs on PTCA use were excluded [13,14,15], and 11 RCTs finally met the inclusion criteria [7,16,17,18,19,20,21,22,23,24,25,26]; the flow diagram of the study selection process is outlined in Figure 1. RCTs for comparison between pVAD and non-pVAD were not found. We also searched for ECMO as a topic but did not find relevant RCTs.

Characteristics of the Studies

The detailed designs of the included studies are listed in Table 1. Among the 11 RCTs, 8 studies including 1340 AMI patients compared IABP with non-IABP (672 in the IABP group and 668 in the non-IABP group) [7,16,19,20,21,22,23,25,26], 3 studies including 115 AMI patients compared pVADs versus IABP (58 in the pVADs group and 57 in the IABP group) [17,18,24]. Seven RCTs included CS patients, among which 4 studies that included 764 patients compared IABP with non-IABP (393 in the IABP group and 371 in the non-IABP group) [7,20,21,23,25], the remaining 3 studies compared pVADs versus IABP. Four RCTs included patients without CS, and all of these studies, which included 576 patients, compared IABP with non-IABP (279 in the IABP group and

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297 in the non-IABP group). The follow-up period ranged from in-hospital only to 6 years. The patient characteristics of the studies are shown in Table 2. Among the 7 studies that reported infarction-related coronary arteries [7,16,17,19,21,22,25], most patients had left coronary artery-related infarction (73.7%). In these studies, 66.5% of patients with CS had left coronary artery-related infarction, and 60.6% of the lesions occurred in the left anterior descending artery [7,17,21,25]; 84.0% of patients without CS had left coronary artery-related infarction, and 93.9% of the lesions occurred in the left anterior descending artery [16,19,22]. The other patient characteristics included age, sex, AMI, CS, hypertension, diabetes mellitus, previous stroke/TIA (transient ischaemic attack), prior myocardial infarction (MI), prior PCI, and prior CABG.

Outcomes of Synthesis Analyses

Of the included RCTs comparing IABP with non-IABP, all studies reported short-term mortality, and 5 studies reported long-term mortality. IABP was not associated with a short-term mortality reduction (IABP=23.8%, non-IABP=24.1%; RR=0.92; 95% CI, 0.57-1.47; p=0.715), with significant heterogeneity among studies (I^2=52.1%; p=0.04) (Figure 2). There was no significant difference between the IABP group and the non-IABP group in long-term mortality (IABP=29.1%, non-IABP=29.9%; RR=0.87; 95% CI, 0.60-1.27; p=0.468), with no significant heterogeneity among studies (I^2=39.0%; p=0.216) (Figure 2). The results of prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 4). For the secondary endpoints, 4 studies reported reinfarction, 2 reported repeat revascularization, 3 reported heart failure, 2 reported stroke-TIA, 5 reported bleeding, 2 reported arrhythmias, and 3 reported vascular complications. The results in online supplementary file 5 show that IABP was associated

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with a decrease in heart failure rate (IABP=3.7%, non-IABP=6.7%; RR=0.56; 95% CI, 0.31-1.00; p=0.049), but there was no effect on other secondary endpoints.

RCTs comparing pVADs with non-pVAD were not found, so we only analysed the 3 RCTs comparing pVADs versus IABP. The pooled analysis revealed that compared with IABP, pVADs was not associated with a decrease in short-term mortality (pVADs=44.8%, IABP=47.4%; RR=0.95; 95% CI, 0.64-1.41; p=0.786), with no significant heterogeneity among studies (I^2=0.0%; p=0.986) (see online supplementary file 6). Long-term mortality was not analysed because only one study reported. For the secondary outcome analysis, only bleeding and vascular complication events met the quantitative requirements of combined analysis. Compared with IABP, pVADs increased the rate of bleeding (pVADs=66.7%, IABP=36.4%; RR=1.86; 95% CI, 1.15-3.00; p=0.011) and trended towards increased vascular complications (pVADs=15.5%, IABP=0%; RR=5.48; 95% CI, 0.96-31.11; p=0.055) (see online supplementary file 6). The results of the prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 7).

Outcomes of Subgroup Analyses

We conducted a subgroup analysis of patients complicated or not complicated with CS. In patients experiencing AMI without CS, IABP did not significantly decrease short-term mortality (IABP=4.3%, non-IABP=8.1%; RR=0.51; 95% CI, 0.22-1.19; p=0.121) (Figure 3A), but long-term mortality was significantly reduced (IABP=5.0%, non-IABP=10.3%; RR=0.49; 95% CI, 0.27-0.90; p=0.020) (Figure 3B). In patients with AMI with CS, IABP did not reduce short-term mortality (IABP=37.7%, non-IABP=36.9%; RR=1.14; 95% CI, 0.67-1.93; p=0.636) (Figure 3A) or long-term

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mortality (IABP=47.5%, non-IABP=46.0%; RR=1.03; 95% CI, 0.88-1.20; p=0.750) (Figure 3B). Due to the limited number of studies included and few reports of the secondary endpoints, subgroup analyses of the secondary outcomes were not performed.

All patients had CS in studies comparing pVADs versus IABP, and the results have been described in the synthesis analyses.

Study Quality and Publication Bias

The results of the literature quality assessment using the Cochrane Collaboration tool are shown in online supplementary file 8. Two trials had a high risk of bias [21,25]. Funnel plots for all-cause mortality did not suggest publication bias (see online supplementary file 9).

Discussion

One of the innovations of this meta-analysis is that we applied strict inclusion criteria in revascularization strategy to explore, for the first time, the safety and efficacy of pMCS in patients with AMI undergoing stent implantation. Compared with non-IABP, IABP does not reduce mortality in AMI patients undergoing stent implantation and can reduce the overall incidence of heart failure. However, we found an interesting finding: compared with non-IABP, IABP reduced long-term mortality by half in patients without CS (5.0% versus 10.3%), and IABP was not associated with an all-cause mortality reduction in patients with CS.

Regarding the principle of IABP, deflation of the pump reduces ventricular workload and helps the ventricle push blood into the aorta during systole, and inflation of the pump improves coronary blood flow during diastole [27]. In our findings, the effect of IABP on reducing the overall incidence

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of heart failure in patients with AMI may be associated with these functions. The evidence of clinical trials also shows the beneficial effect of IABP on haemodynamics, as these pumps can significantly improve myocardial coronary blood flow and myocardial ischaemia [28,29]. IABP might have a greater effect in patients with critically reduced coronary artery perfusion and temporarily severely reduced left ventricular function [30]. In the current meta-analysis, most of the patients without CS had left coronary artery-related infarction, and most of the lesions occurred in the left anterior descending artery [16,19,22]. The haemodynamics of patients without CS may recover more quickly with IABP support, avoiding more serious clinical consequences, such as progression to CS, severe left heart failure, or even death. Some previous meta-analyses showed similar conclusions, but there was no detailed distinction between the revascularization strategies of thrombolytic therapy and PCI (PTCA and stent implantation) [8,9,31]. Stent implantation became the most widely used PCI in 2000 [32]. A recent meta-analysis showed that there was a significant reduction in the cardiac mortality rate, target lesion revascularization rate and adverse cardiac event rate in STEMI patients who underwent stent implantation compared to those who underwent PTCA [33]. According to the current guidelines on revascularization for AMI [5,10], our finding that IABP reduced long-term mortality compared to non-IABP in AMI patients without CS is more applicable to the current stenting era.

In the past, IABP have been the recommended treatment strategy for haemodynamic support in patients with combined AMI and CS [3,34,35]. However, some large-scale RCTs, including IABP-SHOCK II and PAMI-II, and registry data did not show any benefit in terms of mortality reduction in patients with AMI and CS with IABP use [7,20,13,36,37]. One of our findings was that IABP was not associated with an all-cause mortality reduction in patients with CS, which is

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consistent with these studies with AMI patients complicated with CS who underwent stent implantation. Patients with AMI and CS are considered a high-risk group with more impaired cardiac function and haemodynamics and are more likely to develop circulatory failure during PCI [38]. The IABP SHOCK trial showed that IABP treatment did not result in a significant haemodynamic improvement in patients with combined AMI and CS [39]. Despite the implementation of the drug therapy recommended by the guidelines and early revascularization, the in-hospital mortality rate remained close to 50% [40,41,42]. Even though this revascularization strategy is currently recommended in our meta-analysis, these findings are consistent with some previous meta-analyses [43,44,45,46]. Therefore, IABP should be used with caution when patients have a combination of AMI and CS.

We also analyzed the comparisons of the effects of pVAD and IABP, in which all patients had CS, although the results shown that pVADs was not associated with an all-cause mortality reduction but was associated with an increased rate of bleeding. However, due to the small sample size and different types of pVAD of each study, the results should be interpreted with caution.

Therefore, our findings of patients with AMI undergoing stent implantation are highly applicable to the current guidelines on myocardial revascularization. For patients with AMI, the presence of CS should be considered when deciding whether to use IABP during the perioperative period. Our results suggest that patients without CS can benefit from the use of IABP during the perioperative period because the long-term mortality was reduced by half. Most of these patients had left coronary artery-related infarcts. Whether IABP can improve the mortality of patients with other coronary artery-related infarctions needs to be verified by well-designed clinical trials. However, in

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patients with CS, the use of IABP should be more restricted because IABP did not improve mortality. These results are helpful to the clinical decision-making of patients with AMI.

There are several limitations in the current meta-analysis. First, the number of included RCTs and participants was limited; additionally, due to the few reports of secondary endpoints, subsequent subgroup analyses could not be performed. Second, there was high heterogeneity in the pooled CS group, which may be explained by inconsistent timing of initiation, higher incidence of interactions, and inconsistent time points for reporting mortality. Third, there were few RCTs and small sample size comparing pVADs with IABP, all of which included patients with CS, and the types of pVAD were different, Impella 2.5, Impella CP and TandemHeart respectively; thus, the representativeness of the results is not sufficient.

Conclusions

The use of IABP does not reduce mortality but decreases the risk of heart failure in all AMI patients undergoing stent implantation. In subgroup analyses, IABP is associated with a long-term mortality reduction that was reduced by half in patients without CS. However, in patients with CS, IABP is not associated with a mortality reduction. Although pVAD demonstrated no effect in reducting mortality versus IABP in patients with CS, the result requires evidence from larger number of studies and participants.

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Authors' contributions

JC and YL contributed to the conception and revision of the review. YS and YW performed systematic literature search, study selection data extraction and data analyses. YS contributed to writing of the manuscript. XS contributed to review the data and manuscript. YT, MJ, YB, SL and HY checked the manuscript.

Declaration of interest

None.

Data sharing statement

All data associated with this manuscript are included in the main text and supplementary materials.

Supplementary materials

Supplementary data of this article can be found online.

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Online supplementary file 1: PRISMA checklist for the meta-analysis

Online supplementary file 2: Search strategy

Online supplementary file 3: The Endpoint Definitions of the Included Studies

Online supplementary file 4: Sensitivity analyses of the all-cause mortality by comparison between IABP and non-IABP

Online supplementary file 5: Forest plots for the secondary end points of AMI patients treated with IABP compared with non-IABP

Online supplementary file 6: Forest plots for outcomes occurrence comparing pVADs use versus IABP use

Online supplementary file 7: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

Online supplementary file 8: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Online supplementary file 9: Funnel plot of standard error for: A. IABP compared with non-IABP in short-term mortality; B. IABP compared with non-IABP in long-term mortality; C. pVADs compared with IABP in short-term mortality

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Figure legends

Figure 1. Flow diagram of study selection

Randomized clinical trials were included. PTCA indicates percutaneous transluminal coronary angioplasty.

Figure 2. Forest plots for the all-cause mortality

Risk ratio of all-cause mortality of AMI patients treated with IABP compared with non-IABP in short-term or long-term. CI, confidence interval; IABP, intra-aortic balloon pump; non-IABP, no plan of IABP use during peri-operative period; Short-term mortality, occurring within 30 days; long-term mortality, occurring at least 6 months of follow-up.

Figure 3. Forest plots showing the all-cause mortality for the subgroup analyses

A. Risk ratio of short-term mortality of patients with AMI with or without CS treated with IABP compared with non-IABP. **B.** Risk ratio of long-term mortality of patients with AMI with or without CS treated with IABP compared with non-IABP. CS, cardiogenic shock.

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Table 1. Study Design of the Included Studies

Study	Year	Region	Intervention group /control group	No. of Patients	Stent implantation, %	BMS, %	PCI success, %	Timing of pMCS	Time point of reported mortality	Follow-Up
SEMPER FI[13]	2020	Netherlands	IABP+PCI/PCI	50/50	100	NA	90/96	After PCI	30 days, 6 months	6 months
Zhou et al[22]	2017	China	IABP+PCI/PCI	42/42	100	NA	NA	Before PCI	In hospital, 12 months	12 months
IABP-SHOCK II[7,21]	2012, 2013	Germany	IABP+PCI/PCI	301/299	89.8	50.2/51.5	82.2/81.5	Before or after PCI	30 days, 12 months	6 years
CRISP AMI[16]	2011	United State	IABP+PCI/PCI	161/176	96.9	53.7/53.1	92.9/5.3	Before PCI	in hospital, 6 months	6 months
Gu et al[23]	2011	China	IABP+PCI/PCI	51/55	100	NA	NA	Before PCI	30 days, 6 months	6 months
Prondzinsky et al[24]	2010	United State	IABP+PCI/PCI	19/21	85	NA	NA	After PCI	in hospital	in hospital
Arias et al[26]	2005	Mexico	IABP+PCI/PCI	31/9	100	NA	NA	Before PCI	30 days	30 days
Vijayalakshmi et al[27]	2007	Japan	IABP+PCI/PCI	17/16	100	NA	NA	After PCI	in hospital, 30 days	30 days
IMPRESS[14]	2016	Netherlands	pVADs+PCI/IABP+PCI	24/24	97.9	4/8	NA	After PCI	30 days, 6 months	6 months
Seyfarth et al[25]	2008	United State	pVADs+PCI/IABP+PCI	13/13	NA	NA	100/92	After PCI	30 days	30 days
Thiele et al[15]	2005	Germany	pVADs+PCI/IABP+PCI	21/20	95.1	NA	80/74	Before or after PCI	30 days	30 days

IABP indicates intra-aortic balloon pump; PCI, percutaneous coronary intervention; pVADs, percutaneous ventricular assist devices; NA, not available; BMS, bare-metal stent. PCI success was defined as TIMI III flow.

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Table 2. Baseline Clinical Characteristics of the Included Studies

Study (year)	Male, %	Mean age, year	AMI, %	CS, %	Infarct-related artery (LM/LAD/LCX/RCA), %	Hypertension, %	Diabetes mellitus, %	Previous stroke/TIA, %	Prior MI, %	Prior PCI, %	Prior CABG, %
SEMPER FI[13]	74/78	64/63	100	0	1.0/50.0/11.0/33.0	24/32	8/18	2/2	6/10	8/12	2/8
Zhou et al[22]	67/62	57.2/56.7	100	100	0/34.5/38.1/27.4	NA	NA	NA	NA	NA	NA
IABP-SHOCK II[7,21]	70.6/67.1	70/69	100	100	9/42.2/18.7/25.3	72/66.6	35.4/30.1	8/6.7	23.7/20.4	21.1/17.4	6.7/4.0
CRISP AMI[16]	82/81.8	56.1/57.7	100	0	0/97.3/0/0.3	24.2/34.1	16.8/20.5	0/0.6	0/0	1.9/1.1	NA
Gu et al[23]	56.9/65.5	67.4/66.6	100	0	3.8/47.2/11.3/37.7	68.6/60	35.5/34.5	NA	3.9/5.5	68.6/70.9	NA
Prondzinsky et al[24]	74/81	62.1/66.1	100	100	NA	42.1/47.6	52.6/47.6	NA	21.1/23.8	1.9/1.1	NA
Arias et al[26]	NA	NA	100	100	0/60.0/22.5/8.3	NA	NA	NA	NA	NA	NA
Vijayalakshmi et al[27]	82/88	40/45	100	0	NA	25/38	18/25	NA	12/25	NA	NA
IMPRESS[14]	75/83	58/59	100	100	6.3/64.6/18.8/10.4	20/29	9/13	0/4	5/4	5/0	5/0
Seyfarth et al[25]	62/85	65/67	100	100	NA	54/69	39/23	NA	NA	NA	NA
Thiele et al[15]	76/75	63/65	100	100	NA	76/75	52/55	NA	62/45	NA	NA

AMI indicates acute myocardial infarction; CS, cardiogenic shock; LM, left main; LAD, left anterior descending; LCX, left circumflex; RCA, right coronary artery; TIA, transient ischemic attack; MI, myocardial infarction; CABG, coronary artery bypass graft.

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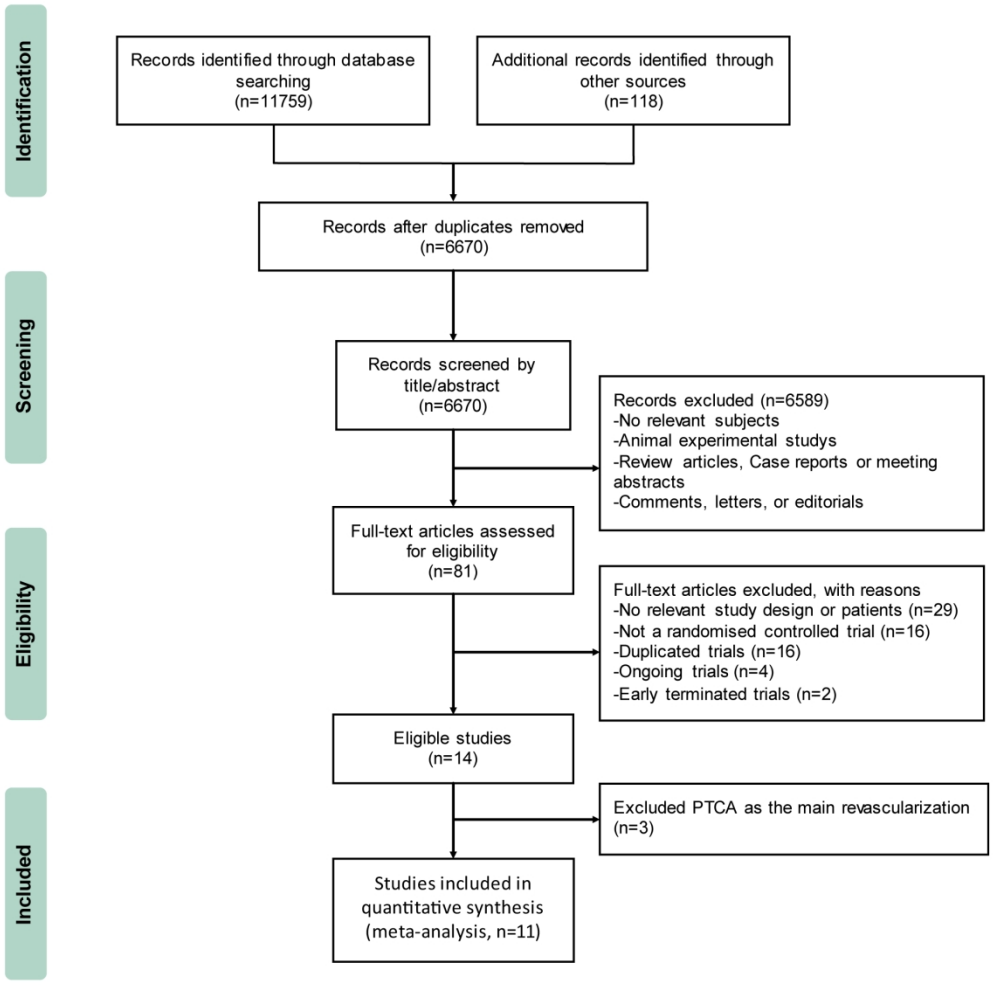


Figure 1. Flow diagram of study selection
Randomized clinical trials were included. PTCA indicates percutaneous transluminal coronary angioplasty.

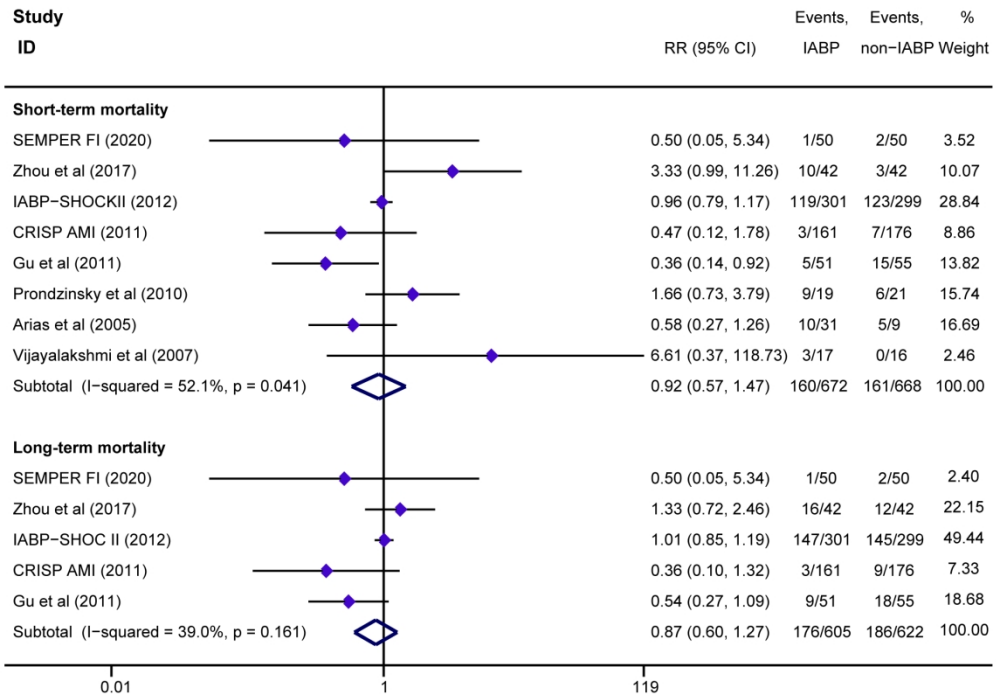


Figure 2. Forest plots for the all-cause mortality
Risk ratio of all-cause mortality of AMI patients treated with IABP compared with non-IABP in short-term or long-term. CI, confidence interval; IABP, intra-aortic balloon pump; non-IABP, no plan of IABP use during peri-operative period; Short-term mortality, occurring within 30 days; long-term mortality, occurring at least 6 months of follow-up.

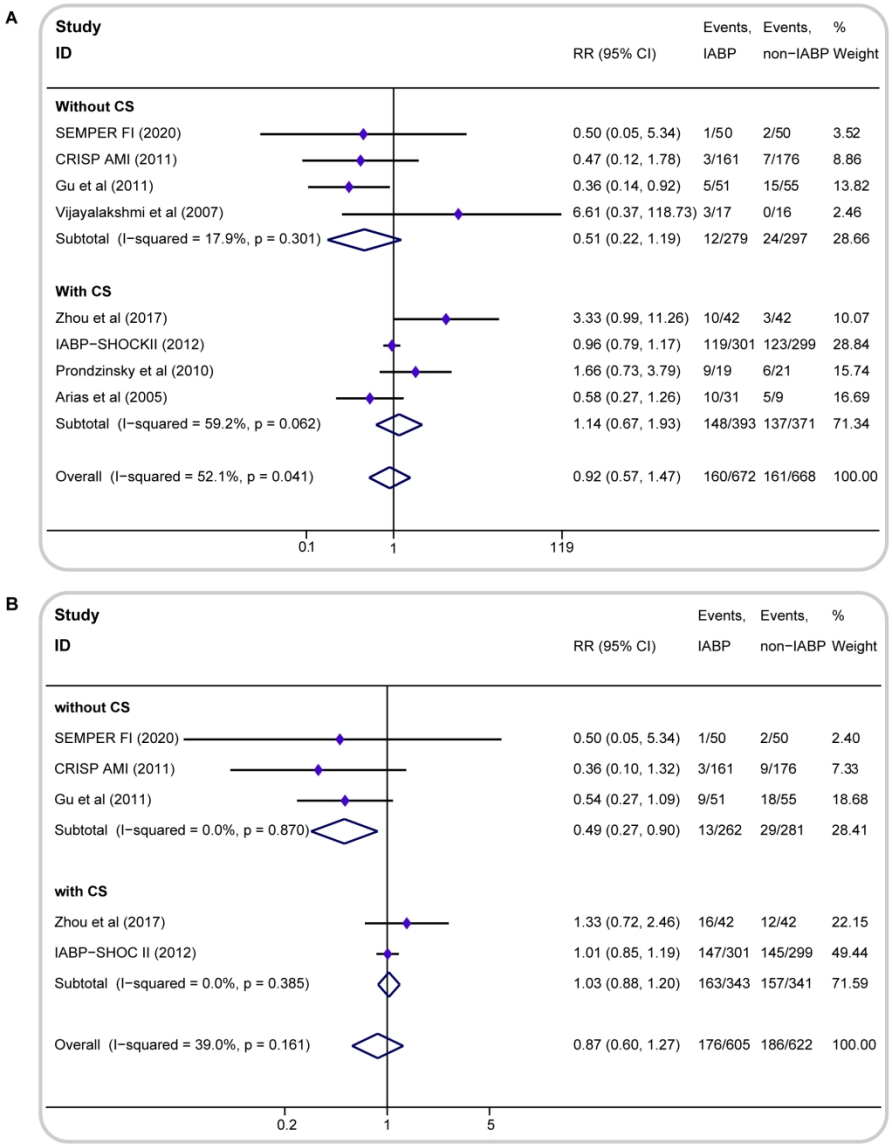


Figure 3. Forest plots showing the all-cause mortality for the subgroup analyses
A. Risk ratio of short-term mortality of patients with AMI with or without CS treated with IABP compared with non-IABP. B. Risk ratio of long-term mortality of patients with AMI with or without CS treated with IABP compared with non-IABP. CS, cardiogenic shock.

PRISMA checklist for the meta-analysis

Section/topic	#	Checklist item	Reported section (top-level heading)
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Both, title, abstract
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Search Strategy and Study Selection
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Search Strategy and Study Selection
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Search Strategy, Study Selection and Supplement Table 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Search Strategy and Study Selection
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Search Strategy, Study Selection, Data Extraction and Quality Assessment

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Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Study Selection and Endpoints
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Data Extraction and Quality Assessment
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Statistical Analysis
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Statistical Analysis
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Data Extraction and Quality Assessment
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Statistical Analysis
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Descriptions of Included Studies, flow chart (Figure 1)
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Characteristics of the Studies, Table 1 and Table 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Study Quality and Publication Bias, Table 5 and Figure 6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Outcomes of Synthesis Analyses, Figure 2, Figure 3 and Figure 4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Outcomes of Synthesis Analyses, Figure 2, Figure 3 and Figure 4
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Study Quality and Publication Bias,

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			Table 5 and Figure 6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Outcomes of Subgroup Analyses, Figure 5, Table 3 and Table 4
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding

NA indicates not available.

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Search strategy

Search strategy in Embase

#	Searches	Results
1	'heart infarction'/exp	389487
2	heart:ab,ti AND infarct*:ab,ti	100833
3	myocard*:ab,ti AND infarct*:ab,ti	291337
4	heart:ab,ti AND attack*:ab,ti	16796
5	cardiovascular:ab,ti AND stroke*:ab,ti	55272
6	stemi:ab,ti	26752
7	nstemi:ab,ti OR 'non stemi':ab,ti OR (non:ab,ti AND stemi:ab,ti)	12922
8	'ischemic heart disease'/exp	676288
9	heart:ab,ti AND ischemi*:ab,ti	113806
10	myocard*:ab,ti AND ischemi*:ab,ti	107041
11	'coronary artery disease'/exp	333302
12	coronary:ab,ti AND artery:ab,ti AND disease*:ab,ti	168210
13	coronary:ab,ti AND atheroscleros*:ab,ti	41830
14	acute:ab,ti AND coronary:ab,ti AND syndrom*:ab,ti	58212
15	acs:ab,ti	39928
16	'angina pectoris'/exp	101922
17	angina*:ab,ti	77333
18	angor:ab,ti AND pectoris:ab,ti	85
19	stenocardia*:ab,ti	921
20	myocardial:ab,ti AND preinfarction:ab,ti AND syndrom*:ab,ti	39
21	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20	968859
22	'heart assist device'/exp	38677
23	heart:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	16686
24	vascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2359
25	cardiovascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	1756
26	artificial:ab,ti AND ventricle*:ab,ti	1684
27	ventricle:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2634
28	ventricular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	22149
29	artificial:ab,ti AND heart:ab,ti AND ventricle*:ab,ti	845
30	blood:ab,ti AND pump*:ab,ti	26035
31	cardiac:ab,ti AND resuscitat*:ab,ti	29834
32	counterpulsation:ab,ti AND device*:ab,ti	607
33	mechanical:ab,ti AND circulat*:ab,ti AND support*:ab,ti	8057

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34	cardiac:ab,ti AND assist*:ab,ti AND system*:ab,ti	5046
35	assist*:ab,ti AND circulat*:ab,ti	12243
36	lvad:ab,ti	11097
37	pvad:ab,ti	333
38	tandemheart*:ab,ti	294
39	heartmate*:ab,ti	3194
40	impella*:ab,ti	1660
41	intra:ab,ti AND aortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	5829
42	intraaortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	2363
43	iabp:ab,ti	4361
44	iabc:ab,ti	164
45	'extracorporeal oxygenation'/exp	22338
46	extracorporeal:ab,ti AND membrane:ab,ti AND oxygenation*:ab,ti	14509
47	extracorporeal:ab,ti AND life:ab,ti AND support*:ab,ti	4304
48	ecmo:ab,ti	14659
49	ecls:ab,ti	1976
50	'heart lung machine'/exp	2071
51	heart:ab,ti AND lung:ab,ti AND machine*:ab,ti	1513
52	'extracorporeal circulation'/exp	70110
53	extracorporeal:ab,ti AND circulat*:ab,ti	14234
54	extracorporeal:ab,ti AND cardiopulmonary:ab,ti AND resuscitation*:ab,ti	1680
55	extracorporeal:ab,ti AND CPR:ab,ti	872
56	ecpr:ab,ti	709
57	#22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56	197482
58	#21 AND #57	36642
59	'animals'/exp NOT 'humans'/exp	5401121
60	'animals, laboratory'/exp	667394
61	'animal experiment'/exp	2504650
62	'animal model'/exp	1346931
63	'rodent'/exp	3758513
64	rat:ti OR rats:ti OR mouse:ti OR mice:ti	1579270
65	#59 OR #60 OR #61 OR #62 OR #63 OR #64	6421085
66	#58 NOT #65	32482
67	'randomized controlled trial':pt OR 'randomized controlled trial'/exp OR 'randomized controlled trials as topic'/exp	761854
68	'controlled clinical trial'/de	429262
69	randomi?ed:ab	808696

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70	placebo:ab	292578
71	randomly:ab	431255
72	'clinical trials as topic'/de	105922
73	trial:ti	297184
74	#67 OR #68 OR #69 OR #70 OR #71 OR #72 OR #73	1799342
75	#66 AND #74	3623

Search strategy in Cochrane Library

#	Searches	Results
1	[mh "Myocardial Infarction"]	10973
2	(Myocard* Infarct*):ti,ab,kw	30947
3	(Cardiovascular Stroke*):ti,ab,kw	9335
4	(Heart Attack*):ti,ab,kw	3698
5	(heart infarct*):ti,ab,kw	21802
6	STEMI:ti,ab,kw	3287
7	(NSTEMI or Non-STEMI or (Non and STEMI)):ti,ab,kw	1213
8	[mh "Myocardial Ischemia"]	27924
9	(Myocard* Ischemi*):ti,ab,kw	13252
10	(heart ischemi*):ti,ab,kw	17368
11	[mh "Acute Coronary Syndrome"]	1923
12	(Acute Coronary Syndrom*):ti,ab,kw	7401
13	ACS:ti,ab,kw	4012
14	[mh "Coronary Artery Disease"]	6317
15	(Coronary Artery AND Disease*):ti,ab,kw	23282
16	(Coronary Arterioscleros*):ti,ab,kw	454
17	[mh "Angina Pectoris"]	4524
18	Stenocardia*:ti,ab,kw	64
19	(Angor Pectoris):ti,ab,kw	21
20	angina*:ti,ab,kw	13810
21	(Myocardial Preinfarction Syndrom*):ti,ab,kw	6
22	{or #1-#21}	75199
23	[mh "Heart-Assist Devices"]	173
24	(Heart and Assist* (Device* or pump*)):ti,ab,kw	1206
25	(Vascular and Assist* (Device* or pump*)):ti,ab,kw	255
26	(cardiovascular and Assist* and (Device* or pump*)):ti,ab,kw	294
27	(Artificial Ventricle*):ti,ab,kw	675
28	(Ventricle and Assist* and (Device* or pump*)):ti,ab,kw	236
29	(Ventricular and Assist* and (Device* OR pump*)):ti,ab,kw	735
30	(Artificial Heart Ventricle*):ti,ab,kw	648
31	(blood pump*):ti,ab,kw	4954
32	(cardiac resuscitat*):ti,ab,kw	2821
33	(counterpulsation device*):ti,ab,kw	53

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34	(mechanical circulat* support*):ti,ab,kw	293
35	(Cardiac Assist* System*):ti,ab,kw	708
36	(Assist* Circulat*):ti,ab,kw	1070
37	LVAD:ti,ab,kw	309
38	pVAD:ti,ab,kw	11
39	TandemHeart*:ti,ab,kw	8
40	HeartMate*:ti,ab,kw	89
41	Impella*:ti,ab,kw	106
42	[mh "Intra-Aortic Balloon Pumping"]	165
43	(Intra and Aortic and Balloon and (Pump* or counterpulsation)):ti,ab,kw	470
44	(Intraaortica and Balloon and (Pump* or counterpulsation)):ti,ab,kw	1
45	IABP:ti,ab,kw	321
46	IABC:ti,ab,kw	16
47	[mh "Extracorporeal Membrane Oxygenation"]	165
48	(Extracorporeal Membrane Oxygenation*):ti,ab,kw	594
49	(Extracorporeal Life Support*):ti,ab,kw	133
50	ECMO:ti,ab,kw	484
51	ECLS:ti,ab,kw	49
52	[mh "Heart-Lung Machine"]	36
53	(Heart Lung Machine*):ti,ab,kw	243
54	[mh "Extracorporeal Circulation"]	3972
55	(Extracorporeal and Circulat*):ti,ab,kw	1356
56	(extracorporeal cardiopulmonary resuscitation*):ti,ab,kw	52
57	(extracorporeal CPR):ti,ab,kw	29
58	ECPR:ti,ab,kw	21
59	{or #23-#58}	16028
60	#22 and #59	3321
61	[mh "animals"] not [mh "humans"]	7306
62	rat or rats or mouse or mice:ti,ab,kw	12317
63	#61 or #62	17577
64	#60 not #63	3289
65	randomized:ti,ab,kw	855276
66	placebo:ti,ab,kw	291542
67	randomly:ti,ab,kw	235571
68	clinical trials:ti,ab,kw	170973
69	{or #65-#68}	1062122
70	#64 and #69	2575

Search strategy in Medline

#	Searches	Results
1	Myocardial Infarction/	163805

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2	(Myocard$ and infarct$).ab,ti.	198069
3	(heart and infarct$).ab,ti.	62062
4	(Heart and Attack$).ab,ti.	10872
5	(Cardiovascular and Stroke$).ab,ti.	32638
6	STEMI.ab,ti.	10207
7	(NSTEMI or Non-STEMI or (Non and STEMI)).ab,ti.	4087
8	Myocardial Ischemia/	38230
9	(Myocard$ and Ischemi$).ab,ti.	72673
10	(heart and ischemi$).ab,ti.	69531
11	Acute Coronary Syndrome/	14964
12	(Acute and Coronary and Syndrom$).ab,ti.	32777
13	ACS.ab,ti.	20938
14	Coronary Artery Disease/	60303
15	(Coronary and Artery and Disease$).ab,ti.	106514
16	(Coronary and Arterioscleros$).ab,ti.	1887
17	Angina Pectoris/	32366
18	Stenocardia$.ab,ti.	944
19	(Angor and Pectoris).ab,ti.	42
20	Angina$.ab,ti.	53771
21	(Myocardial and Preinfarction and Syndrom$).ab,ti.	21
22	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21	498670
23	Heart Assist Devices/	13515
24	(Heart and Assist$ and (Device$ or pump$)).ab,ti.	8926
25	(Vascular and Assist$ and (Device$ or pump$)).ab,ti.	1214
26	(Cardiovascular and Assist$ and (Device$ or pump$)).ab,ti.	989
27	(Artificial and Ventricle$).ab,ti.	1148
28	(Ventricle and Assist$ and (Device$ or pump$)).ab,ti.	1536
29	(Ventricular and Assist$ and (Device$ or pump$)).ab,ti.	12632
30	(Artificial and Heart and Ventricle$).ab,ti.	552
31	(blood and pump$).ab,ti.	16722
32	(cardiac and resuscitat$).ab,ti.	18729
33	(counterpulsation and device$).ab,ti.	398
34	(mechanical and circulat$ and support$).ab,ti.	4542
35	(Cardiac and Assist$ and System$).ab,ti.	2976
36	(Assist$ and Circulat$).ab,ti.	7912
37	LVAD.ab,ti.	4313
38	pVAD.ab,ti.	146
39	TandemHeart$.ab,ti.	177
40	HeartMate$.ab,ti.	1359
41	Impella$.ab,ti.	712
42	Intra Aortic Balloon Pumping/	3902
43	(Intra and Aortic and Balloon AND (Pump* or counterpulsation)).ab,ti.	3712

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44	(Intraaortic and Balloon and (Pump* or counterpulsation)).ab,ti.	1643
45	IABP.ab,ti.	2069
46	IABC.ab,ti.	118
47	Extracorporeal Membrane Oxygenation/	9975
48	(Extracorporeal and Membrane and Oxygenation$).ab,ti.	9932
49	(Extracorporeal and Life and Support$).ab,ti.	2724
50	ECMO.ab,ti.	6947
51	ECLS.ab,ti.	1159
52	Heart Lung Machine/	2087
53	(Heart and Lung and Machine$).ab,ti.	1011
54	Extracorporeal Circulation/	12854
55	(Extracorporeal and Circulat*).ab,ti.	10536
56	(extracorporeal and cardiopulmonary and resuscitation*).ab,ti.	1024
57	(extracorporeal and CPR).ab,ti.	361
58	ECPR.ab,ti.	282
59	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58	93046
60	22 and 59	11906
61	animals/ not humans/	4637565
62	exp Animals, Laboratory/	862220
63	exp Animal Experimentation/	9307
64	exp Models, Animal/	556859
65	exp rodentia/	3186357
66	(rat or rats or mouse or mice).ti.	1317731
67	61 or 62 or 63 or 64 or 65 or 66	5552545
68	60 not 67	10670
69	randomized controlled trial.pt. or randomized controlled trial/ or Randomized Controlled Trials as Topic/	625259
70	controlled clinical trial.pt.	93531
71	placebo.ab.	204798
72	randomly.ab.	326929
73	clinical trials as topic.sh.	190121
74	randomi?ed.ab.	561389
75	trial.ti.	212789
76	69 or 70 or 71 or 72 or 73 or 74 or 75	1353796
77	68 and 76	1236

Search strategy in Pubmed

#	Searches	Results
1	Search "Myocardial Infarction"[Mesh]	172645

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2	Search (Myocard*[Title/Abstract]) AND Infarct*[Title/Abstract]	202868
3	Search (Cardiovascular[Title/Abstract]) AND Stroke*[Title/Abstract]	33545
4	Search (Heart[Title/Abstract]) AND Attack*[Title/Abstract]	11092
5	Search (heart[Title/Abstract]) AND infarct*[Title/Abstract]	65211
6	Search STEMI[Title/Abstract]	10480
7	Search (((NSTEMI[Title/Abstract]) OR Non-STEMI[Title/Abstract])) OR ((Non[Title/Abstract]) AND STEMI[Title/Abstract])	4238
8	Search "Myocardial Ischemia"[Mesh]	423899
9	Search (Myocard*[Title/Abstract]) AND Ischemi*[Title/Abstract]	73961
10	Search (heart[Title/Abstract]) AND ischemi*[Title/Abstract]	70950
11	Search "Coronary Artery Disease"[Mesh]	60304
12	Search ((Coronary[Title/Abstract]) AND Artery[Title/Abstract]) AND Disease*[Title/Abstract]	110960
13	Search (Coronary[Title/Abstract]) AND Arterioscleros*[Title/Abstract]	2483
14	Search "Acute Coronary Syndrome"[Mesh]	14963
15	Search ((Acute[Title/Abstract]) AND Coronary[Title/Abstract]) AND Syndrom*[Title/Abstract]	34099
16	Search ACS[Title/Abstract]	21331
17	Search "Angina Pectoris"[Mesh]	43141
18	Search Stenocardia*[Title/Abstract]	944
19	Search (Angor[Title/Abstract]) AND Pectoris[Title/Abstract]	53
20	Search angina*[Title/Abstract]	55072
21	Search ((Myocardial[Title/Abstract]) AND Preinfarction[Title/Abstract]) AND Syndrom*[Title/Abstract]	23
22	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	590373
23	Search "Heart-Assist Devices"[Mesh]	13516
24	Search ((Heart[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	9625
25	Search ((Vascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1284
26	Search ((cardiovascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1073
27	Search (Artificial[Title/Abstract]) AND Ventricle*[Title/Abstract]	1203
28	Search ((Ventricle[Title/Abstract]) AND Assist*[Title/Abstract]) AND Device*[Title/Abstract]	1478
29	Search ((Ventricular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	12971
30	Search ((Artificial[Title/Abstract]) AND Heart[Title/Abstract]) AND Ventricle*[Title/Abstract]	590

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31	Search (blood[Title/Abstract]) AND pump*[Title/Abstract]	16957
32	Search (cardiac[Title/Abstract]) AND resuscitat*[Title/Abstract]	19772
33	Search (counterpulsation[Title/Abstract]) AND device*[Title/Abstract]	409
34	Search ((mechanical[Title/Abstract]) AND circulat*[Title/Abstract]) AND support*[Title/Abstract]	4961
35	Search ((Cardiac[Title/Abstract]) AND Assist*[Title/Abstract]) AND System*[Title/Abstract]	3083
36	Search ((Assist*[Title/Abstract])) AND Circulat*[Title/Abstract]	8436
37	Search LVAD[Title/Abstract]	4466
38	Search pVAD[Title/Abstract]	150
39	Search TandemHeart*[Title/Abstract]	184
40	Search HeartMate*[Title/Abstract]	1386
41	Search Impella*[Title/Abstract]	747
42	Search "Intra-Aortic Balloon Pumping"[Mesh]	3903
43	Search (((Intra[Title/Abstract]) AND Aortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	3754
44	Search ((Intraaortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	1656
46	Search IABP[Title/Abstract]	2108
47	Search IABC[Title/Abstract]	119
48	Search "Extracorporeal Membrane Oxygenation"[Mesh]	9976
49	Search ((Extracorporeal[Title/Abstract]) AND Membrane[Title/Abstract]) AND Oxygenation*[Title/Abstract]	10302
50	Search ((Extracorporeal[Title/Abstract]) AND Life[Title/Abstract]) AND Support*[Title/Abstract]	2907
51	Search ECMO[Title/Abstract]	7248
52	Search ECLS[Title/Abstract]	1255
53	Search "Heart-Lung Machine"[Mesh]	2087
54	Search ((Heart[Title/Abstract]) AND Lung[Title/Abstract]) AND Machine*[Title/Abstract]	1026
55	Search "Extracorporeal Circulation"[Mesh]	81849
56	Search (Extracorporeal[Title/Abstract]) AND Circulat*[Title/Abstract]	10838
57	Search ((extracorporeal[Title/Abstract]) AND cardiopulmonary[Title/Abstract]) AND resuscitation*[Title/Abstract]	1085
58	Search (extracorporeal[Title/Abstract]) AND CPR[Title/Abstract]	392
59	Search ECPR[Title/Abstract]	306
60	23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33 OR 34 OR 35 OR 36 OR 37 OR 38 OR 39 OR 40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48 OR 49 OR 50 OR 51 OR 52 OR 53 OR 54 OR 55 OR 56 OR 57 OR 58 OR 59	149904

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61	22 AND 60	18894
62	Search ((animals[mh]) NOT humans[mh])	4670734
63	Search Animals, Laboratory[mh]	862222
64	Search Animal Experimentation[mh]	9307
65	Search Models, Animal[mh]	556860
66	Search rodentia[mh]	3186378
67	Search ((rat[Title] OR rats[Title] OR mouse[Title] OR mice[Title]))	1315929
68	Search (62 OR 63 OR 64 OR 65 OR 66 OR 67)	5568284
69	Search (61 NOT 68)	16857
70	Search ((randomized controlled trial[Publication Type]) OR randomized controlled trial[mh]) OR Randomized Controlled Trials as Topic[mh]	626933
71	Search controlled clinical trial[Publication Type]	589588
72	Search ((randomized[Title/Abstract]) OR randomised[Title/Abstract])	605914
73	Search placebo[Title/Abstract]	210526
74	Search randomly[Title/Abstract]	327640
75	Search trial[Title]	212653
76	Search (70 OR 71 OR 72 OR 73 OR 74 OR 75)	1230471
77	Search (69 and 76)	1960

Search strategy in Web of Science

#	Searches	Results
1	TS=(heart AND infarct*)	342744
2	TS=(myocard* AND infarct*)	445216
3	TS=(heart AND attack*)	42923
4	TS=(cardiovascular AND stroke*)	225229
5	TS=STEMI	18399
6	TS=(NSTEMI OR Non-STEMI OR (Non AND STEMI))	7289
7	TS=(Myocard* AND Ischemi*)	187296
8	TS=(heart AND ischemi*)	271370
9	TS=(Acute AND Coronary AND Syndrom*)	70891
10	TS=ACS	40200
11	TS=(Coronary AND Artery AND Disease*)	346098
12	TS=(Coronary AND Arterioscleros*)	49371
13	TS=Angina*	102862
14	TS=Stenocardia*	1165
15	TS=(Angor AND Pectoris)	471
16	TS=(Myocardial AND Preinfarction AND Syndrom*)	61
17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16	1073401
18	TS=(Heart AND Assist* AND (Device* OR pump*))	33113

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19	TS=(Vascular AND Assist* AND (Device* OR pump*))	9542
20	TS=(cardiovascular AND Assist* AND (Device* OR pump*))	20738
21	TS=(Artificial AND Ventricle*)	9885
22	TS=((Ventricular OR Ventricle) AND Assist* AND (Device* OR pump*))	26361
23	TS=(Artificial AND Heart AND Ventricle*)	8704
24	TS=(blood AND pump*)	58214
25	TS=(cardiac AND resuscitat*)	37874
26	TS=(counterpulsation AND device*)	1151
27	TS=(mechanical AND circulat* AND support*)	20555
28	TS=(Cardiac AND Assist* AND System*)	24348
29	TS=(Assist* AND Circulat*)	57139
30	TS=LVAD	8035
31	TS=pVAD	228
32	TS=TandemHeart*	252
33	TS=HeartMate*	2455
34	TS=Impella*	1277
35	TS=(Intra AND Aortic AND Balloon AND (Pump* OR counterpulsation))	7211
36	TS=(Intraaortic AND Balloon AND (Pump* OR counterpulsation))	2904
37	TS=IABP	2699
38	TS=IABC	262
39	TS=(Extracorporeal AND Membrane AND Oxygenation*)	18902
40	TS=(Extracorporeal AND Life AND Support*)	5624
41	TS=ECMO	10616
42	TS=ECLS	1918
43	TS=(Heart AND Lung AND Machine*)	3410
44	TS=(Extracorporeal AND Circulat*)	27421
45	TS=(extracorporeal AND cardiopulmonary AND resuscitation*)	2273
46	TS=(extracorporeal AND CPR)	647
47	TS=ECPR	497
48	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47	240395
49	#17 AND #48	45195
50	TS= clinical trial* OR TS=research design OR TS=comparative stud* OR TS=evaluation stud* OR TS=controlled trial* OR TS=follow-up stud* OR TS=prospective stud* OR TS=random* OR TS=placebo* OR TS=(single blind*) OR TS=(double blind*)	10164129
51	#49 AND #50	16279
52	#49 AND #50 limit clinical trials	2365

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Search strategy in ClinicalTrials.gov

Basic search

86 Studies found for: Ventricular Assist Device OR Intra Aortic Balloon Pump OR Extracorporeal Membrane Oxygenation | MI OR Myocardial Infarction OR Coronary Artery Disease

Search strategy in Clinicaltrialsregister.eu

Basic search

32 Studies found for: "ventricular assist device" OR "intra aortic balloon pump" OR "Extracorporeal Membrane Oxygenation"

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The Endpoint Definitions of the Included Studies

Study	Endpoint	Definition
SEMPER FI[13]	Reinfarction	NA.
	Repeat revascularization	NA.
	Heart failure	Congestive heart failure was defined as a hospital admission due to clinical signs of heart failure and the necessity of administration of intravenous diuretics or inotropes.
	Stroke-TIA	NA.
	Bleeding	Major bleeding events according to the Global Use of Strategies to Open Coronary Arteries (GUSTO) trial.
	Arrhythmias	NA.
	Vascular complications	NA.
Zhou et al[22]	Reinfarction	NA.
	Repeat revascularization	NA.
	Heart failure	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	Malignant arrhythmia (not detailed).
	Vascular complications	NA.
IABP-SHOCK II[7,21]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Heart failure	NA.
	Stroke-TIA	Stroke identified by the presence of new neurologic symptoms in conjunction with signs of ischemia or bleeding on computed tomography.
	Bleeding	Severe or life-threatening bleeding and moderate bleeding as assessed according to the Global Use of Strategies to Open Occluded Coronary Arteries (GUSTO) criteria1.
	Arrhythmias	NA.
	Vascular complications	Peripheral ischemic vascular complications requiring surgical or interventional therapy.
CRISP AMI[16]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Heart failure	Heart failure (not detailed).
	Stroke-TIA	Stroke (not detailed).
	Bleeding	Major bleeding events defined by theGlobal Use of Strategies To Open Coronary Arteries (GUSTO) trial and transfusions.
	Arrhythmias	NA.

	Vascular complications	Major vascular complications defined to include limb ischemia requiring intervention, distal embolization, major dissection, pseudoaneurysm or arteriovenousfistula, hematoma larger than 5 cm, and amputations.
Gu et al[23]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Heart failure	NA.
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Vascular complications (not detailed).
Prondzinsky et al[24]	Reinfarction	NA.
	Repeat revascularization	NA.
	Heart failure	NA.
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmia (not detailed).
	Vascular complications	NA.
Arias et al[26]	Reinfarction	NA.
	Repeat revascularization	NA.
	Heart failure	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.
Vijayalakshmi et al[27]	Reinfarction	Reinfarction was defined as chest pain associated with ST-segment re-elevation and an increase in cardiac enzymes (3 times creatine kinase release).
	Repeat revascularization	Repeat revascularization (not detailed).
	Heart failure	Heart failure (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmias including ventricular tachycardia, ventricular fibrillation or atrial fibrillation.
	Vascular complications	Vascular complications (not detailed).
IMPRESS[14]	Reinfarction	Reinfarction Defined by 3rd definition MI. Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following: 1. Symptoms of ischemia. 2. New or presumed new significant ST-segment–T wave

		(ST–T) changes or new left bundle branch block (LBBB). 3. Development of pathological Q waves in the ECG. 4. Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. 5. Identification of an intracoronary thrombus by angiography or autopsy.
	Repeat revascularization	Repeat PCI or CABG.
	Heart failure	NA.
	Stroke-TIA	Any stroke confirmed by a neurologist and a concurring CT scan.
	Bleeding	1.Bleeding with associated serum hemoglobin level decrease of at least 5 g/dL (=3.1 mmol/L). The decrease in hemoglobin will be calculated as the last recorded Hb measurement preceding the onset of the bleeding, subtracted by the nadir Hb measurement (associated with the bleeding). 2. Bleeding necessitating a minimum of 2 packed cells of blood product transfusion (only blood transfusions that are explicitly related to the bleeding are taken into account). 3. The need for surgery to control the bleeding.
	Arrhythmias	Placement of implantable cardioverter-defibrillator.
	Vascular complications	1.A major bleed arising at the arterial access site (for major bleeding definition see above) requiring extraction of the device. 2. A thrombotic occlusion of the femoral artery. 3. Limb ischemia requiring extraction of either of the study devices. 4. The need for vascular surgery to correct a vascular complication.
Seyfarth et al[25]	Reinfarction	NA.
	Repeat revascularization	NA.
	Heart failure	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Limb ischemia (not detailed).
Thiele et al[15]	Reinfarction	NA.
	Repeat revascularization	NA.
	Heart failure	NA.
	Stroke-TIA	Cerebrovascular accidents with neurological dysfunction.
	Bleeding	Major bleedings requiring transfusion of blood components.
	Arrhythmias	NA.
	Vascular complications	Lower extremity ischaemia requiring surgical or interventional action.

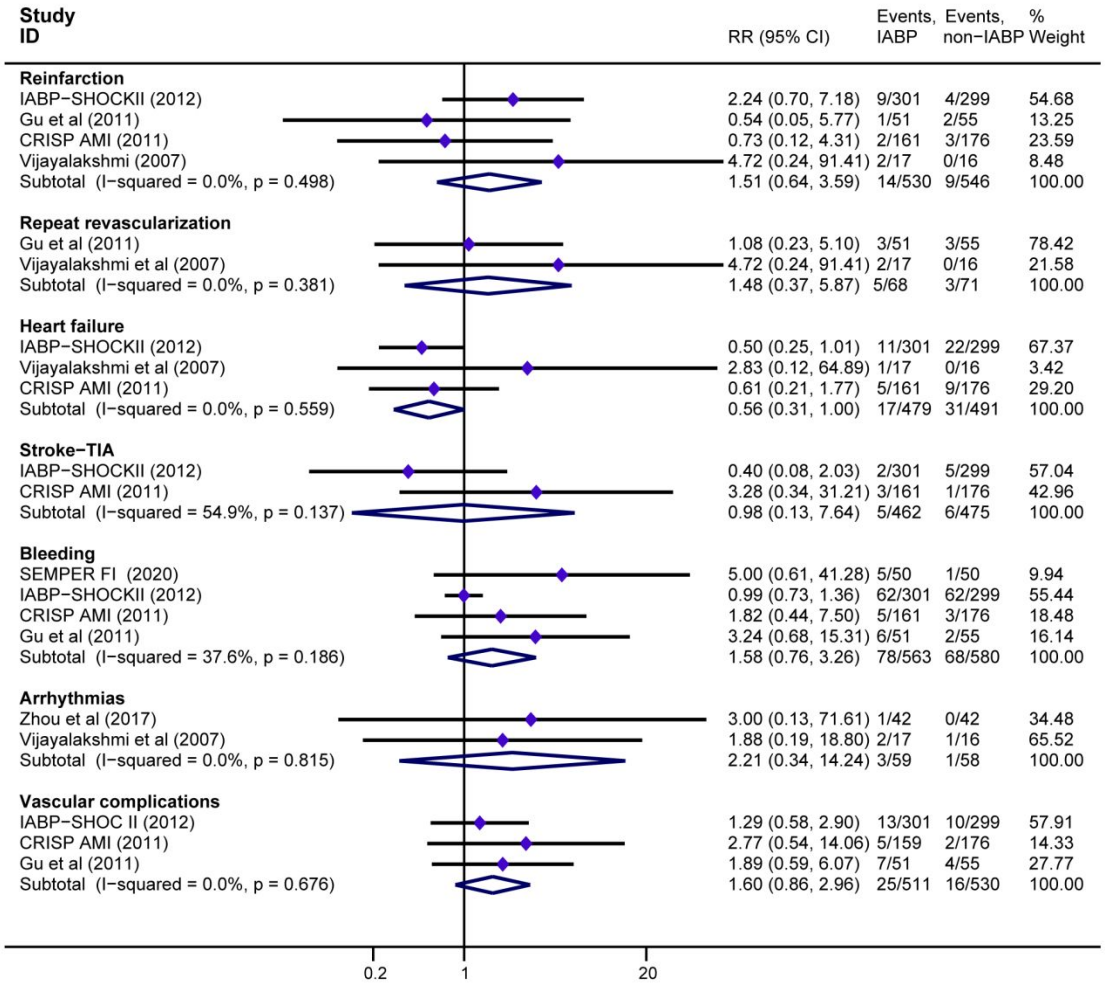
NA indicates not available.

Sensitivity analyses of the all-cause mortality by comparison between IABP and non-IABP

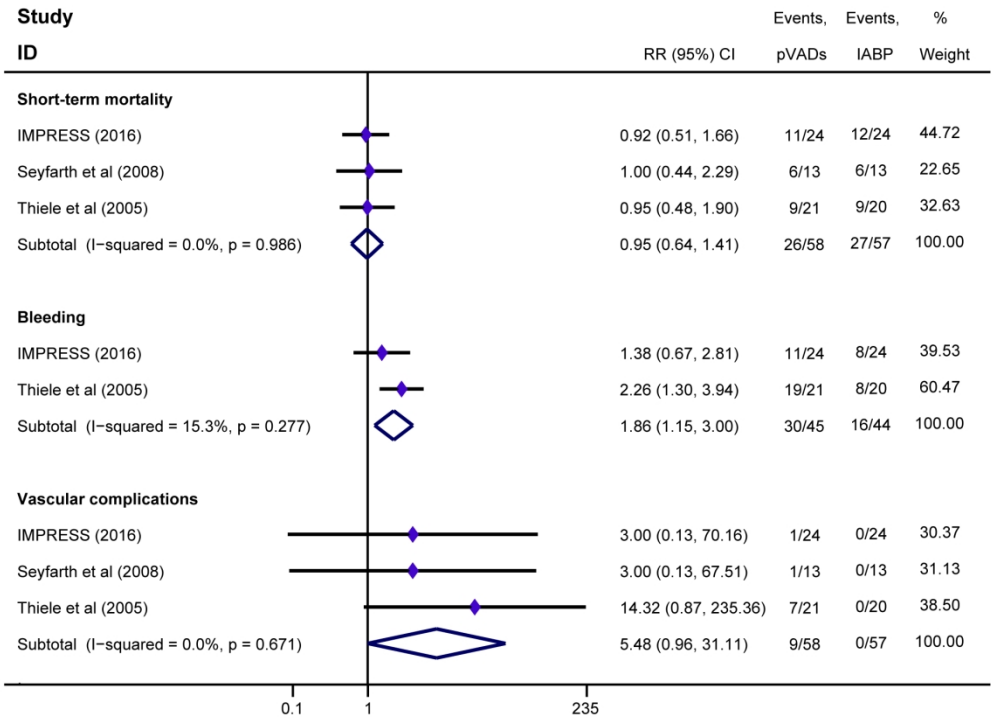
		Risk Ratio (95%CI)
Short-term mortality		
	Fixed effects model	0.950 (0.796-1.132)
	Random effects model	0.916 (0.571-1.469)
	Excluding high risk bias studies	
	Zhou et al[22] and Arias et al[26]	0.855 (0.505-1.447)
	Excluding each of the studies	
	SEMPER FI[13]	0.939 (0.570-1.546)
	Zhou et al[22]	0.804 (0.518-1.246)
	IABP-SHOCK II[7]	0.926 (0.456-1.881)
	CRISP AMI[16]	0.980 (0.590-1.629)
	Gu et al[23]	1.051 (0.664-1.666)
	Prondzinsky et al[24]	0.821 (0.480-1.405)
	Arias et al[26]	1.008 (0.579-1.754)
	Vijayalakshmi et al[27]	0.872 (0.547-1.391)
Long-term mortality		
	Fixed effects model	0.949 (0.813-1.108)
	Random effects model	0.870 (0.598-1.266)
	Excluding high risk bias studies	
	Zhou et al[22]	0.723 (0.433-1.207)
	Excluding each of the studies	
	SEMPER FI[13]	0.871 (0.579-1.309)
	Zhou et al[22]	0.723 (0.433-1.207)
	IABP-SHOCK II[21]	0.713 (0.368-1.380)
	CRISP AMI[16]	0.953 (0.697-1.302)
	Gu et al[23]	0.999 (0.749-1.333)

CI indicates confidence interval.

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Study
ID
RR (95%) CI
Events, pVADs
Events, IABP
% Weight
Short-term mortality
IMPRESS (2016) 0.92 (0.51, 1.66) 11/24 12/24 44.72
Seyfarth et al (2008) 1.00 (0.44, 2.29) 6/13 6/13 22.65
Thiele et al (2005) 0.95 (0.48, 1.90) 9/21 9/20 32.63
Subtotal (I−squared = 0.0%, p = 0.986) 0.95 (0.64, 1.41) 26/58 27/57 100.00
Bleeding
IMPRESS (2016) 1.38 (0.67, 2.81) 11/24 8/24 39.53
Thiele et al (2005) 2.26 (1.30, 3.94) 19/21 8/20 60.47
Subtotal (I−squared = 15.3%, p = 0.277) 1.86 (1.15, 3.00) 30/45 16/44 100.00
Vascular complications
IMPRESS (2016) 3.00 (0.13, 70.16) 1/24 0/24 30.37
Seyfarth et al (2008) 3.00 (0.13, 67.51) 1/13 0/13 31.13
Thiele et al (2005) 14.32 (0.87, 235.36) 7/21 0/20 38.50
Subtotal (I−squared = 0.0%, p = 0.671) 5.48 (0.96, 31.11) 9/58 0/57 100.00
0.1
1
235

	Risk Ratio (95%CI)
Fixed effects model	0.947 (0.638-1.406)
Random effects model	0.947 (0.638-1.405)
Excluding high risk bias studies	None
Excluding each of the studies	
IMPRESS[14]	0.972 (0.571-1.653)
Seyfarth et al[25]	0.932 (0.594-1.460)
Thiele et al[15]	0.944 (0.583-1.528)

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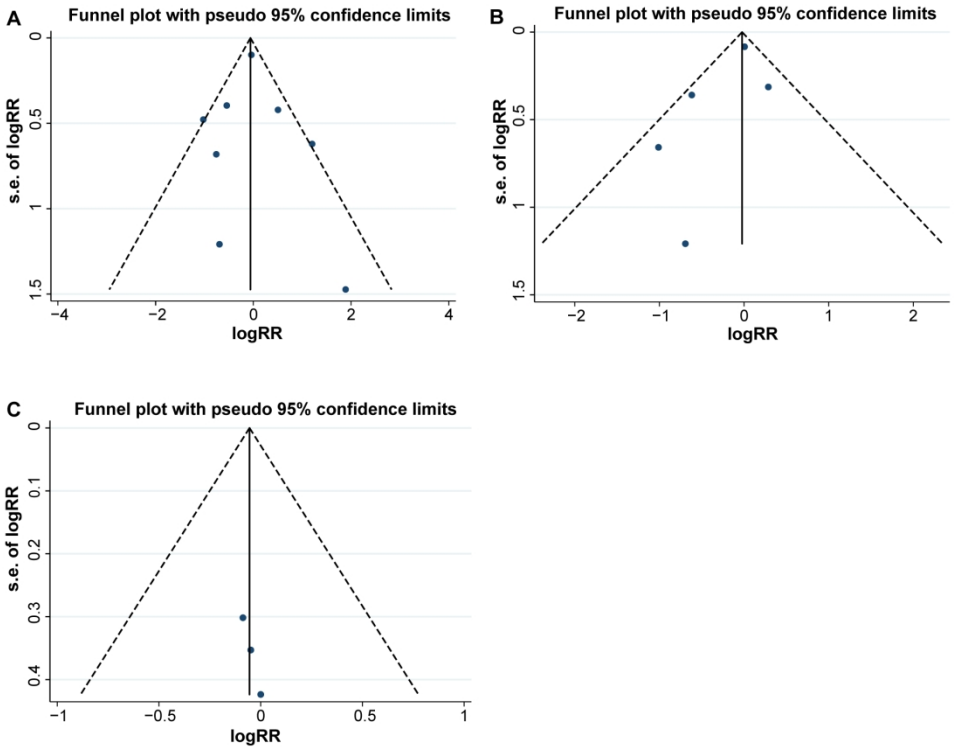
Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Study	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other bias
SEMPER FI[13]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhou et al[22]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IABP-SHOCK II[7,21]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
CRISP AMI[16]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Gu et al[23]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Prondzinsky et al[24]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Arias et al[26]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Vijayalakshmi et al[27]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IMPRESS[14]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Seyfarth et al[25]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Thiele et al[15]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk

Low risk

Unclear risk

High risk


A
Funnel plot with pseudo 95% confidence limits
s.e. of logRR
0
0.5
1
1.5
−4
−2
0
2
4
logRR
B
Funnel plot with pseudo 95% confidence limits
s.e. of logRR
0
0.5
1
1.5
−2
−1
0
1
2
logRR
C
Funnel plot with pseudo 95% confidence limits
s.e. of logRR
0
0.1
0.2
0.3
0.4
−1
−0.5
0
0.5
1
logRR

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Effects of Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients Undergoing Stent Implantation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Effects of Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients Undergoing Stent Implantation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Abstract

Objective

The survival benefit of using mechanical circulatory support (MCS) in acute myocardial infarction (AMI) patients is still controversial. It is necessary to explore the impact on clinical outcomes of MCS in AMI patients with stenting.

Design

Systematic review and meta-analysis.

Data sources

The Embase, Cochrane Library, Medline, PubMed, Web of Science, ClinicalTrials.gov and Clinicaltrialsregister.eu databases were searched from database inception to February 2021.

Eligibility criteria

Randomized clinical trials (RCTs) on MCS use in AMI patients undergoing stent implantation were included.

Data extraction and synthesis

Data were extracted and summarized independently by two reviewers. Risk ratios (RRs) and 95% confidence intervals (CIs) were calculated for clinical outcomes according to random effects model.

Results

Twelve studies of 1497 AMI patients were included, nine studies including 1382 patients compared MCS with non-MCS, and three studies including 115 patients compared percutaneous ventricular assist devices (pVADs) versus intra-aortic balloon pump (IABP). Compared with non-MCS, MCS was not associated with short-term (within 30 days) (RR=0.90; 95% CI, 0.57-1.41; I^2=46.8%) and long-term (at least 6 months) (RR=0.82; 95% CI, 0.57-1.17; I^2=37.6%) mortality reductions. In the subset of patients without cardiogenic shock (CS), compared with non-MCS, the patients with IABP

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treatment significantly decrease the long-term mortality (RR=0.49; 95% CI, 0.27-0.90; I^2=0), but without the short-term mortality reductions (RR=0.51; 95% CI, 0.22-1.19; I^2=17.9%). While, in the patients with CS, the patients with MCS did not benefit the short-term (RR=1.09; 95% CI, 0.67-1.79; I^2=46.6%) or long-term (RR=1.00; 95% CI, 0.75-1.33; I^2=22.1%) survival. Moreover, the application of pVADs increased risk of bleeding (RR=1.86; 95% CI, 1.15-3.00; I^2=15.3%) compared to IABP treatment (RR=1.86; 95% CI, 1.15-3.00; I^2=15.3%).

Conclusions

In all AMI patients undergoing stent implantation, the MCS use does not reduce all-cause mortality. Patients without CS can benefit from MCS regarding to long-term survival, while patients with CS seem not.

Keywords: Acute myocardial infarction (AMI), Percutaneous coronary intervention (PCI), Stent implantation, Mechanical circulatory support (MCS), Intra-aortic balloon pump (IABP), Percutaneous ventricular assist devices (pVADs)

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Article Summary

Strengths and limitations of this study

- We conducted a comprehensive systematic search of literature from various databases to find out the application of MCS in AMI patients with stent implementation.
- To our knowledge, this meta-analysis is the first to explore the effect of MCS in AMI patients undergoing stent implantation.
- Limited number of RCTs comparing the effect of pVADs with IABP, and different types of pVADs, the results should be interpreted with cautious.

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Introduction

Acute myocardial infarction (AMI) is the most common cause of hospitalization in elderly patients [1,2]. Emerging mechanical circulatory support (MCS), including intra-aortic balloon pump (IABP), percutaneous ventricular assist devices (pVADs), and extracorporeal life support (ECLS) are important haemodynamic supportive strategy for maintaining a stable haemodynamic and organ perfusion in the acute phase of myocardial infarction. However, the survival benefit of using MCS in AMI patients is still controversial.

Although previous guidelines from various committees have recommended the necessity of using MCS in patients with severe cardiac ischaemia [3,4], accumulating evidence from RCT and meta-analysis has shown that neither MCS devices have significantly improved the survival in AMI patients with CS[5,6,7,8,9]. Therefore, in AMI patients with cardiogenic shock (CS), IABP and pVADs were recommended to class IIa and class IIb in US guidelines respectively, and IABP to class IIaC-IIIA in European guidelines [10,11]. Recent meta-analyses have focused on the use of MCS in patients with CS, but no benefits for survival was found [12,13,14]. Even if the conclusions from MCS studies were something of disappointment, these data still worth to be further explored in detail.

As rapid advances in percutaneous coronary intervention (PCI) technology, stenting is better for reperfusion after AMI than balloon angioplasty [15]. However, it is noteworthy that both patient with stenting and balloon angioplasty were included in the studies of MCS [16,17]. As the revascularization strategy may bring large heterogeneous, the choice of the strategy must take into account the evidence. Therefore, in this study, we conduct a systematic review and meta-analysis to highlight the effect of MCS in AMI patients with stent implantation and further exploration on the effect of MCS are performed in the patients with or without CS.

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Methods

The current meta-analysis was conducted in accordance with the PRISMA guidelines (PRISMA checklist see online supplementary file 1) [18].

Search Strategy

We conducted a systematic search of the literature in February 2021 without restrictions on region, publication type, or language. The Embase, Cochrane Library, Medline, PubMed, and Web of Science databases were included. ClinicalTrials.gov and Clinicaltrialsregister.eu were searched as well. The MeSH search terms included the following: myocardial infarction, acute coronary syndrome, heart-assist devices, intra-aortic balloon pump and mechanical circulatory support. We excluded non-human studies. Details of the search strategy were presented in online supplementary file 2.

Study Selection

In the current meta-analysis, only RCTs in which stent implantation was the dominant form of revascularization and MCS was used during the perioperative period in patients with AMI were included. We excluded nonrandomized trials, cohort studies, cross-sectional studies, ongoing trials, trials terminated early and other publication types, including review, case reports or case series, editorials, letters, and meeting abstracts. Studies in which patients underwent coronary artery bypass grafting (CABG), balloon angioplasty or systemic thrombolysis as the main revascularization method were also excluded.

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Endpoints

The primary outcome was all-cause mortality, including short-term mortality and long-term mortality. Short-term mortality was defined as mortality occurring within 30 days. Long-term mortality occurred at least 6 months of follow-up. The secondary endpoints included reinfarction, repeat revascularization, stroke-TIA (transient ischaemic attack), bleeding (moderate to major bleeding, blood transfusion or surgery to control the bleeding), arrhythmias and vascular complications (peripheral ischaemic vascular complications, major dissection, pseudoaneurysm or arteriovenous fistula) occurring within 30 days. The endpoint definitions as applied in each study were presented in online supplementary file 3.

Data Extraction and Quality Assessment

Data from the included studies were extracted and summarized independently by two reviewers (Shi and Wang). All extracted data from the included studies were independently collected in duplicate using a standardized data extraction form, which contained authors or trails' name, year of publication, study design (interventions, sample size, percentage of stenting and bare-metal stent, PCI success, timing of MCS, time point of reported mortality, follow-up), baseline characteristics of study population (sex, age, AMI, CS, infarct-related artery, hypertension, diabetes mellitus, previous stroke/TIA and MI, prior PCI and CABG), the endpoint definitions, the centre and the time of the study enrolment. Any disagreements were resolved by a third reviewer (Sun). The quality of the included studies was assessed using the Cochrane Collaborations tool [19], which including selection

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bias, performance bias, detection bias, attrition bias, reporting bias, other possible bias. Each source of bias was classified as low, unclear, and high risk.

Statistical Analysis

We applied risk ratios (RRs) and 95% confidence intervals (CIs) to summarize the statistics. Statistical heterogeneity was assessed using Cochran's Q test and quantified using the I^2 statistic; $p < 0.10$. Heterogeneity was categorized by I^2 as follows [20]: insignificant heterogeneity, $I^2<25\%$; low heterogeneity, $25\% \leq I^2<50\%$; moderate heterogeneity, $50\% \leq I^2<75\%$; high heterogeneity, $I^2 \geq 75\%$. Considering the clinical differences among the included studies, a random-effects model was applied for statistical analyses. Sensitivity analyses were performed for primary outcomes with the following: (1) inconsistent length of follow-up, (2) different types of MCS use, (3) exclusion of research with a high risk of bias, and (4) the one-study-out method. Funnel plots, Begg and Mazumdar rank correlation and Egger's regression intercept were used to evaluate publication bias[20]. If publication bias existed, the trim-and-fill method was performed for estimating the number of missing studies and RR adjustment [21]. Statistical analyses were performed using STATA/SE 12.0 (Stata Corp LP, College Station, Texas, USA). $P<0.05$ was considered statistically significant.

Results

There are 12574 citations were identified through the search criteria. The full texts of 107 potentially eligible articles were scrutinized in detail. Among these, 15 RCTs used PCI as the main revascularization therapy, 3 RCTs on balloon angioplasty use were excluded [22,23,24], and 12 RCTs

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finally met the inclusion criteria [8,25,26,27,28,9,29,30,31,32,33,34,35]. The study selection process is outlined in Figure 1.

Characteristics of the Studies

The centre and the time of enrolment in each study were shown that no duplicate of the sample population of included studies were found (see online supplementary file 4). The detailed designs of the included studies are listed in Table 1. Among the 12 RCTs, 1497 AMI patients were included, 94.5% of patients (1391/1471) underwent stent implantation, 86.9% of patients (959/1104) were with PCI success. The follow-up period ranged from in-hospital only to 6 years. Nine studies including 1382 AMI patients compared MCS with non-MCS (693 in the MCS group and 690 in the non-MCS group) [8,25,28,9,29,30,31,33,34,35], 3 studies including 115 AMI patients compared pVADs versus IABP (58 in the pVADs group and 57 in the IABP group) [26,27,32]. Eight RCTs included CS patients, among which 5 studies that included 806 patients compared MCS with non-MCS (414 in the MCS group and 392 in the non-MCS group) [8,9,29,31,33,35], the remaining 3 studies compared pVADs versus IABP. Four RCTs included patients without CS, and all types of MCS use were IABP, which included 576 patients, compared MCS with non-MCS (279 in the IABP group and 297 in the non-IABP group) [25,28,30,34]. The summarized patients' characteristics of included studies are shown in Table 2. The mean age of the population was 63.8, and 73.3% were male, 77.1% (1046/1357) had left coronary artery-related infarction, 52.9% (726/1373) had hypertension, 27.7% (381/1373) were diabetes mellitus, 4.3% (49/1172) had a previous diagnosis of stroke/TIA, 14.3% (193/1347) had a previous diagnosis of MI, 17.6% (208/1185) underwent PCI previously and 5.0% (37/742) underwent CABG previously.

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Outcomes of Synthesis Analyses

Of the included 9 RCTs comparing the survival benefit of MCS with non-MCS, all studies reported short-term mortality, and 6 studies reported long-term mortality. MCS was not associated with a short-term mortality reduction compared to the non-MCS (MCS=23.1%, 160/693; non-MCS=23.5%, 162/689; RR=0.90; 95% CI, 0.57-1.41; p=0.640), with low heterogeneity among studies (I^2=46.8%; p=0.059) (Figure 2). Similarly, there was no significant difference between the MCS group and the non-MCS group in the long-term mortality (MCS=28.6%, 176/626; non-MCS=30.3%, 195/643; RR=0.82; 95% CI, 0.57-1.17; p=0.276), with low heterogeneity among studies (I^2=37.6%; p=0.156) (Figure 2). The results of prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 5).

For the secondary endpoints, MCS was no effect the incidence of reinfarction (MCS=2.6%, 14/530; non-MCS=1.6%, 9/546; RR=1.51; 95% CI, 0.64-3.59; p=0.346; I^2=0, p=0.498), repeat revascularization (MCS=7.9%, 7/89; non-MCS=4.3%, 4/92; RR=1.60; 95% CI, 0.49-5.24; p=0.435; I^2=0, p=0.668), stroke-TIA (MCS=1.1%, 5/462; non-MCS=1.3%, 6/475; RR=1.51; 95% CI, 0.64-3.59; p=0.988; I^2=54.9%, p=0.137), bleeding (MCS=13.9%, 78/563; non-MCS=11.7%, 68/580; RR=1.58; 95% CI, 0.76-3.26; p=0.218; I^2=37.6%, p=0.186), arrhythmias (MCS=5.1%, 3/59; non-MCS=1.7%, 1/58; RR=2.21; 95% CI, 0.34-14.24; p=0.404; I^2=0, p=0.815), and vascular complications (MCS=4.9%, 25/511; non-MCS=3.0%, 16/530; RR=1.60; 95% CI, 0.86-2.96; p=0.134; I^2=0, p=0.676) (see online supplementary file 6).

Head-to-head comparisons between different type of MCS were also performed. There were 3 RCTs comparing pVADs versus IABP included. The pooled analysis revealed that pVADs was not

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associated with a decrease in short-term mortality compared with IABP (pVADs=44.8%, 26/58; IABP=47.4%, 27/57; RR=0.95; 95% CI, 0.64-1.41; p=0.786), with insignificant heterogeneity among studies (I^2=0; p=0.986) (see online supplementary file 7). Long-term mortality was not analysed because only one study reported. For the secondary outcome analysis, only bleeding and vascular complications met the quantitative requirements for the analysis. Compared with IABP, pVADs increased the rate of bleeding (pVADs=66.7%, 30/45; IABP=36.4%, 16/44; RR=1.86; 95% CI, 1.15-3.00; p=0.011; I^2=15.3%, p=0.227) and trended towards increased vascular complications (pVADs=15.5%, 9/58; IABP=0, 0/57; RR=5.48; 95% CI, 0.96-31.11; p=0.055; I^2=0, p=0.671) (see online supplementary file 7). The results of the prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 8).

Outcomes of Subgroup Analyses

In the subset of patients without cardiogenic shock (CS), compared with the patients without MCS, the patients with MCS treatment significantly decrease the long-term (MCS=5.0%, non-MCS=10.3%; RR=0.49; 95% CI, 0.27-0.90; p=0.020; I^2=0, p=0.870) (Figure 3B), but without changed the outcome in the short-term (MCS=4.3%, non-MCS=8.1%; RR=0.51; 95% CI, 0.22-1.19; p=0.121; I^2=17.9%, p=0.301) (Figure 3A). In patients with CS, MCS did not reduce short-term mortality (MCS=35.7%, non-MCS=35.2%; RR=1.09; 95% CI, 0.67-1.79; p=0.726; I^2=46.6%, p=0.100) (Figure 3A) or long-term mortality (MCS=45.6%, non-MCS=45.9%; RR=1.00; 95% CI, 0.75-1.33; p=0.984; I^2=22.1%, p=0.227) (Figure 3B). Due to the limited number of studies, subgroup analyses of the secondary outcomes were not performed.

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All patients had CS in studies comparing pVADs versus IABP, and the results have been described in the synthesis analyses.

Study Quality and Publication Bias

The results of the literature quality assessment using the Cochrane Collaboration tool are shown in online supplementary file 9. Three trials had a high risk of bias [29,33,35]. There was no evidence of publication bias for included studies except the Egger's test for short-term mortality of studies comparing pVADs use versus IABP use (P<0.05) (online supplementary file 10 and online supplementary file 11). Then the trim-and-fill method was performed for the estimation of the number of missing studies (pVADs vs IABP) that might exist. Result suggested 2 theoretically missing studies with an adjusted RR of short-term mortality (RR=0.92; 95% CI, 0.67-1.26; p=0.59).

Discussion

Summary of the evidence

To our knowledge, it is the first study that evaluate the efficacy and safety of MCS in patients with AMI undergoing stent implantation. Compared with non-MCS, MCS does not reduce mortality in AMI patients undergoing stent implantation and other adverse events (reinfarction, repeat revascularization, stroke-TIA, bleeding, arrhythmias and vascular complications) in AMI patients undergoing stent implantation. Surprisingly, the results shows that MCS reduced long-term mortality by half in patients without CS compared to the non-MCS (5.0% versus 10.3%). This benefit does not exist in the patients with CS. The strategy of using pVADs did not outperformed IABP regarding to all-cause mortality in the patients with CS, but increased rate of bleeding compared to IABP.

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Efficacy of MCS in patients without CS

In the analysis of MCS in patients without CS, only IABP related studies were found in included RCTs. Regarding the work mechanism of IABP, deflation of the pump reduces ventricular workload and helps the ventricle push blood into the aorta during systole, and inflation of the pump improves coronary blood flow during diastole [36]. The evidence from clinical trials also shows the beneficial effect of IABP on haemodynamic, as these pumps can significantly improve myocardial coronary blood flow and myocardial ischaemia [37,38]. IABP might have a greater effect in patients with critically reduced coronary artery perfusion and temporarily impaired left ventricular function [39]. The haemodynamic of patients without CS may recover more quickly with IABP support, avoiding more serious clinical consequences, such as progression to CS, severe left heart failure, or even death. In the last decades, the application of PCI technology significantly reduces mortality in STEMI patients [11,40]. In particular, stenting reduces acute risk of major complications and restenosis [15]. One of our findings shows that IABP can benefit long-term mortality reduction in AMI patients without CS. This suggests that the use of IABP in AMI patients without CS with contemporary recommended reperfusion strategy (stent implantation) is reasonable. Some previous meta-analyses showed similar conclusions, but there was no detailed distinction between the revascularization strategies of thrombolytic therapy or PCI (balloon angioplasty and stent implantation) [16,17,41]. This result from the meta-analysis generates a promising hypothesis that IABP reduced long-term mortality in the non-CS AMI patients receiving stenting treatment. Studies with randomized control and decent sample size are warranted to confirm this finding.

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Efficacy of MCS in patients with CS

Previous guidelines have recommended IABP as the complementary strategy to maintaining a stable haemodynamic in patients with combined AMI and CS [3,4,42]. However, accumulating evidence from large-scale RCTs, including IABP-SHOCK II and PAMI-II, as well as registry data, did not show that the use of IABP bring benefit to the mortality reduction in the AMI patients with CS [8,9,22,43,44]. The findings from our study also showed that MCS was not associated with an all-cause mortality reduction in patients with CS undergoing stent implantation. Patients with AMI and CS are considered a high-risk group with more impaired cardiac function and haemodynamic instability and are more likely to develop circulatory failure during PCI [45]. The IABP SHOCK trial showed that IABP treatment do not significantly improve the hemodynamic profile in patients with combined AMI and CS [46]. Even though AMI patients with CS with currently advanced reperfusion strategy and MCS combined use, mortality was not reduced in our meta-analysis, which is consistent with the results from previous meta-analyses focusing on the impact of IABP in CS patients [14,47,48,49]. Therefore, IABP should be used with caution when patients have a combination of AMI and CS.

Several clinical trials exploring the efficacy of using different types of MCS in patients with CS, including ECMO, pVADs or the combination of two types MCS, are ongoing (NCT04184635, NCT03637205, NCT01633502, NCT03813134, NCT02301819, NCT03729765, NCT03431467, NCT03947619). We look forward to seeing those studies provide new insights into better MCS strategies in improving survival in patients with AMI and CS. Moreover, we compared the efficacy of survival between pVADs and IABP in patients combined with AMI and CS. The results showed that pVADs do not reduce the all-cause mortality, but significantly increased the rate of bleeding. These results are consistent with previous which focus on the impact of pVADs in CS patients [6,13].

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Clinical implications

In AMI patients without CS and received stenting implementation, the application of IABP may improve the long-term survival. However, in the patients with CS, neither IABP nor pVADs improves the survival outcome. Moreover, pVADs may bring higher risk of bleeding in the AMI patients with CS. This result generates an interesting hypothesis that IABP reduced long-term mortality in the non-CS AMI patients who receiving stenting treatment. RCT studies with a large number of patients are warranted to confirm this finding.

Limitations

There are several limitations in the current meta-analysis. First, the number of included RCTs and participants was limited. Second, due to few reports of secondary endpoints, subsequent subgroup analyses could not be performed. Third, there was high heterogeneity in the pooled CS group, which may be explained by inconsistent timing of initiation, higher incidence of interactions, and inconsistent time points for reporting mortality. Fourth, there were very limited number of RCTs comparing the efficacy between pVADs with IABP and the pVADs devices have large heterogeneities.

Conclusions

The use of MCS does not reduce mortality in overall AMI patients undergoing stent implantation. However, IABP demonstrated a promising effect in reducing the long-term mortality in AMI patents

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without CS. However, in patients with CS, MCS is not associated with a mortality reduction and pVADs further increased risk of bleeding.

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Authors' contributions

JC and YL contributed to the conception and revision of the review. YS and YW performed systematic literature search, study selection data extraction and data analyses. YS contributed to writing of the manuscript. XS contributed to review the data and manuscript. YT, MJ, YB, SL, WJ and HY checked the manuscript.

Patient and public involvement

No patient involved.

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Ethical Approval Statement

Not applicable.

Declaration of interest

None.

Data sharing statement

All data associated with this manuscript are included in the main text and supplementary materials.

Supplementary materials

Supplementary data of this article can be found online.

Online supplementary file 1: PRISMA checklist for the meta-analysis

Online supplementary file 2: Search strategy

Online supplementary file 3: The Endpoint Definitions of the Included Studies

Online supplementary file 4: The Enrollment Information of the Included Studies

Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

Online supplementary file 6: Forest plots for the secondary end points of AMI patients treated with MCS compared with non-MCS. TIA, transient ischaemic attack.

Online supplementary file 7: Forest plots for outcomes occurrence comparing pVADs use versus IABP use

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Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Online supplementary file 10: Funnel plot of standard error for: A. MCS compared with non-MCS in short-term mortality; B. MCS compared with non-MCS in long-term mortality; C. pVADs compared with IABP in short-term mortality

Online supplementary file 11: Publication bias on the association of primary outcome

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Figure legends

Figure 1. Flow diagram of study selection

Randomized clinical trials were included.

Figure 2. Forest plots for the all-cause mortality

Risk ratio of all-cause mortality of AMI patients treated with MCS compared with non-MCS in short-term or long-term. CI, confidence interval; MCS, mechanical circulatory support; non-MCS, no plan of MCS use during peri-operative period; Short-term mortality, occurring within 30 days; long-term mortality, occurring at least 6 months of follow-up.

Figure 3. Forest plots showing the all-cause mortality for the subgroup analyses

A. Risk ratio of short-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. **B.** Risk ratio of long-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. CS, cardiogenic shock.

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Table 1. Study Design of the Included Studies

Study	Year	MCS group / non-MCS group	No. of Patients	Stenting, %	BMS, %	PCI success, %	Timing of MCS	Time point of reported mortality	Follow-Up
SEMPER FI[25]	2020	IABP+ stenting/stenting	50/50	100	NA	90/96	After Stenting	30 days, 6 months	6 months
Zhou et al[29]	2017	IABP+ stenting/stenting	42/42	100	NA	NA	Before Stenting	In hospital, 12 months	12 months
IABP-SHOCK II[8,9]	2012, 2013	IABP+ stenting/stenting	301/299	89.8	50.2/51.5	82.2/81.5	Before or after Stenting	30 days, 6 months	6 years
CRISP AMI[28]	2011	IABP+ stenting/stenting	161/176	96.9	53.7/53.1	92.9/95.3	Before Stenting	30 days, 6 months	6 months
Gu et al[30]	2011	IABP+ stenting/stenting	51/55	100	NA	NA	Before Stenting	30 days, 6 months	6 months
Prondzinsky et al[31]	2010	IABP+ stenting/stenting	19/21	85	NA	NA	After Stenting	In hospital	In hospital
Arias et al[33]	2005	IABP+ stenting/stenting	31/9	100	NA	NA	Before Stenting	In hospital	In hospital
Vijayalakshmi et al[34]	2007	IABP+ stenting/stenting	17/16	100	NA	NA	After Stenting	In hospital, 30 days	30 days
IMPRESS[26]	2017	pVADs+ stenting /IABP+ stenting	24/24	97.9	4/8	NA	After Stenting	30 days, 6 months	6 months
Seyfarth et al[32]	2008	pVADs+ stenting /IABP+ stenting	13/13	NA	NA	100/92	After Stenting	30 days	30 days
Thiele et al[27]	2005	pVADs+ stenting /IABP+ stenting	21/20	95.1	NA	80/74	Before or after Stenting	30 days	30 days
Korbinian et al[35]	2020	ECLS+ stenting / stenting	21/21	100	NA	NA	After stenting	30 days, 12 months	12 months

MCS, indicates mechanical circulatory support; IABP, intra-aortic balloon pump; PCI, percutaneous coronary intervention, including stenting and balloon angioplasty; pVADs, percutaneous ventricular assist devices; ECLS, extracorporeal life support; BMS, bare-metal stent; NA, not available. PCI success was defined as TIMI III flow.

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Table 2. Baseline Clinical Characteristics of the Included Studies

Study (year)	Male, %	Mean age, year	AMI, %	CS, %	Infarct-related artery (LM/LAD/LCX/RCA), %	Hypertension, %	Diabetes mellitus, %	Previous stroke/TIA, %	Prior MI, %	Prior PCI, %	Prior CABG, %
SEMPER FI[25]	74/78	64/63	100	0	1.0/50.0/11.0/33.0	24/32	8/18	2/2	6/10	8/12	2/8
Zhou et al[29]	67/62	57.2/56.7	100	100	0/34.5/38.1/27.4	NA	NA	NA	NA	NA	NA
IABP-SHOCK II[8,9]	70.6/67.1	70/69	100	100	9/42.2/18.7/25.3	72/66.6	35.4/30.1	8/6.7	23.7/20.4	21.1/17.4	6.7/4.0
CRISP AMI[28]	82/81.8	56.1/57.7	100	0	0/97.3/0/0.3	24.2/34.1	16.8/20.5	0/0.6	0/0	1.9/1.1	NA
Gu et al[30]	56.9/65.5	67.4/66.6	100	0	3.8/47.2/11.3/37.7	68.6/60	35.5/34.5	NA	3.9/5.5	68.6/70.9	NA
Prondzinsky et al[31]	74/81	62.1/66.1	100	100	NA	42.1/47.6	52.6/47.6	NA	21.1/23.8	NA	NA
Arias et al[33]	NA	NA	100	100	0/60.0/22.5/8.3	NA	NA	NA	NA	NA	NA
Vijayalakshmi et al[34]	82/88	40/45	100	0	NA	25/38	18/25	NA	12/25	NA	NA
IMPRESS[26]	75/83	58/59	100	100	6.3/64.6/18.8/10.4	20/29	9/13	0/4	5/4	NA	NA
Seyfarth et al[32]	62/85	65/67	100	100	NA	54/69	39/23	NA	NA	NA	NA
Thiele et al[27]	76/75	63/65	100	100	NA	76/75	52/55	NA	62/45	NA	NA
Korbinian et al[35]	76.2/ 95.2	62/ 70	100	100	9.5/54.8/16.7/19.0	55/75	15/30	0/10	15/30	20/20	0/0

AMI indicates acute myocardial infarction; CS, cardiogenic shock; LM, left main; LAD, left anterior descending; LCX, left circumflex; RCA, right coronary artery; TIA, transient ischemic attack; MI, myocardial infarction; CABG, coronary artery bypass graft.

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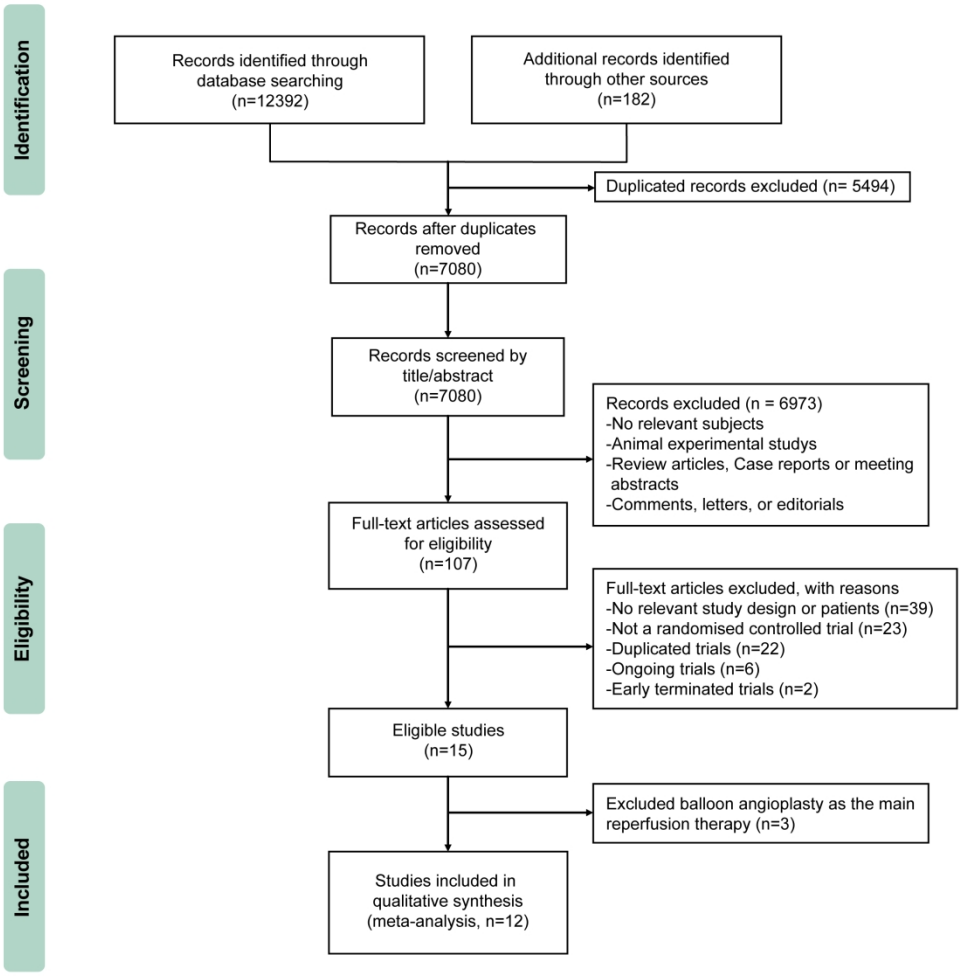


Figure 1. Flow diagram of study selection
Randomized clinical trials were included.

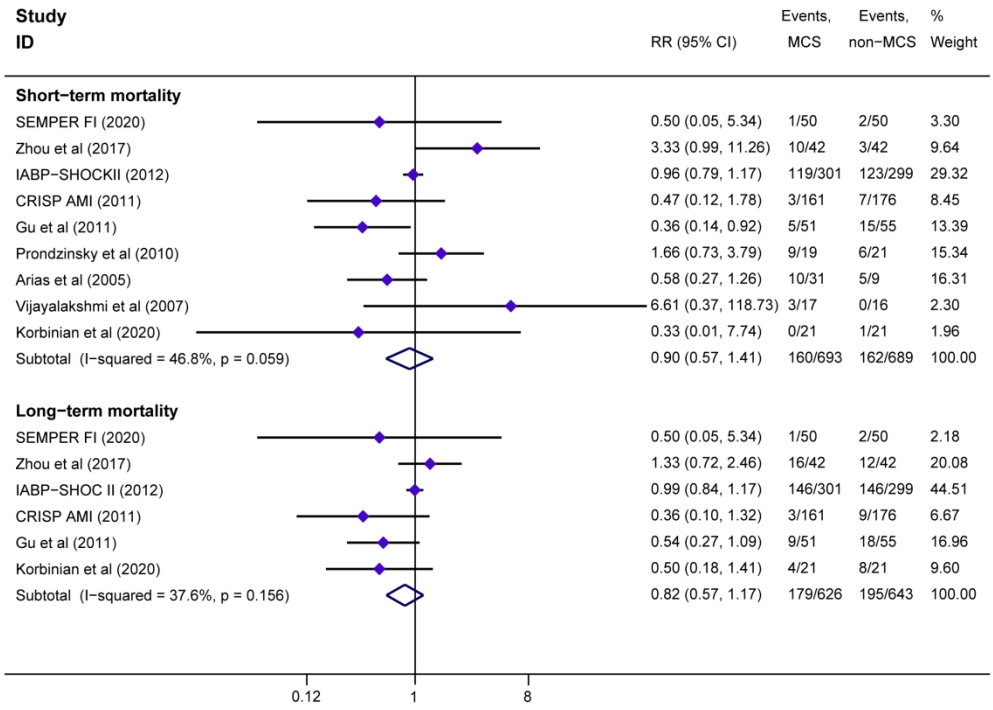


Figure 2. Forest plots for the all-cause mortality
Risk ratio of all-cause mortality of AMI patients treated with MCS compared with non-MCS in short-term or long-term. CI, confidence interval; MCS, mechanical circulatory support; non-MCS, no plan of MCS use during peri-operative period; Short-term mortality, occurring within 30 days; long-term mortality, occurring at least 6 months of follow-up.

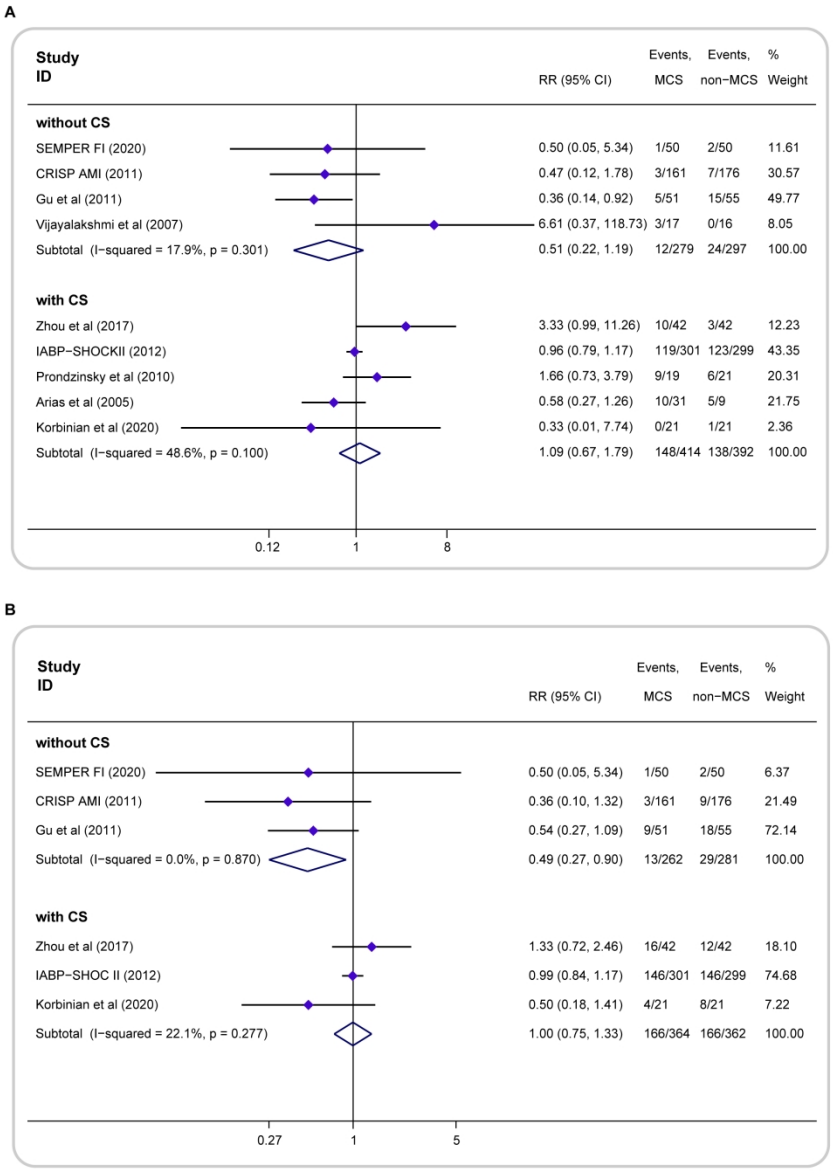


Figure 3. Forest plots showing the all-cause mortality for the subgroup analyses
A. Risk ratio of short-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. B. Risk ratio of long-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. CS, cardiogenic shock.

Online supplementary file 1: PRISMA checklist for the meta-analysis

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6 and Online supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7, and Online supplementary file 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used	7

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		in any data synthesis.	
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9, Table 1, Table 2 and Online supplementary file 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 and Online supplementary file 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12, Online supplementary file 9, Online supplementary file 10 and Online supplementary file 11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-11, Figure 3, Online supplementary file 5 and Online supplementary file 8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15

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Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

NA indicates not available.

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Online supplementary file 2: Search strategy

Search strategy in Embase

#	Searches	Results
1	'heart infarction'/exp	409027
2	heart:ab,ti AND infarct*:ab,ti	106516
3	myocard*:ab,ti AND infarct*:ab,ti	306544
4	heart:ab,ti AND attack*:ab,ti	17766
5	cardiovascular:ab,ti AND stroke*:ab,ti	60439
6	stemi:ab,ti	29180
7	nstemi:ab,ti OR 'non stemi':ab,ti OR (non:ab,ti AND stemi:ab,ti)	14111
8	'ischemic heart disease'/exp	709611
9	heart:ab,ti AND ischemi*:ab,ti	120006
10	myocard*:ab,ti AND ischemi*:ab,ti	112169
11	'coronary artery disease'/exp	353023
12	coronary:ab,ti AND artery:ab,ti AND disease*:ab,ti	178679
13	coronary:ab,ti AND atheroscleros*:ab,ti	43833
14	acute:ab,ti AND coronary:ab,ti AND syndrom*:ab,ti	62869
15	acs:ab,ti	43936
16	'angina pectoris'/exp	105403
17	angina*:ab,ti	79942
18	angor:ab,ti AND pectoris:ab,ti	86
19	stenocardia*:ab,ti	927
20	myocardial:ab,ti AND preinfarction:ab,ti AND syndrom*:ab,ti	41
21	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20	1020302
22	'heart assist device'/exp	43026
23	heart:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	18671
24	vascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2623
25	cardiovascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	1949
26	artificial:ab,ti AND ventricle*:ab,ti	1772
27	ventricle:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2918
28	ventricular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	24992
29	artificial:ab,ti AND heart:ab,ti AND ventricle*:ab,ti	890
30	blood:ab,ti AND pump*:ab,ti	27560
31	cardiac:ab,ti AND resuscitat*:ab,ti	32204
32	counterpulsation:ab,ti AND device*:ab,ti	646
33	mechanical:ab,ti AND circulat*:ab,ti AND support*:ab,ti	9334
34	cardiac:ab,ti AND assist*:ab,ti AND system*:ab,ti	5468
35	assist*:ab,ti AND circulat*:ab,ti	13513

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36	lvad:ab,ti	12954
37	pvad:ab,ti	375
38	tandemheart*:ab,ti	332
39	heartmate*:ab,ti	3663
40	impella*:ab,ti	2192
41	intra:ab,ti AND aortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	6288
42	intraaortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	2414
43	iabp:ab,ti	4778
44	iabc:ab,ti	172
45	'extracorporeal oxygenation'/exp	27601
46	extracorporeal:ab,ti AND membrane:ab,ti AND oxygenation*:ab,ti	17345
47	extracorporeal:ab,ti AND life:ab,ti AND support*:ab,ti	4998
48	ecmo:ab,ti	17782
49	ecls:ab,ti	2325
50	'heart lung machine'/exp	2137
51	heart:ab,ti AND lung:ab,ti AND machine*:ab,ti	1596
52	'extracorporeal circulation'/exp	73515
53	extracorporeal:ab,ti AND circulat*:ab,ti	15079
54	extracorporeal:ab,ti AND cardiopulmonary:ab,ti AND resuscitation*:ab,ti	1987
55	extracorporeal:ab,ti AND CPR:ab,ti	993
56	ecpr:ab,ti	901
57	#22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56	215006
58	#21 AND #57	39754
59	'animals'/exp NOT 'humans'/exp	5572218
60	'animals, laboratory'/exp	716834
61	'animal experiment'/exp	2667375
62	'animal model'/exp	1461488
63	'rodent'/exp	3906464
64	rat:ti OR rats:ti OR mouse:ti OR mice:ti	1627604
65	#59 OR #60 OR #61 OR #62 OR #63 OR #64	6677800
66	#58 NOT #65	35401
67	'randomized controlled trial':pt OR 'randomized controlled trial'/exp OR 'randomized controlled trials as topic'/exp	839617
68	'controlled clinical trial'/de	432791
69	randomi?ed:ab	880556
70	placebo:ab	310582
71	randomly:ab	466369

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72	'clinical trials as topic'/de	109908
73	trial:ti	327712
74	#67 OR #68 OR #69 OR #70 OR #71 OR #72 OR #73	1929647
75	#66 AND #74	3852

Search strategy in Cochrane Library

#	Searches	Results
1	[mh "Myocardial Infarction"]	11198
2	(Myocard* Infarct*):ti,ab,kw	31737
3	(Cardiovascular Stroke*):ti,ab,kw	9663
4	(Heart Attack*):ti,ab,kw	3717
5	(heart infarct*):ti,ab,kw	21977
6	STEMI:ti,ab,kw	3415
7	(NSTEMI or Non-STEMI or (Non and STEMI)):ti,ab,kw	1241
8	[mh "Myocardial Ischemia"]	28522
9	(Myocard* Ischemi*):ti,ab,kw	13500
10	(heart ischemi*):ti,ab,kw	17249
11	[mh "Acute Coronary Syndrome"]	2051
12	(Acute Coronary Syndrom*):ti,ab,kw	7589
13	ACS:ti,ab,kw	4150
14	[mh "Coronary Artery Disease"]	6571
15	(Coronary Artery AND Disease*):ti,ab,kw	23566
16	(Coronary Arterioscleros*):ti,ab,kw	503
17	[mh "Angina Pectoris"]	4588
18	Stenocardia*:ti,ab,kw	64
19	(Angor Pectoris):ti,ab,kw	21
20	angina*:ti,ab,kw	14048
21	(Myocardial Preinfarction Syndrom*):ti,ab,kw	7
22	{or #1-#21}	76205
23	[mh "Heart-Assist Devices"]	180
24	(Heart and Assist* (Device* or pump*)):ti,ab,kw	1280
25	(Vascular and Assist* (Device* or pump*)):ti,ab,kw	265
26	(cardiovascular and Assist* and (Device* or pump*)):ti,ab,kw	307
27	(Artificial Ventricle*):ti,ab,kw	661
28	(Ventricle and Assist* and (Device* or pump*)):ti,ab,kw	241
29	(Ventricular and Assist* and (Device* OR pump*)):ti,ab,kw	764
30	(Artificial Heart Ventricle*):ti,ab,kw	636
31	(blood pump*):ti,ab,kw	5249
32	(cardiac resuscitat*):ti,ab,kw	2945
33	(counterpulsation device*):ti,ab,kw	51
34	(mechanical circulat* support*):ti,ab,kw	340
35	(Cardiac Assist* System*):ti,ab,kw	710

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36	(Assist* Circulat*):ti,ab,kw	1077
37	LVAD:ti,ab,kw	346
38	pVAD:ti,ab,kw	14
39	TandemHeart*:ti,ab,kw	9
40	HeartMate*:ti,ab,kw	107
41	Impella*:ti,ab,kw	112
42	[mh "Intra-Aortic Balloon Pumping"]	167
43	(Intra and Aortic and Balloon and (Pump* or counterpulsation)):ti,ab,kw	478
44	(Intraaortica and Balloon and (Pump* or counterpulsation)):ti,ab,kw	4
45	IABP:ti,ab,kw	337
46	IABC:ti,ab,kw	15
47	[mh "Extracorporeal Membrane Oxygenation"]	178
48	(Extracorporeal Membrane Oxygenation*):ti,ab,kw	660
49	(Extracorporeal Life Support*):ti,ab,kw	130
50	ECMO:ti,ab,kw	619
51	ECLS:ti,ab,kw	47
52	[mh "Heart-Lung Machine"]	37
53	(Heart Lung Machine*):ti,ab,kw	262
54	[mh "Extracorporeal Circulation"]	4388
55	(Extracorporeal and Circulat*):ti,ab,kw	1392
56	(extracorporeal cardiopulmonary resuscitation*):ti,ab,kw	48
57	(extracorporeal CPR):ti,ab,kw	23
58	ECPR:ti,ab,kw	20
59	{or #23-#58}	17090
60	#22 and #59	3418
61	[mh "animals"] not [mh "humans"]	61
62	rat or rats or mouse or mice:ti,ab,kw	13725
63	#61 or #62	13782
64	#60 not #63	3379
65	randomized:ti,ab,kw	915787
66	placebo:ti,ab,kw	310693
67	randomly:ti,ab,kw	251577
68	clinical trials:ti,ab,kw	176838
69	{or #65-#68}	1131307
70	#64 and #69	2684

Search strategy in Medline

#	Searches	Results
1	Myocardial Infarction/	167860
2	(Myocard$ and infarct$).ab,ti.	208035
3	(heart and infarct$).ab,ti.	65507

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4	(Heart and Attack$).ab,ti.	11461
5	(Cardiovascular and Stroke$).ab,ti.	35842
6	STEMI.ab,ti.	11486
7	(NSTEMI or Non-STEMI or (Non and STEMI)).ab,ti.	4639
8	Myocardial Ischemia/	39284
9	(Myocard$ and Ischemi$).ab,ti.	76017
10	(heart and ischemi$).ab,ti.	73007
11	Acute Coronary Syndrome/	16517
12	(Acute and Coronary and Syndrom$).ab,ti.	35635
13	ACS.ab,ti.	23142
14	Coronary Artery Disease/	64571
15	(Coronary and Artery and Disease$).ab,ti.	112543
16	(Coronary and Arterioscleros$).ab,ti.	1916
17	Angina Pectoris/	32616
18	Stenocardia$.ab,ti.	951
19	(Angor and Pectoris).ab,ti.	45
20	Angina$.ab,ti.	55365
21	(Myocardial and Preinfarction and Syndrom$).ab,ti.	23
22	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21	523082
23	Heart Assist Devices/	14753
24	(Heart and Assist$ and (Device$ or pump$)).ab,ti.	9718
25	(Vascular and Assist$ and (Device$ or pump$)).ab,ti.	1324
26	(Cardiovascular and Assist$ and (Device$ or pump$)).ab,ti.	1090
27	(Artificial and Ventricle$).ab,ti.	1183
28	(Ventricle and Assist$ and (Device$ or pump$)).ab,ti.	1635
29	(Ventricular and Assist$ and (Device$ or pump$)).ab,ti.	13854
30	(Artificial and Heart and Ventricle$).ab,ti.	563
31	(blood and pump$).ab,ti.	17426
32	(cardiac and resuscitat$).ab,ti.	20141
33	(counterpulsation and device$).ab,ti.	414
34	(mechanical and circulat$ and support$).ab,ti.	5182
35	(Cardiac and Assist$ and System$).ab,ti.	3192
36	(Assist$ and Circulat$).ab,ti.	8583
37	LVAD.ab,ti.	4852
38	pVAD.ab,ti.	162
39	TandemHeart$.ab,ti.	193
40	HeartMate$.ab,ti.	1480
41	Impella$.ab,ti.	925
42	Intra Aortic Balloon Pumping/	4056
43	(Intra and Aortic and Balloon AND (Pump* or counterpulsation)).ab,ti.	3937
44	(Intraaortic and Balloon and (Pump* or counterpulsation)).ab,ti.	1667
45	IABP.ab,ti.	2199

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46	IABC.ab,ti.	120
47	Extracorporeal Membrane Oxygenation/	11447
48	(Extracorporeal and Membrane and Oxygenation$).ab,ti.	11804
49	(Extracorporeal and Life and Support$).ab,ti.	3112
50	ECMO.ab,ti.	8267
51	ECLS.ab,ti.	1293
52	Heart Lung Machine/	2095
53	(Heart and Lung and Machine$).ab,ti.	1060
54	Extracorporeal Circulation/	12973
55	(Extracorporeal and Circulat*).ab,ti.	11010
56	(extracorporeal and cardiopulmonary and resuscitation*).ab,ti.	1242
57	(extracorporeal and CPR).ab,ti.	427
58	ECPR.ab,ti.	380
59	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58	100032
60	22 and 59	12723
61	animals/ not humans/	4757950
62	exp Animals, Laboratory/	892881
63	exp Animal Experimentation/	9614
64	exp Models, Animal/	584650
65	exp rodentia/	3282798
66	(rat or rats or mouse or mice).ti.	1355142
67	61 or 62 or 63 or 64 or 65 or 66	5725191
68	60 not 67	11457
69	randomized controlled trial.pt. or randomized controlled trial/ or Randomized Controlled Trials as Topic/	658872
70	controlled clinical trial.pt.	94082
71	placebo.ab.	215311
72	randomly.ab.	351488
73	clinical trials as topic.sh.	194813
74	randomi?ed.ab.	609920
75	trial.ti.	234697
76	69 or 70 or 71 or 72 or 73 or 74 or 75	1439354
77	68 and 76	1339

Search strategy in Pubmed

#	Searches	Results
1	"Myocardial Infarction"[Mesh]	178446
2	(Myocard*[Title/Abstract]) AND Infarct*[Title/Abstract]	213501
3	(Cardiovascular[Title/Abstract]) AND Stroke*[Title/Abstract]	37057
4	(Heart[Title/Abstract]) AND Attack*[Title/Abstract]	11736

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5	(heart[Title/Abstract]) AND infarct*[Title/Abstract]	69148
6	STEMI[Title/Abstract]	11856
7	(((NSTEMI[Title/Abstract]) OR Non-STEMI[Title/Abstract])) OR ((Non[Title/Abstract]) AND STEMI[Title/Abstract])	4838
8	"Myocardial Ischemia"[Mesh]	437366
9	(Myocard*[Title/Abstract]) AND Ischemi*[Title/Abstract]	77593
10	(heart[Title/Abstract]) AND ischemi*[Title/Abstract]	74762
11	"Coronary Artery Disease"[Mesh]	64579
12	((Coronary[Title/Abstract]) AND Artery[Title/Abstract]) AND Disease*[Title/Abstract]	117973
13	(Coronary[Title/Abstract]) AND Arterioscleros*[Title/Abstract]	2525
14	"Acute Coronary Syndrome"[Mesh]	16526
15	((Acute[Title/Abstract]) AND Coronary[Title/Abstract]) AND Syndrom*[Title/Abstract]	37318
16	ACS[Title/Abstract]	23712
17	"Angina Pectoris"[Mesh]	43697
18	Stenocardia*[Title/Abstract]	952
19	(Angor[Title/Abstract]) AND Pectoris[Title/Abstract]	58
20	angina*[Title/Abstract]	56696
21	((Myocardial[Title/Abstract]) AND Preinfarction[Title/Abstract]) AND Syndrom*[Title/Abstract]	44
22	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21	617614
23	"Heart-Assist Devices"[Mesh]	14752
24	((Heart[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	10556
25	((Vascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1407
26	((cardiovascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1202
27	(Artificial[Title/Abstract]) AND Ventricle*[Title/Abstract]	1255
28	((Ventricle[Title/Abstract]) AND Assist*[Title/Abstract]) AND Device*[Title/Abstract]	1601
29	((Ventricular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	14282
30	((Artificial[Title/Abstract]) AND Heart[Title/Abstract]) AND Ventricle*[Title/Abstract]	610
31	(blood[Title/Abstract]) AND pump*[Title/Abstract]	17702
32	(cardiac[Title/Abstract]) AND resuscitat*[Title/Abstract]	21366
33	(counterpulsation[Title/Abstract]) AND device*[Title/Abstract]	451
34	((mechanical[Title/Abstract]) AND circulat*[Title/Abstract]) AND support*[Title/Abstract]	5748
35	((Cardiac[Title/Abstract]) AND Assist*[Title/Abstract]) AND	3326

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	System*[Title/Abstract]	
36	((Assist*[Title/Abstract])) AND Circulat*[Title/Abstract]	9223
37	LVAD[Title/Abstract]	5058
38	pVAD[Title/Abstract]	173
39	TandemHeart*[Title/Abstract]	204
40	HeartMate*[Title/Abstract]	1515
41	Impella*[Title/Abstract]	972
42	"Intra-Aortic Balloon Pumping"[Mesh]	4056
43	(((Intra[Title/Abstract]) AND Aortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	4026
44	((Intraaortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	5487
45	IABP[Title/Abstract]	2253
46	IABC[Title/Abstract]	127
47	"Extracorporeal Membrane Oxygenation"[Mesh]	11452
48	((Extracorporeal[Title/Abstract]) AND Membrane[Title/Abstract]) AND Oxygenation*[Title/Abstract]	12432
49	((Extracorporeal[Title/Abstract]) AND Life[Title/Abstract]) AND Support*[Title/Abstract]	3413
50	ECMO[Title/Abstract]	8706
51	ECLS[Title/Abstract]	1416
52	"Heart-Lung Machine"[Mesh]	2095
53	((Heart[Title/Abstract]) AND Lung[Title/Abstract]) AND Machine*[Title/Abstract]	1081
54	"Extracorporeal Circulation"[Mesh]	84949
55	(Extracorporeal[Title/Abstract]) AND Circulat*[Title/Abstract]	11650
56	((extracorporeal[Title/Abstract]) AND cardiopulmonary[Title/Abstract]) AND resuscitation*[Title/Abstract]	1349
57	(extracorporeal[Title/Abstract]) AND CPR[Title/Abstract]	477
58	ECPR[Title/Abstract]	482
59	#23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56 OR #57 OR #58	158913
60	#22 AND #59	20135
61	((animals[mh]) NOT humans[mh])	4791677
62	Animals, Laboratory[mh]	892913
63	Animal Experimentation[mh]	9615
64	Models, Animal[mh]	584678
65	rodentia[mh]	3282948
66	((rat[Title] OR rats[Title] OR mouse[Title] OR mice[Title]))	1355108

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67	#61 OR #62 OR #63 OR #64 OR #65 OR #66	5741699
68	#60 NOT #67	18053
69	((randomized controlled trial[Publication Type]) OR randomized controlled trial[mh]) OR Randomized Controlled Trials as Topic[mh]	661250
70	controlled clinical trial[Publication Type]	614085
71	((randomized[Title/Abstract]) OR randomised[Title/Abstract])	660183
72	placebo[Title/Abstract]	221488
73	randomly[Title/Abstract]	352402
74	trial[Title]	234625
75	#69 OR #70 OR #71 OR #72 OR #73 OR #74	1313992
76	#68 and #75	2105

Search strategy in Web of Science

#	Searches	Results
1	TS=(heart AND infarct*)	345636
2	TS=(myocard* AND infarct*)	452517
3	TS=(heart AND attack*)	44545
4	TS=(cardiovascular AND stroke*)	232512
5	TS=STEMI	20441
6	TS=(NSTEMI OR Non-STEMI OR (Non AND STEMI))	8228
7	TS=(Myocard* AND Ischemi*)	190331
8	TS=(heart AND ischemi*)	276064
9	TS=(Acute AND Coronary AND Syndrom*)	74970
10	TS=ACS	43248
11	TS=(Coronary AND Artery AND Disease*)	351532
12	TS=(Coronary AND Arterioscleros*)	48592
13	TS=Angina*	103224
14	TS=Stenocardia*	1163
15	TS=(Angor AND Pectoris)	472
16	TS=(Myocardial AND Preinfarction AND Syndrom*)	61
17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16	1094596
18	TS=(Heart AND Assist* AND (Device* OR pump*))	34749
19	TS=(Vascular AND Assist* AND (Device* OR pump*))	10298
20	TS=(cardiovascular AND Assist* AND (Device* OR pump*))	21034
21	TS=(Artificial AND Ventricle*)	10067
22	TS=((Ventricular OR Ventricle) AND Assist* AND (Device* OR pump*))	28212
23	TS=(Artificial AND Heart AND Ventricle*)	8861
24	TS=(blood AND pump*)	58329
25	TS=(cardiac AND resuscitat*)	40430

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26	TS=(counterpulsation AND device*)	1133
27	TS=(mechanical AND circulat* AND support*)	21978
28	TS=(Cardiac AND Assist* AND System*)	25077
29	TS=(Assist* AND Circulat*)	59084
30	TS=LVAD	9139
31	TS=pVAD	254
32	TS=TandemHeart*	276
33	TS=HeartMate*	2711
34	TS=Impella*	1654
35	TS=(Intra AND Aortic AND Balloon AND (Pump* OR counterpulsation))	7553
36	TS=(Intraaortic AND Balloon AND (Pump* OR counterpulsation))	2998
37	TS=IABP	2862
38	TS=IABC	283
39	TS=(Extracorporeal AND Membrane AND Oxygenation*)	21874
40	TS=(Extracorporeal AND Life AND Support*)	6398
41	TS=ECMO	12728
42	TS=ECLS	2140
43	TS=(Heart AND Lung AND Machine*)	3515
44	TS=(Extracorporeal AND Circulat*)	28102
45	TS=(extracorporeal AND cardiopulmonary AND resuscitation*)	2647
46	TS=(extracorporeal AND CPR)	764
47	TS=ECPR	664
48	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47	250465
49	#17 AND #48	47026
50	TS= clinical trial* OR TS=research design OR TS=comparative stud* OR TS=evaluation stud* OR TS=controlled trial* OR TS=follow-up stud* OR TS=prospective stud* OR TS=random* OR TS=placebo* OR TS=(single blind*) OR TS=(double blind*)	10462135
51	#49 AND #50	16870
52	#49 AND #50 limit clinical trials	2412

Search strategy in ClinicalTrials.gov

Basic search

104 Studies found for: Ventricular Assist Device OR Intra Aortic Balloon Pump OR Extracorporeal

Membrane Oxygenation | MI OR Myocardial Infarction OR Coronary Artery Disease

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Search strategy in Clinicaltrialsregister.eu

Basic search

78 Studies found for: "ventricular assist device" OR "intra aortic balloon pump" OR "Extracorporeal Membrane Oxygenation"

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Online supplementary file 3: The Endpoint Definitions of the Included Studies

Study	Endpoint	Definition
SEMPER FI[25]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding events according to the Global Use of Strategies to Open Coronary Arteries (GUSTO) trial.
	Arrhythmias	NA.
	Vascular complications	NA.
Zhou et al[29]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	Malignant arrhythmia (not detailed).
	Vascular complications	NA.
IABP-SHOCK II[8,9]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke identified by the presence of new neurologic symptoms in conjunction with signs of ischemia or bleeding on computed tomography.
	Bleeding	Severe or life-threatening bleeding and moderate bleeding as assessed according to the Global Use of Strategies to Open Occluded Coronary Arteries (GUSTO) criteria1.
	Arrhythmias	NA.
	Vascular complications	Peripheral ischemic vascular complications requiring surgical or interventional therapy.
CRISP AMI[28]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke (not detailed).
	Bleeding	Major bleeding events defined by theGlobal Use of Strategies To Open Coronary Arteries (GUSTO) trial and transfusions.
	Arrhythmias	NA.
	Vascular complications	Major vascular complications defined to include limb ischemia requiring intervention, distal embolization, major dissection, pseudoaneurysm or arteriovenousfistula, hematoma larger than 5 cm, and amputations.
Gu et al[30]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Vascular complications (not detailed).
Prondzinsky et al[31]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmia (not detailed).
	Vascular complications	NA.

Arias et al[33]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.
Vijayalakshmi et al[34]	Reinfarction	Reinfarction was defined as chest pain associated with ST-segment re-elevation and an increase in cardiac enzymes (3 times creatine kinase release).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmias including ventricular tachycardia, ventricular fibrillation or atrial fibrillation.
	Vascular complications	Vascular complications (not detailed).
IMPRESS[26]	Reinfarction	Reinfarction Defined by 3rd definition MI. Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following: 1. Symptoms of ischemia. 2. New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB). 3. Development of pathological Q waves in the ECG. 4. Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. 5. Identification of an intracoronary thrombus by angiography or autopsy.
	Repeat revascularization	Repeat PCI or CABG.
	Stroke-TIA	Any stroke confirmed by a neurologist and a concurring CT scan.
	Bleeding	1.Bleeding with associated serum hemoglobin level decrease of at least 5 g/dL (=3.1 mmol/L). The decrease in hemoglobin will be calculated as the last recorded Hb measurement preceding the onset of the bleeding, subtracted by the nadir Hb measurement (associated with the bleeding). 2. Bleeding necessitating a minimum of 2 packed cells of blood product transfusion (only blood transfusions that are explicitly related to the bleeding are taken into account). 3. The need for surgery to control the bleeding.
	Arrhythmias	Placement of implantable cardioverter-defibrillator.
	Vascular complications	1.A major bleed arising at the arterial access site (for major bleeding definition see above) requiring extraction of the device. 2. A thrombotic occlusion of the femoral artery. 3. Limb ischemia requiring extraction of either of the study devices. 4. The need for vascular surgery to correct a vascular complication.
Seyfarth et al[32]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Limb ischemia (not detailed).
Thiele et al[27]	Reinfarction	NA.

	Repeat revascularization	NA.
	Stroke-TIA	Cerebrovascular accidents with neurological dysfunction.
	Bleeding	Major bleedings requiring transfusion of blood components.
	Arrhythmias	NA.
	Vascular complications	Lower extremity ischaemia requiring surgical or interventional action.
Korbinian et al[35]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.

NA indicates not available.

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Online supplementary file 4: The Enrollment Information of the Included Studies

Study	Center of Enrollment	Time of Enrollment
SEMPER FI[25]	Catharina Hospital, Eindhoven, Netherlands	August 2014 to December 2018
Zhou et al[29]	Department of Cardiology of Liaocheng People's Hospital, Liaocheng, China	January 2014 to January 2016
IABP-SHOCK II[8,9]	University of Leipzig-Heart Center, Leipzig, Germany Heart Center Bad Krozingen, Bad Krozingen, Germany Asklepios Klinik Langen, Langen, Germany German Heart Center Munich, Munich, Germany Heart Center-Segeberger Kliniken, Bad Segeberg, Germany SLK Kliniken Heilbronn, Heilbronn, Germany University of Greifswald, Greifswald, Germany Martin-Luther-University Halle-Wittenberg, Halle (Saale), Germany Klinikum Links der Weser, Bremen, Germany Heart Center Ludwigshafen, Ludwigshafen, Germany Zentralklinik Bad Berka, Bad Berka, Germany University Clinic of Saarland, Homburg/Saar, Germany Friedrich-Schiller University Jena, Jena, Germany University of Lübeck, Lübeck, Germany Klinik Hennigsdorf, Hennigsdorf, Germany Otto-von-Guericke-University Magdeburg, Magdeburg, Germany Sana Heart Center Cottbus, Cottbus, Germany Heart Center Dresden, Dresden, Germany University of Würzburg, Würzburg, Germany Hospital München-Neuperlach, Germany Krankenhaus der Barmherzigen Brüder, Regensburg, Germany	June 2009 to March 2012

	Charité Berlin Campus Benjamin Franklin, Berlin, Germany Vivantes Klinikum am Urban, Berlin, Germany Carl-Thiem-Klinikum Cottbus, Cottbus, Germany University Rostock, Germany Kerckhoff Klinik Bad Nauheim, Bad Nauheim, Germany Helios Klinikum Erfurt, Erfurt, Germany Charité Berlin-Campus Mitte, Berlin, Germany Ludwigs-Maximilian University Munich-Hospital Großhadern, Germany Vivantes Auguste-Viktoria-Klinikum, Berlin, Germany Klinikum Weiden, Weiden, Germany Klinikum Fulda, Fulda, Germany Katharinenhospital Unna, Unna, Germany Philippusstift Essen, Essen, Germany Carl-von-Basedow-Klinikum Merseburg, Merseburg, Germany Klinikum Lippe-Detmold, Detmold, Germany Schwarzwald-Baar Klinikum VillingenSchwenningen, Villingen-Schwenningen, Germany	
CRISP AMI[28]	Flinders Medical Centre, Bedford Park, Australia Virga Jesse Hospital, Hasselt, Belgium Leeds General Infirmary, Leeds, England Saint Thomas Hospital, London, England Universitatsklinikum Leipzig, Leipzig, Germany University of Freiburg, Freiburg, Germany Klinik III Fur Innere Medizin der Universitat zu Koln, Dresden, Germany CARE Hospital, Hyderabad, India Baroda Heart Institute and Research Centre, Vadodara, India Sir Ganga Ram Hospital, New Delhi, India Gurunanak CARE Hospital, Hyderabad, India Medanta-The Medicity, Gurgaon, India	December 2008 to March 2011

	Tagore Hospital and Heart Care Centre, Jalandhar, India Indraprastha Apollo Hospital, New Delhi, India Apollo Gleneagles Hospital, Kolkata, India Saint John's Medical College and Hospital, Bangalore, India Mater Misercordiae Dublin, Dublin, Ireland Catherina Hospital, Eindhoven, Netherlands Royal Infirmary of Edinburgh, Edinburgh, Scotland Duke University Medical Center, Durham, United States University of Texas Health Science Center, Houston, United States Temple University Hospital, Philadelphia, United States Tufts Medical Center, Boston, United States Newark Beth Israel Medical Center, Newark, United States University of Rochester Medical Center-Strong Memorial Hospital, Rochester, United States Lancaster General Hospital, Lancaster, United States University of Alabama Medical Center, Birmingham, United States Washington Hospital Center, Washington, United States Allegheny General Hospital, Pittsburgh, United States University of Southern California University Hospital, Los Angeles, United States	
Gu et al[30]	Min Hang Hospital, Shanghai, China	January 2005 to February 2009
Prondzinsky et al[31]	Department of Medicine III, Department of Biometrics and Statistics (SU), Halle-Wittenberg, Germany	March 2003 to June 2004
Arias et al[33]	Departamento de Medicina Interna, Fundación Clínica Médica Sur, México	February 2001 to February 2003
Vijayalakshmi et al[34]	The James Cook University Hospital, Middlesbrough, United Kingdom	NA
IMPRESS[26]	Academic Medical Center, Amsterdam, Netherlands Haukeland University Hospital, Bergen, Norway	May 2012 to September 2015
Seyfarth et al[32]	Deutsches Herzzentrum München, Munich, Germany	August 2004 to March

	Medizinische Klinik rechts der Isar, Munich, Germany	2006
Thiele et al[27]	University of Leipzig-Heart Centre, Leipzig, Germany	August 2000 to December 2003
Korbinian et al[35]	University Hospital Munich, Munich, Germany	November 2015 to March 2018

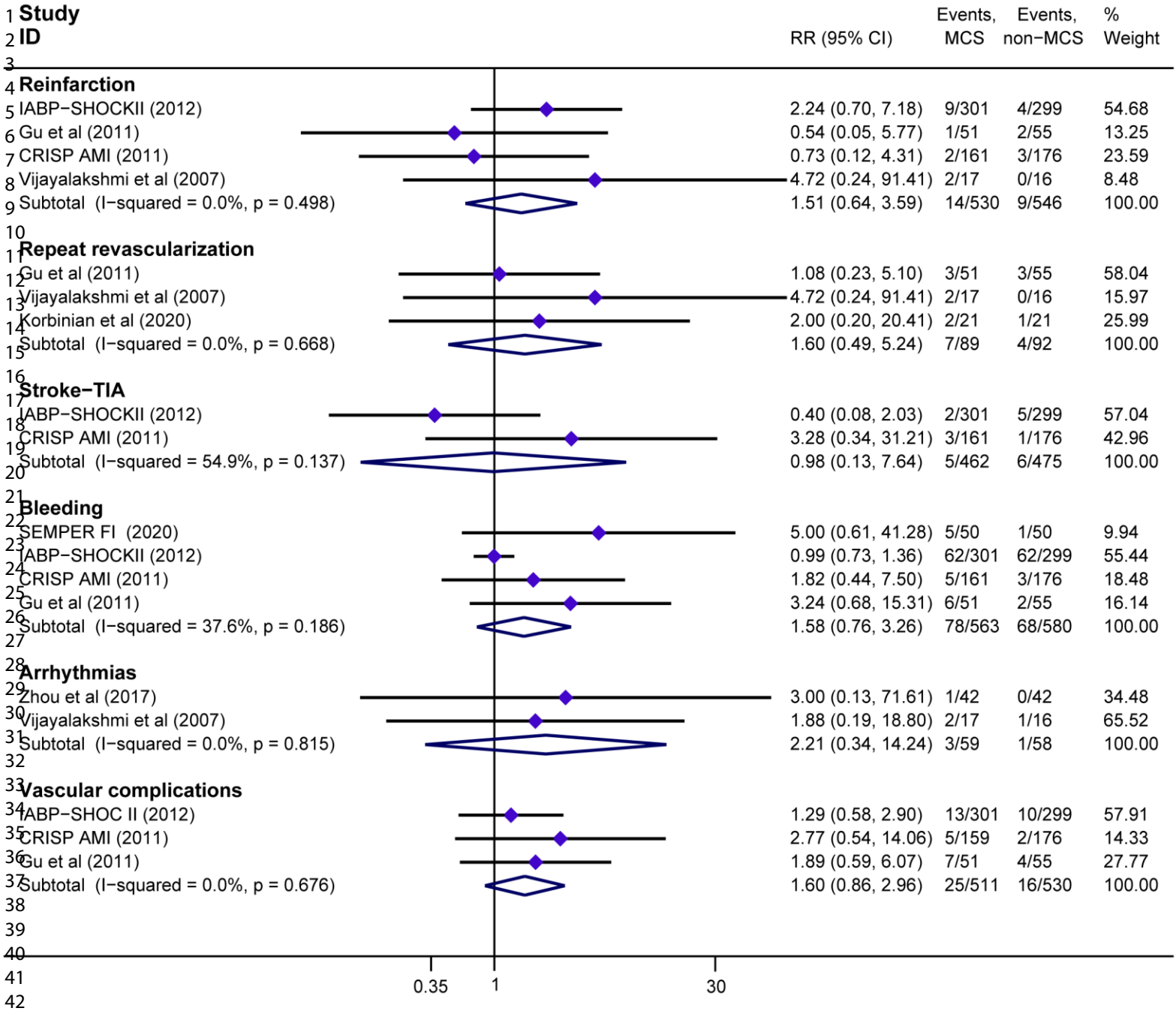
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Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

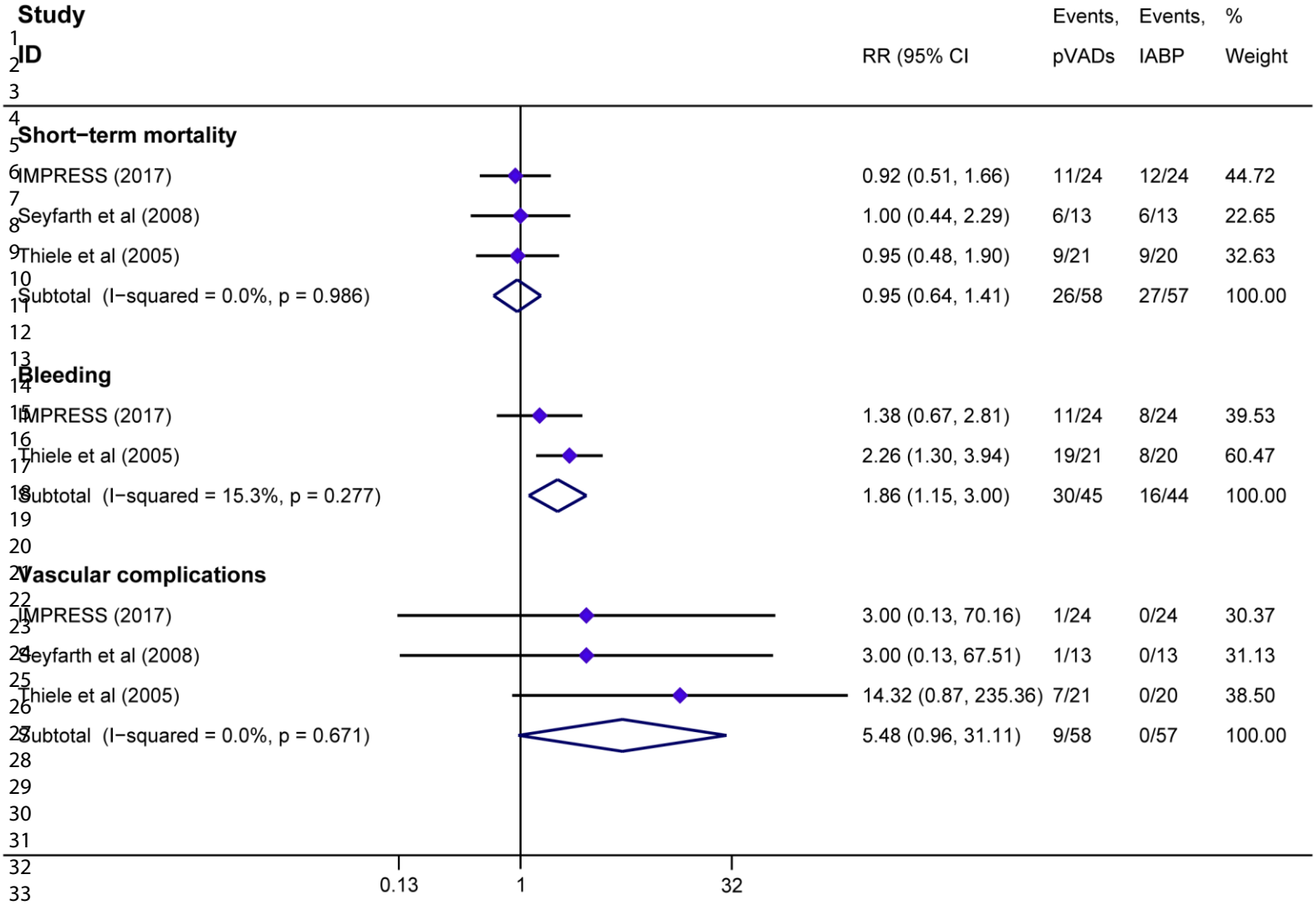
		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	
	In hospital	1.37 (0.50-3.78)
	30 days	0.70 (0.39-1.25)
	Different types of MCS use	
	IABP vs non-IABP	0.92 (0.57-1.47)
	ECLS vs non-ECLS	0.33 (0.01-1.47)
	Excluding high risk bias studies	
	Zhou et al[29], Arias et al[33] and Korbinian et al[35]	0.85 (0.51-1.45)
	Excluding each of the studies	
	SEMPER FI[25]	0.92 (0.57-1.48)
	Zhou et al[29]	0.80 (0.53-1.21)
	IABP-SHOCK II[8]	0.89 (0.45-1.74)
	CRISP AMI[28]	0.95 (0.59-1.55)
	Gu et al[30]	1.02 (0.66-1.58)
	Prondzinsky et al[31]	0.80 (0.48-1.34)
	Arias et al[33]	0.98 (0.58-1.66)
	Vijayalakshmi et al[34]	0.86 (0.55-1.34)
	Korbinian et al[35]	0.92 (0.57-1.47)
Long-term mortality		
	Inconsistent length of follow-up	
	6 months	0.72 (0.43-1.20)
	12 months	0.90 (0.35-2.31)
	Different types of MCS use	
	IABP vs non-IABP	0.87 (0.60-1.25)
	ECLS vs non-ECLS	0.50 (0.18-1.41)
	Excluding high risk bias studies	
	Zhou et al[29] and Korbinian et al[35]	0.72 (0.43-1.20)
	Excluding each of the studies	
	SEMPER FI[25]	0.82 (0.56-1.20)
	Zhou et al[29]	0.70 (0.45-1.09)
	IABP-SHOCK II[9]	0.68 (0.40-1.17)
	CRISP AMI[28]	0.89 (0.65-1.23)
	Gu et al[30]	0.92 (0.66-1.28)
	Korbinian et al[35]	0.87 (0.61-1.25)

CI indicates confidence interval.

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Online supplementary file 6: Forest plots for the secondary end points of AMI patients treated with MCS compared with non-MCS. TIA, transient ischaemic attack.



Online supplementary file 7: Forest plots for outcomes occurrence comparing pVADs use versus IABP use

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Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	
	IMPRESS[26]	0.97 (0.57-1.65)
	Seyfarth et al[32]	0.93 (0.59-1.46)
	Thiele et al[27]	0.94 (0.58-1.53)
Long-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	Not applicable

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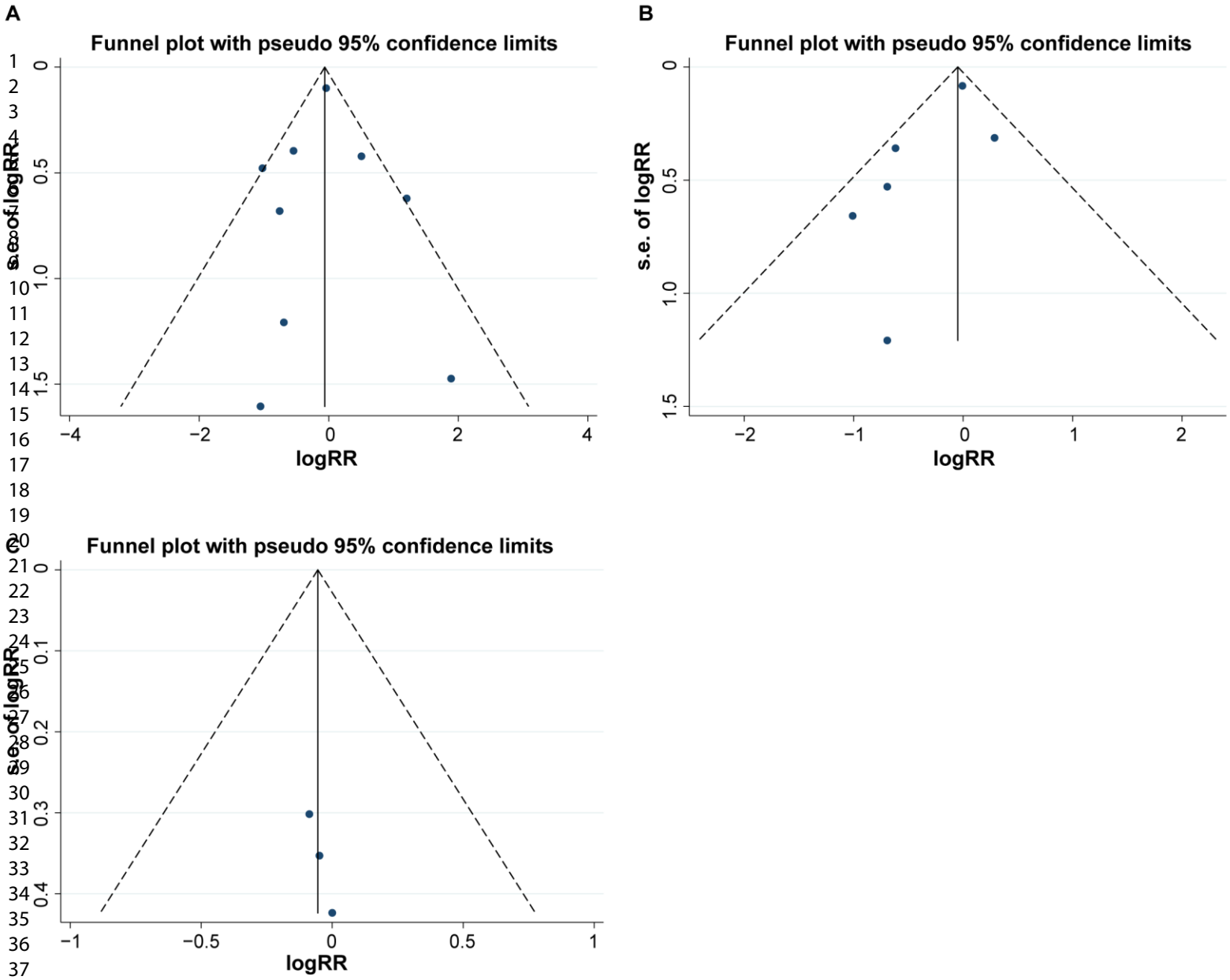
Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
SEMPER FI[25]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhou et al[29]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IABP-SHOCK II[8,9]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
CRISP AMI[28]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Gu et al[30]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Prondzinsky et al[31]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Arias et al[33]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Vijayalakshmi et al[34]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IMPRESS[26]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Seyfarth et al[32]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Thiele et al[27]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Korbinian et al[35]	Low risk	Low risk	Low risk	Low risk	Low risk	High risk	Low risk

Low risk

Unclear risk

High risk



Online supplementary file 10: Funnel plot of standard error for: A. MCS compared with non-MCS in short-term mortality; B. MCS compared with non-MCS in long-term mortality; C. pVADs compared with IABP in short-term mortality.

Online supplementary file 11: Publication bias on the association of primary outcome

		Egger's test (P)	Begg's test (P)
MCS vs non-MCS			
	Short-term mortality	0.886	0.917
	Long-term mortality	0.154	0.452
pVAD vs IABP			
	Short-term mortality	0.015*	0.296
	Long-term mortality	Not applicable	Not applicable

* Have statistical difference.

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PRISMA checklist for the meta-analysis

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6 and Online supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7

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Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7, and Online supplementary file 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9, Table 1, Table 2 and Online supplementary file 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 and Online supplementary file 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12, Online supplementary file 9, Online supplementary

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			file 10 and Online supplementary file 11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-11, Figure 3, Online supplementary file 5 and Online supplementary file 8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

NA indicates not available.

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Effects of Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients Undergoing Stent Implantation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Primary Subject Heading:	Cardiovascular medicine
Secondary Subject Heading:	Cardiovascular medicine, Evidence based practice
Keywords:	Adult cardiology < CARDIOLOGY, Myocardial infarction < CARDIOLOGY, Coronary intervention < CARDIOLOGY

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Effects of Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients Undergoing Stent Implantation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Abstract

Objective

The survival benefit of using mechanical circulatory support (MCS) in acute myocardial infarction (AMI) patients is still controversial. It is necessary to explore the impact on clinical outcomes of MCS in AMI patients with stenting.

Design

Systematic review and meta-analysis.

Data sources

The Embase, Cochrane Library, Medline, PubMed, Web of Science, ClinicalTrials.gov and Clinicaltrialsregister.eu databases were searched from database inception to February 2021.

Eligibility criteria

Randomized clinical trials (RCTs) on MCS use in AMI patients undergoing stent implantation were included.

Data extraction and synthesis

Data were extracted and summarized independently by two reviewers. Risk ratios (RRs) and 95% confidence intervals (CIs) were calculated for clinical outcomes according to random effects model.

Results

Twelve studies of 1497 AMI patients were included, nine studies including 1382 patients compared MCS with non-MCS, and three studies including 115 patients compared percutaneous ventricular assist devices (pVADs) versus intra-aortic balloon pump (IABP). Compared with non-MCS, MCS was not associated with short-term (within 30 days) (RR=0.90; 95% CI, 0.57-1.41; I^2=46.8%) and long-term (at least 6 months) (RR=0.82; 95% CI, 0.57-1.17; I^2=37.6%) mortality reductions. In the subset of patients without cardiogenic shock (CS), compared with non-MCS, the patients with IABP

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treatment significantly decrease the long-term mortality (RR=0.49; 95% CI, 0.27-0.90; I^2=0), but without the short-term mortality reductions (RR=0.51; 95% CI, 0.22-1.19; I^2=17.9%). While, in the patients with CS, the patients with MCS did not benefit the short-term (RR=1.09; 95% CI, 0.67-1.79; I^2=46.6%) or long-term (RR=1.00; 95% CI, 0.75-1.33; I^2=22.1%) survival. Moreover, the application of pVADs increased risk of bleeding (RR=1.86; 95% CI, 1.15-3.00; I^2=15.3%) compared to IABP treatment (RR=1.86; 95% CI, 1.15-3.00; I^2=15.3%).

Conclusions

In all AMI patients undergoing stent implantation, the MCS use does not reduce all-cause mortality. Patients without CS can benefit from MCS regarding to long-term survival, while patients with CS seem not.

Keywords: Acute myocardial infarction (AMI), Percutaneous coronary intervention (PCI), Stent implantation, Mechanical circulatory support (MCS), Intra-aortic balloon pump (IABP), Percutaneous ventricular assist devices (pVADs)

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Article Summary

Strengths and limitations of this study

- This meta-analysis is focusing on the effect of MCS in AMI patients with stenting.
- This meta-analysis is the first to identity that MCS benefits AMI patients with stenting intervention without cardiogenic shock.
- Limited number of RCTs comparing the effect of pVADs with IABP, and different types of pVADs, the results should be interpreted with cautious.

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Introduction

Acute myocardial infarction (AMI) is the most common cause of hospitalization in elderly patients [1,2]. Emerging mechanical circulatory support (MCS), including intra-aortic balloon pump (IABP), percutaneous ventricular assist devices (pVADs), and extracorporeal life support (ECLS) are important haemodynamic supportive strategy for maintaining a stable haemodynamic and organ perfusion in the acute phase of myocardial infarction. However, the survival benefit of using MCS in AMI patients is still controversial.

Although previous guidelines from various committees have recommended the necessity of using MCS in patients with severe cardiac ischaemia [3,4], accumulating evidence from RCT and meta-analysis has shown that neither MCS devices have significantly improved the survival in AMI patients with CS[5,6,7,8,9]. Therefore, in AMI patients with cardiogenic shock (CS), IABP and pVADs were recommended to class IIa and class IIb in US guidelines respectively, and IABP to class IIaC-IIIA in European guidelines [10,11]. Recent meta-analyses have focused on the use of MCS in patients with CS, but no benefits for survival was found [12,13,14]. Even if the conclusions from MCS studies were something of disappointment, these data still worth to be further explored in detail.

As rapid advances in percutaneous coronary intervention (PCI) technology, stenting is better for reperfusion after AMI than balloon angioplasty [15]. However, it is noteworthy that both patient with stenting and balloon angioplasty were included in the studies of MCS [16,17]. As the revascularization strategy may bring large heterogeneous, the choice of the strategy must take into account the evidence. Therefore, in this study, we conduct a systematic review and meta-analysis to highlight the effect of MCS in AMI patients with stent implantation and further exploration on the effect of MCS are performed in the patients with or without CS.

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Methods

The current meta-analysis was conducted in accordance with the PRISMA guidelines (PRISMA checklist see online supplementary file 1) [18].

Search Strategy

We conducted a systematic search of the literature in February 2021 without restrictions on region, publication type, or language. The Embase, Cochrane Library, Medline, PubMed, and Web of Science databases were included. ClinicalTrials.gov and Clinicaltrialsregister.eu were searched as well. The MeSH search terms included the following: myocardial infarction, acute coronary syndrome, heart-assist devices, intra-aortic balloon pump and mechanical circulatory support. We excluded non-human studies. Details of the search strategy were presented in online supplementary file 2.

Study Selection

In the current meta-analysis, only RCTs in which stent implantation was the dominant form of revascularization and MCS was used during the perioperative period in patients with AMI were included. We excluded nonrandomized trials, cohort studies, cross-sectional studies, ongoing trials, trials terminated early and other publication types, including review, case reports or case series, editorials, letters, and meeting abstracts. Studies in which patients underwent coronary artery bypass grafting (CABG), balloon angioplasty or systemic thrombolysis as the main revascularization method were also excluded.

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Endpoints

The primary outcome was all-cause mortality, including short-term mortality and long-term mortality. Short-term mortality was defined as mortality occurring within 30 days. Long-term mortality occurred at least 6 months of follow-up. The secondary endpoints included reinfarction, repeat revascularization, stroke-TIA (transient ischaemic attack), bleeding (moderate to major bleeding, blood transfusion or surgery to control the bleeding), arrhythmias and vascular complications (peripheral ischaemic vascular complications, major dissection, pseudoaneurysm or arteriovenous fistula) occurring within 30 days. The endpoint definitions as applied in each study were presented in online supplementary file 3.

Data Extraction and Quality Assessment

Data from the included studies were extracted and summarized independently by two reviewers (Shi and Wang). All extracted data from the included studies were independently collected in duplicate using a standardized data extraction form, which contained authors or trails' name, year of publication, study design (interventions, sample size, percentage of stenting and bare-metal stent, PCI success, timing of MCS, time point of reported mortality, follow-up), baseline characteristics of study population (sex, age, AMI, CS, infarct-related artery, hypertension, diabetes mellitus, previous stroke/TIA and MI, prior PCI and CABG), the endpoint definitions, the centre and the time of the study enrolment. Any disagreements were resolved by a third reviewer (Sun). The quality of the included studies was assessed using the Cochrane Collaborations tool [19], which including selection

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bias, performance bias, detection bias, attrition bias, reporting bias, other possible bias. Each source of bias was classified as low, unclear, and high risk.

Statistical Analysis

We applied risk ratios (RRs) and 95% confidence intervals (CIs) to summarize the statistics. Statistical heterogeneity was assessed using Cochran's Q test and quantified using the I^2 statistic; $p < 0.10$. Heterogeneity was categorized by I^2 as follows [20]: insignificant heterogeneity, $I^2<25\%$; low heterogeneity, $25\% \leq I^2<50\%$; moderate heterogeneity, $50\% \leq I^2<75\%$; high heterogeneity, $I^2 \geq 75\%$. Considering the clinical differences among the included studies, a random-effects model was applied for statistical analyses. Sensitivity analyses were performed for primary outcomes with the following: (1) inconsistent length of follow-up, (2) different types of MCS use, (3) exclusion of research with a high risk of bias, and (4) the one-study-out method. Funnel plots, Begg and Mazumdar rank correlation and Egger's regression intercept were used to evaluate publication bias[20]. If publication bias existed, the trim-and-fill method was performed for estimating the number of missing studies and RR adjustment [21]. Statistical analyses were performed using STATA/SE 12.0 (Stata Corp LP, College Station, Texas, USA). $P<0.05$ was considered statistically significant.

Results

There are 12574 citations were identified through the search criteria. The full texts of 107 potentially eligible articles were scrutinized in detail. Among these, 15 RCTs used PCI as the main revascularization therapy, 3 RCTs on balloon angioplasty use were excluded [22,23,24], and 12 RCTs

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finally met the inclusion criteria [8,25,26,27,28,9,29,30,31,32,33,34,35]. The study selection process is outlined in Figure 1.

Characteristics of the Studies

The centre and the time of enrolment in each study were shown that no duplicate of the sample population of included studies were found (see online supplementary file 4). The detailed designs of the included studies are listed in Table 1. Among the 12 RCTs, 1497 AMI patients were included, 94.5% of patients (1391/1471) underwent stent implantation, 86.9% of patients (959/1104) were with PCI success. The follow-up period ranged from in-hospital only to 6 years. Nine studies including 1382 AMI patients compared MCS with non-MCS (693 in the MCS group and 690 in the non-MCS group) [8,25,28,9,29,30,31,33,34,35], 3 studies including 115 AMI patients compared pVADs versus IABP (58 in the pVADs group and 57 in the IABP group) [26,27,32]. Eight RCTs included CS patients, among which 5 studies that included 806 patients compared MCS with non-MCS (414 in the MCS group and 392 in the non-MCS group) [8,9,29,31,33,35], the remaining 3 studies compared pVADs versus IABP. Four RCTs included patients without CS, and all types of MCS use were IABP, which included 576 patients, compared MCS with non-MCS (279 in the IABP group and 297 in the non-IABP group) [25,28,30,34]. The summarized patients' characteristics of included studies are shown in Table 2. The mean age of the population was 63.8, and 73.3% were male, 77.1% (1046/1357) had left coronary artery-related infarction, 52.9% (726/1373) had hypertension, 27.7% (381/1373) were diabetes mellitus, 4.3% (49/1172) had a previous diagnosis of stroke/TIA, 14.3% (193/1347) had a previous diagnosis of MI, 17.6% (208/1185) underwent PCI previously and 5.0% (37/742) underwent CABG previously.

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Outcomes of Synthesis Analyses

Of the included 9 RCTs comparing the survival benefit of MCS with non-MCS, all studies reported short-term mortality, and 6 studies reported long-term mortality. MCS was not associated with a short-term mortality reduction compared to the non-MCS (MCS=23.1%, 160/693; non-MCS=23.5%, 162/689; RR=0.90; 95% CI, 0.57-1.41; p=0.640), with low heterogeneity among studies (I^2=46.8%; p=0.059) (Figure 2). Similarly, there was no significant difference between the MCS group and the non-MCS group in the long-term mortality (MCS=28.6%, 176/626; non-MCS=30.3%, 195/643; RR=0.82; 95% CI, 0.57-1.17; p=0.276), with low heterogeneity among studies (I^2=37.6%; p=0.156) (Figure 2). The results of prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 5).

For the secondary endpoints, MCS was no effect the incidence of reinfarction (MCS=2.6%, 14/530; non-MCS=1.6%, 9/546; RR=1.51; 95% CI, 0.64-3.59; p=0.346; I^2=0, p=0.498), repeat revascularization (MCS=7.9%, 7/89; non-MCS=4.3%, 4/92; RR=1.60; 95% CI, 0.49-5.24; p=0.435; I^2=0, p=0.668), stroke-TIA (MCS=1.1%, 5/462; non-MCS=1.3%, 6/475; RR=1.51; 95% CI, 0.64-3.59; p=0.988; I^2=54.9%, p=0.137), bleeding (MCS=13.9%, 78/563; non-MCS=11.7%, 68/580; RR=1.58; 95% CI, 0.76-3.26; p=0.218; I^2=37.6%, p=0.186), arrhythmias (MCS=5.1%, 3/59; non-MCS=1.7%, 1/58; RR=2.21; 95% CI, 0.34-14.24; p=0.404; I^2=0, p=0.815), and vascular complications (MCS=4.9%, 25/511; non-MCS=3.0%, 16/530; RR=1.60; 95% CI, 0.86-2.96; p=0.134; I^2=0, p=0.676) (see online supplementary file 6).

Head-to-head comparisons between different type of MCS were also performed. There were 3 RCTs comparing pVADs versus IABP included. The pooled analysis revealed that pVADs was not

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associated with a decrease in short-term mortality compared with IABP (pVADs=44.8%, 26/58; IABP=47.4%, 27/57; RR=0.95; 95% CI, 0.64-1.41; p=0.786), with insignificant heterogeneity among studies (I^2=0; p=0.986) (see online supplementary file 7). Long-term mortality was not analysed because only one study reported. For the secondary outcome analysis, only bleeding and vascular complications met the quantitative requirements for the analysis. Compared with IABP, pVADs increased the rate of bleeding (pVADs=66.7%, 30/45; IABP=36.4%, 16/44; RR=1.86; 95% CI, 1.15-3.00; p=0.011; I^2=15.3%, p=0.227) and trended towards increased vascular complications (pVADs=15.5%, 9/58; IABP=0, 0/57; RR=5.48; 95% CI, 0.96-31.11; p=0.055; I^2=0, p=0.671) (see online supplementary file 7). The results of the prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 8).

Outcomes of Subgroup Analyses

In the subset of patients without cardiogenic shock (CS), compared with the treatment without MCS, the MCS treatment did not changed the short-term mortality (MCS=4.3%, 12/279; non-MCS=8.1%, 24/297; RR=0.51; 95% CI, 0.22-1.19; p=0.121; I^2=17.9%, p=0.301) (Figure 3A), but significantly decrease the long-term mortality (MCS=5.0%, 13/262; non-MCS=10.3%, 29/281; RR=0.49; 95% CI, 0.27-0.90; p=0.020; I^2=0, p=0.870) (Figure 3B). In patients with CS, MCS did not reduce short-term mortality (MCS=35.7%, 148/414; non-MCS=35.2%, 138/392; RR=1.09; 95% CI, 0.67-1.79; p=0.726; I^2=46.6%, p=0.100) (Figure 3A) or long-term mortality (MCS=45.6%, 166/364; non-MCS=45.9%, 166/362; RR=1.00; 95% CI, 0.75-1.33; p=0.984; I^2=22.1%, p=0.227) (Figure 3B). Due to the limited number of studies, subgroup analyses of the secondary outcomes were not performed.

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All patients had CS in studies comparing pVADs versus IABP, and the results have been described in the synthesis analyses.

Study Quality and Publication Bias

The results of the literature quality assessment using the Cochrane Collaboration tool are shown in online supplementary file 9. Three trials had a high risk of bias [29,33,35]. There was no evidence of publication bias for included studies except the Egger's test for short-term mortality of studies comparing pVADs use versus IABP use (P<0.05) (online supplementary file 10 and online supplementary file 11). Then the trim-and-fill method was performed for the estimation of the number of missing studies (pVADs vs IABP) that might exist. Result suggested 2 theoretically missing studies with an adjusted RR of short-term mortality (RR=0.92; 95% CI, 0.67-1.26; p=0.59).

Discussion

Summary of the evidence

It is the first study that evaluate the efficacy and safety of MCS in patients with AMI undergoing stent implantation. Compared with non-MCS, MCS does not reduce mortality in AMI patients undergoing stent implantation and other adverse events (reinfarction, repeat revascularization, stroke-TIA, bleeding, arrhythmias and vascular complications) in AMI patients undergoing stent implantation. Surprisingly, the results shows that MCS reduced long-term mortality by half in patients without CS compared to the non-MCS (5.0% versus 10.3%). This benefit does not exist in the patients with CS. The strategy of using pVADs did not outperformed IABP regarding to all-cause mortality in the patients with CS, but increased rate of bleeding compared to IABP.

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Efficacy of MCS in patients without CS

In the analysis of MCS in patients without CS, only IABP related studies were found in included RCTs. Regarding the work mechanism of IABP, deflation of the pump reduces ventricular workload and helps the ventricle push blood into the aorta during systole, and inflation of the pump improves coronary blood flow during diastole [36]. The evidence from clinical trials also shows the beneficial effect of IABP on haemodynamic, as these pumps can significantly improve myocardial coronary blood flow and myocardial ischaemia [37,38]. IABP might have a greater effect in patients with critically reduced coronary artery perfusion and temporarily impaired left ventricular function [39]. The haemodynamic of patients without CS may recover more quickly with IABP support, avoiding more serious clinical consequences, such as progression to CS, severe left heart failure, or even death. In the last decades, the application of PCI technology significantly reduces mortality in STEMI patients [11,40]. In particular, stenting reduces acute risk of major complications and restenosis [15]. One of our findings shows that IABP can benefit long-term mortality reduction in AMI patients without CS. This suggests that the use of IABP in AMI patients without CS with contemporary recommended reperfusion strategy (stent implantation) is reasonable. Some previous meta-analyses showed similar conclusions, but there was no detailed distinction between the revascularization strategies of thrombolytic therapy or PCI (balloon angioplasty and stent implantation) [16,17,41]. This result from the meta-analysis generates a promising hypothesis that IABP reduced long-term mortality in the non-CS AMI patients receiving stenting treatment. Studies with randomized control and decent sample size are warranted to confirm this finding.

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Efficacy of MCS in patients with CS

Previous guidelines have recommended IABP as the complementary strategy to maintaining a stable haemodynamic in patients with combined AMI and CS [3,4,42]. However, accumulating evidence from large-scale RCTs, including IABP-SHOCK II and PAMI-II, as well as registry data, did not show that the use of IABP bring benefit to the mortality reduction in the AMI patients with CS [8,9,22,43,44]. The findings from our study also showed that MCS was not associated with an all-cause mortality reduction in patients with CS undergoing stent implantation. Patients with AMI and CS are considered a high-risk group with more impaired cardiac function and haemodynamic instability and are more likely to develop circulatory failure during PCI [45]. The IABP SHOCK trial showed that IABP treatment do not significantly improve the hemodynamic profile in patients with combined AMI and CS [46]. Even though AMI patients with CS with currently advanced reperfusion strategy and MCS combined use, mortality was not reduced in our meta-analysis, which is consistent with the results from previous meta-analyses focusing on the impact of IABP in CS patients [14,47,48,49]. Therefore, IABP should be used with caution when patients have a combination of AMI and CS.

Several clinical trials exploring the efficacy of using different types of MCS in patients with CS, including ECMO, pVADs or the combination of two types MCS, are ongoing (NCT04184635, NCT03637205, NCT01633502, NCT03813134, NCT02301819, NCT03729765, NCT03431467, NCT03947619). We look forward to seeing those studies provide new insights into better MCS strategies in improving survival in patients with AMI and CS. Moreover, we compared the efficacy of survival between pVADs and IABP in patients combined with AMI and CS. The results showed that pVADs do not reduce the all-cause mortality, but significantly increased the rate of bleeding. These results are consistent with previous which focus on the impact of pVADs in CS patients [6,13].

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Clinical implications

In AMI patients without CS and received stenting implementation, the application of IABP may improve the long-term survival. However, in the patients with CS, neither IABP nor pVADs improves the survival outcome. Moreover, pVADs may bring higher risk of bleeding in the AMI patients with CS. This result generates an interesting hypothesis that IABP reduced long-term mortality in the non-CS AMI patients who receiving stenting treatment. RCT studies with a large number of patients are warranted to confirm this finding.

Limitations

There are several limitations in the current meta-analysis. First, the number of included RCTs and participants was limited. Second, due to few reports of secondary endpoints, subsequent subgroup analyses could not be performed. Third, there was high heterogeneity in the pooled CS group, which may be explained by inconsistent timing of initiation, higher incidence of interactions, and inconsistent time points for reporting mortality. Fourth, there were very limited number of RCTs comparing the efficacy between pVADs with IABP and the pVADs devices have large heterogeneities.

Conclusions

The use of MCS does not reduce mortality in overall AMI patients undergoing stent implantation. However, IABP demonstrated a promising effect in reducing the long-term mortality in AMI patents

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without CS. However, in patients with CS, MCS is not associated with a mortality reduction and pVADs further increased risk of bleeding.

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Authors' contributions

JC contributed to the conception and revision of the review. YS and YW performed systematic literature search, study selection data extraction and data analyses. YS contributed to writing and revision of the manuscript. XS contributed to review the data and manuscript. YT, MJ, YB, SL, WJ, YL and HY checked the manuscript.

Patient and public involvement

No patient involved.

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Ethical Approval Statement

Not applicable.

Declaration of interest

None.

Data sharing statement

All data associated with this manuscript are included in the main text and supplementary materials.

Supplementary materials

Supplementary data of this article can be found online.

Online supplementary file 1: PRISMA checklist for the meta-analysis

Online supplementary file 2: Search strategy

Online supplementary file 3: The Endpoint Definitions of the Included Studies

Online supplementary file 4: The Enrollment Information of the Included Studies

Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

Online supplementary file 6: Forest plots for the secondary end points of AMI patients treated with MCS compared with non-MCS. TIA, transient ischaemic attack.

Online supplementary file 7: Forest plots for outcomes occurrence comparing pVADs use versus IABP use

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Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Online supplementary file 10: Funnel plot of standard error for: A. MCS compared with non-MCS in short-term mortality; B. MCS compared with non-MCS in long-term mortality; C. pVADs compared with IABP in short-term mortality

Online supplementary file 11: Publication bias on the association of primary outcome

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Figure legends

Figure 1. Flow diagram of study selection

Randomized clinical trials were included.

Figure 2. Forest plots for the all-cause mortality

Risk ratio of all-cause mortality of AMI patients treated with MCS compared with non-MCS in short-term or long-term. CI, confidence interval; MCS, mechanical circulatory support; non-MCS, no plan of MCS use during peri-operative period; Short-term mortality, occurring within 30 days; long-term mortality, occurring at least 6 months of follow-up.

Figure 3. Forest plots showing the all-cause mortality for the subgroup analyses

A. Risk ratio of short-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. **B.** Risk ratio of long-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. CS, cardiogenic shock.

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Table 1. Study Design of the Included Studies

Study	Year	MCS group / non-MCS group	No. of Patients	Stenting, %	BMS, %	PCI success, %	Timing of MCS	Time point of reported mortality	Follow-Up
SEMPER FI[25]	2020	IABP+ stenting/stenting	50/50	100	NA	90/96	After Stenting	30 days, 6 months	6 months
Zhou et al[29]	2017	IABP+ stenting/stenting	42/42	100	NA	NA	Before Stenting	In hospital, 12 months	12 months
IABP-SHOCK II[8,9]	2012, 2013	IABP+ stenting/stenting	301/299	89.8	50.2/51.5	82.2/81.5	Before or after Stenting	30 days, 6 months	6 years
CRISP AMI[28]	2011	IABP+ stenting/stenting	161/176	96.9	53.7/53.1	92.9/95.3	Before Stenting	30 days, 6 months	6 months
Gu et al[30]	2011	IABP+ stenting/stenting	51/55	100	NA	NA	Before Stenting	30 days, 6 months	6 months
Prondzinsky et al[31]	2010	IABP+ stenting/stenting	19/21	85	NA	NA	After Stenting	In hospital	In hospital
Arias et al[33]	2005	IABP+ stenting/stenting	31/9	100	NA	NA	Before Stenting	In hospital	In hospital
Vijayalakshmi et al[34]	2007	IABP+ stenting/stenting	17/16	100	NA	NA	After Stenting	In hospital, 30 days	30 days
IMPRESS[26]	2017	pVADs+ stenting /IABP+ stenting	24/24	97.9	4/8	NA	After Stenting	30 days, 6 months	6 months
Seyfarth et al[32]	2008	pVADs+ stenting /IABP+ stenting	13/13	NA	NA	100/92	After Stenting	30 days	30 days
Thiele et al[27]	2005	pVADs+ stenting /IABP+ stenting	21/20	95.1	NA	80/74	Before or after Stenting	30 days	30 days
Korbinian et al[35]	2020	ECLS+ stenting / stenting	21/21	100	NA	NA	After stenting	30 days, 12 months	12 months

MCS, indicates mechanical circulatory support; IABP, intra-aortic balloon pump; PCI, percutaneous coronary intervention, including stenting and balloon angioplasty; pVADs, percutaneous ventricular assist devices; ECLS, extracorporeal life support; BMS, bare-metal stent; NA, not available. PCI success was defined as TIMI III flow.

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Table 2. Baseline Clinical Characteristics of the Included Studies

Study (year)	Male, %	Mean age, year	AMI, %	CS, %	Infarct-related artery (LM/LAD/LCX/RCA), %	Hypertension, %	Diabetes mellitus, %	Previous stroke/TIA, %	Prior MI, %	Prior PCI, %	Prior CABG, %
SEMPER FI[25]	74/78	64/63	100	0	1.0/50.0/11.0/33.0	24/32	8/18	2/2	6/10	8/12	2/8
Zhou et al[29]	67/62	57.2/56.7	100	100	0/34.5/38.1/27.4	NA	NA	NA	NA	NA	NA
IABP-SHOCK II[8,9]	70.6/67.1	70/69	100	100	9/42.2/18.7/25.3	72/66.6	35.4/30.1	8/6.7	23.7/20.4	21.1/17.4	6.7/4.0
CRISP AMI[28]	82/81.8	56.1/57.7	100	0	0/97.3/0/0.3	24.2/34.1	16.8/20.5	0/0.6	0/0	1.9/1.1	NA
Gu et al[30]	56.9/65.5	67.4/66.6	100	0	3.8/47.2/11.3/37.7	68.6/60	35.5/34.5	NA	3.9/5.5	68.6/70.9	NA
Prondzinsky et al[31]	74/81	62.1/66.1	100	100	NA	42.1/47.6	52.6/47.6	NA	21.1/23.8	NA	NA
Arias et al[33]	NA	NA	100	100	0/60.0/22.5/8.3	NA	NA	NA	NA	NA	NA
Vijayalakshmi et al[34]	82/88	40/45	100	0	NA	25/38	18/25	NA	12/25	NA	NA
IMPRESS[26]	75/83	58/59	100	100	6.3/64.6/18.8/10.4	20/29	9/13	0/4	5/4	NA	NA
Seyfarth et al[32]	62/85	65/67	100	100	NA	54/69	39/23	NA	NA	NA	NA
Thiele et al[27]	76/75	63/65	100	100	NA	76/75	52/55	NA	62/45	NA	NA
Korbinian et al[35]	76.2/ 95.2	62/ 70	100	100	9.5/54.8/16.7/19.0	55/75	15/30	0/10	15/30	20/20	0/0

AMI indicates acute myocardial infarction; CS, cardiogenic shock; LM, left main; LAD, left anterior descending; LCX, left circumflex; RCA, right coronary artery; TIA, transient ischemic attack; MI, myocardial infarction; CABG, coronary artery bypass graft.

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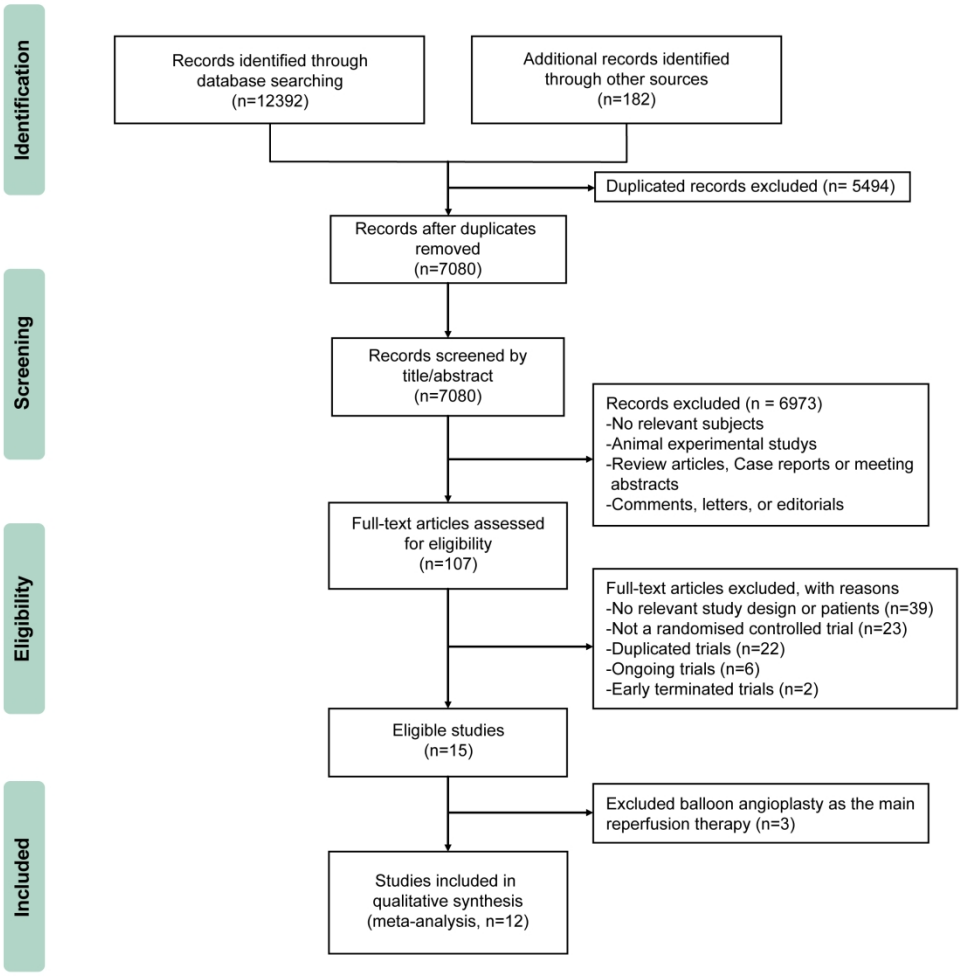


Figure 1. Flow diagram of study selection
Randomized clinical trials were included.

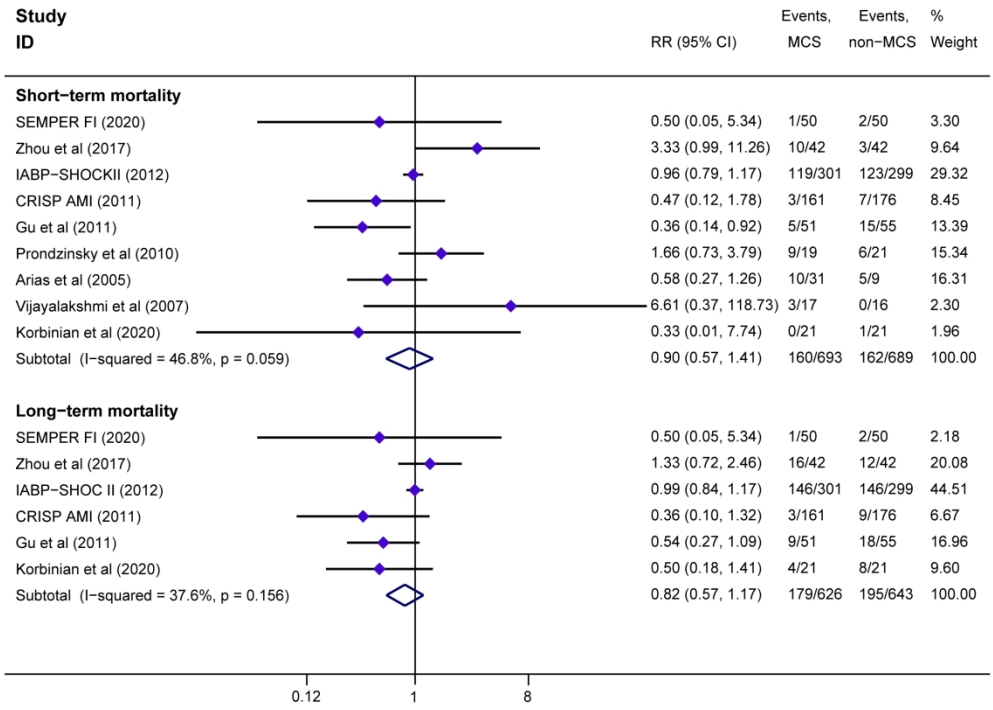


Figure 2. Forest plots for the all-cause mortality
Risk ratio of all-cause mortality of AMI patients treated with MCS compared with non-MCS in short-term or long-term. CI, confidence interval; MCS, mechanical circulatory support; non-MCS, no plan of MCS use during peri-operative period; Short-term mortality, occurring within 30 days; long-term mortality, occurring at least 6 months of follow-up.

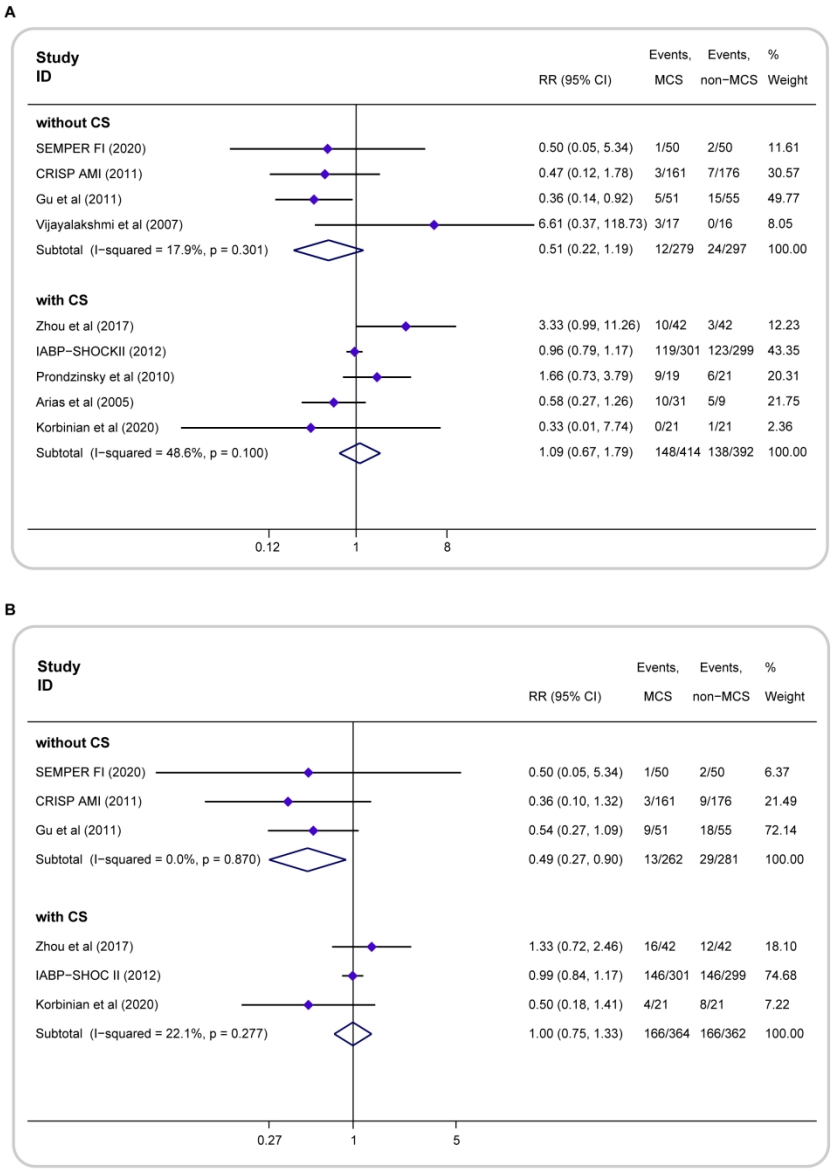


Figure 3. Forest plots showing the all-cause mortality for the subgroup analyses
A. Risk ratio of short-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. B. Risk ratio of long-term mortality of patients with AMI with or without CS treated with MCS compared with non-MCS. CS, cardiogenic shock.

Online supplementary file 1: PRISMA checklist for the meta-analysis

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6 and Online supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7, and Online supplementary file 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used	7

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		in any data synthesis.	
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9, Table 1, Table 2 and Online supplementary file 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 and Online supplementary file 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12, Online supplementary file 9, Online supplementary file 10 and Online supplementary file 11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-11, Figure 3, Online supplementary file 5 and Online supplementary file 8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15

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Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

NA indicates not available.

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Online supplementary file 2: Search strategy

Search strategy in Embase

#	Searches	Results
1	'heart infarction'/exp	409027
2	heart:ab,ti AND infarct*:ab,ti	106516
3	myocard*:ab,ti AND infarct*:ab,ti	306544
4	heart:ab,ti AND attack*:ab,ti	17766
5	cardiovascular:ab,ti AND stroke*:ab,ti	60439
6	stemi:ab,ti	29180
7	nstemi:ab,ti OR 'non stemi':ab,ti OR (non:ab,ti AND stemi:ab,ti)	14111
8	'ischemic heart disease'/exp	709611
9	heart:ab,ti AND ischemi*:ab,ti	120006
10	myocard*:ab,ti AND ischemi*:ab,ti	112169
11	'coronary artery disease'/exp	353023
12	coronary:ab,ti AND artery:ab,ti AND disease*:ab,ti	178679
13	coronary:ab,ti AND atheroscleros*:ab,ti	43833
14	acute:ab,ti AND coronary:ab,ti AND syndrom*:ab,ti	62869
15	acs:ab,ti	43936
16	'angina pectoris'/exp	105403
17	angina*:ab,ti	79942
18	angor:ab,ti AND pectoris:ab,ti	86
19	stenocardia*:ab,ti	927
20	myocardial:ab,ti AND preinfarction:ab,ti AND syndrom*:ab,ti	41
21	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20	1020302
22	'heart assist device'/exp	43026
23	heart:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	18671
24	vascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2623
25	cardiovascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	1949
26	artificial:ab,ti AND ventricle*:ab,ti	1772
27	ventricle:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2918
28	ventricular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	24992
29	artificial:ab,ti AND heart:ab,ti AND ventricle*:ab,ti	890
30	blood:ab,ti AND pump*:ab,ti	27560
31	cardiac:ab,ti AND resuscitat*:ab,ti	32204
32	counterpulsation:ab,ti AND device*:ab,ti	646
33	mechanical:ab,ti AND circulat*:ab,ti AND support*:ab,ti	9334
34	cardiac:ab,ti AND assist*:ab,ti AND system*:ab,ti	5468
35	assist*:ab,ti AND circulat*:ab,ti	13513

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36	lvad:ab,ti	12954
37	pvad:ab,ti	375
38	tandemheart*:ab,ti	332
39	heartmate*:ab,ti	3663
40	impella*:ab,ti	2192
41	intra:ab,ti AND aortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	6288
42	intraaortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	2414
43	iabp:ab,ti	4778
44	iabc:ab,ti	172
45	'extracorporeal oxygenation'/exp	27601
46	extracorporeal:ab,ti AND membrane:ab,ti AND oxygenation*:ab,ti	17345
47	extracorporeal:ab,ti AND life:ab,ti AND support*:ab,ti	4998
48	ecmo:ab,ti	17782
49	ecls:ab,ti	2325
50	'heart lung machine'/exp	2137
51	heart:ab,ti AND lung:ab,ti AND machine*:ab,ti	1596
52	'extracorporeal circulation'/exp	73515
53	extracorporeal:ab,ti AND circulat*:ab,ti	15079
54	extracorporeal:ab,ti AND cardiopulmonary:ab,ti AND resuscitation*:ab,ti	1987
55	extracorporeal:ab,ti AND CPR:ab,ti	993
56	ecpr:ab,ti	901
57	#22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56	215006
58	#21 AND #57	39754
59	'animals'/exp NOT 'humans'/exp	5572218
60	'animals, laboratory'/exp	716834
61	'animal experiment'/exp	2667375
62	'animal model'/exp	1461488
63	'rodent'/exp	3906464
64	rat:ti OR rats:ti OR mouse:ti OR mice:ti	1627604
65	#59 OR #60 OR #61 OR #62 OR #63 OR #64	6677800
66	#58 NOT #65	35401
67	'randomized controlled trial':pt OR 'randomized controlled trial'/exp OR 'randomized controlled trials as topic'/exp	839617
68	'controlled clinical trial'/de	432791
69	randomi?ed:ab	880556
70	placebo:ab	310582
71	randomly:ab	466369

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72	'clinical trials as topic'/de	109908
73	trial:ti	327712
74	#67 OR #68 OR #69 OR #70 OR #71 OR #72 OR #73	1929647
75	#66 AND #74	3852

Search strategy in Cochrane Library

#	Searches	Results
1	[mh "Myocardial Infarction"]	11198
2	(Myocard* Infarct*):ti,ab,kw	31737
3	(Cardiovascular Stroke*):ti,ab,kw	9663
4	(Heart Attack*):ti,ab,kw	3717
5	(heart infarct*):ti,ab,kw	21977
6	STEMI:ti,ab,kw	3415
7	(NSTEMI or Non-STEMI or (Non and STEMI)):ti,ab,kw	1241
8	[mh "Myocardial Ischemia"]	28522
9	(Myocard* Ischemi*):ti,ab,kw	13500
10	(heart ischemi*):ti,ab,kw	17249
11	[mh "Acute Coronary Syndrome"]	2051
12	(Acute Coronary Syndrom*):ti,ab,kw	7589
13	ACS:ti,ab,kw	4150
14	[mh "Coronary Artery Disease"]	6571
15	(Coronary Artery AND Disease*):ti,ab,kw	23566
16	(Coronary Arterioscleros*):ti,ab,kw	503
17	[mh "Angina Pectoris"]	4588
18	Stenocardia*:ti,ab,kw	64
19	(Angor Pectoris):ti,ab,kw	21
20	angina*:ti,ab,kw	14048
21	(Myocardial Preinfarction Syndrom*):ti,ab,kw	7
22	{or #1-#21}	76205
23	[mh "Heart-Assist Devices"]	180
24	(Heart and Assist* (Device* or pump*)):ti,ab,kw	1280
25	(Vascular and Assist* (Device* or pump*)):ti,ab,kw	265
26	(cardiovascular and Assist* and (Device* or pump*)):ti,ab,kw	307
27	(Artificial Ventricle*):ti,ab,kw	661
28	(Ventricle and Assist* and (Device* or pump*)):ti,ab,kw	241
29	(Ventricular and Assist* and (Device* OR pump*)):ti,ab,kw	764
30	(Artificial Heart Ventricle*):ti,ab,kw	636
31	(blood pump*):ti,ab,kw	5249
32	(cardiac resuscitat*):ti,ab,kw	2945
33	(counterpulsation device*):ti,ab,kw	51
34	(mechanical circulat* support*):ti,ab,kw	340
35	(Cardiac Assist* System*):ti,ab,kw	710

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36	(Assist* Circulat*):ti,ab,kw	1077
37	LVAD:ti,ab,kw	346
38	pVAD:ti,ab,kw	14
39	TandemHeart*:ti,ab,kw	9
40	HeartMate*:ti,ab,kw	107
41	Impella*:ti,ab,kw	112
42	[mh "Intra-Aortic Balloon Pumping"]	167
43	(Intra and Aortic and Balloon and (Pump* or counterpulsation)):ti,ab,kw	478
44	(Intraaortica and Balloon and (Pump* or counterpulsation)):ti,ab,kw	4
45	IABP:ti,ab,kw	337
46	IABC:ti,ab,kw	15
47	[mh "Extracorporeal Membrane Oxygenation"]	178
48	(Extracorporeal Membrane Oxygenation*):ti,ab,kw	660
49	(Extracorporeal Life Support*):ti,ab,kw	130
50	ECMO:ti,ab,kw	619
51	ECLS:ti,ab,kw	47
52	[mh "Heart-Lung Machine"]	37
53	(Heart Lung Machine*):ti,ab,kw	262
54	[mh "Extracorporeal Circulation"]	4388
55	(Extracorporeal and Circulat*):ti,ab,kw	1392
56	(extracorporeal cardiopulmonary resuscitation*):ti,ab,kw	48
57	(extracorporeal CPR):ti,ab,kw	23
58	ECPR:ti,ab,kw	20
59	{or #23-#58}	17090
60	#22 and #59	3418
61	[mh "animals"] not [mh "humans"]	61
62	rat or rats or mouse or mice:ti,ab,kw	13725
63	#61 or #62	13782
64	#60 not #63	3379
65	randomized:ti,ab,kw	915787
66	placebo:ti,ab,kw	310693
67	randomly:ti,ab,kw	251577
68	clinical trials:ti,ab,kw	176838
69	{or #65-#68}	1131307
70	#64 and #69	2684

Search strategy in Medline

#	Searches	Results
1	Myocardial Infarction/	167860
2	(Myocard$ and infarct$).ab,ti.	208035
3	(heart and infarct$).ab,ti.	65507

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4	(Heart and Attack$).ab,ti.	11461
5	(Cardiovascular and Stroke$).ab,ti.	35842
6	STEMI.ab,ti.	11486
7	(NSTEMI or Non-STEMI or (Non and STEMI)).ab,ti.	4639
8	Myocardial Ischemia/	39284
9	(Myocard$ and Ischemi$).ab,ti.	76017
10	(heart and ischemi$).ab,ti.	73007
11	Acute Coronary Syndrome/	16517
12	(Acute and Coronary and Syndrom$).ab,ti.	35635
13	ACS.ab,ti.	23142
14	Coronary Artery Disease/	64571
15	(Coronary and Artery and Disease$).ab,ti.	112543
16	(Coronary and Arterioscleros$).ab,ti.	1916
17	Angina Pectoris/	32616
18	Stenocardia$.ab,ti.	951
19	(Angor and Pectoris).ab,ti.	45
20	Angina$.ab,ti.	55365
21	(Myocardial and Preinfarction and Syndrom$).ab,ti.	23
22	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21	523082
23	Heart Assist Devices/	14753
24	(Heart and Assist$ and (Device$ or pump$)).ab,ti.	9718
25	(Vascular and Assist$ and (Device$ or pump$)).ab,ti.	1324
26	(Cardiovascular and Assist$ and (Device$ or pump$)).ab,ti.	1090
27	(Artificial and Ventricle$).ab,ti.	1183
28	(Ventricle and Assist$ and (Device$ or pump$)).ab,ti.	1635
29	(Ventricular and Assist$ and (Device$ or pump$)).ab,ti.	13854
30	(Artificial and Heart and Ventricle$).ab,ti.	563
31	(blood and pump$).ab,ti.	17426
32	(cardiac and resuscitat$).ab,ti.	20141
33	(counterpulsation and device$).ab,ti.	414
34	(mechanical and circulat$ and support$).ab,ti.	5182
35	(Cardiac and Assist$ and System$).ab,ti.	3192
36	(Assist$ and Circulat$).ab,ti.	8583
37	LVAD.ab,ti.	4852
38	pVAD.ab,ti.	162
39	TandemHeart$.ab,ti.	193
40	HeartMate$.ab,ti.	1480
41	Impella$.ab,ti.	925
42	Intra Aortic Balloon Pumping/	4056
43	(Intra and Aortic and Balloon AND (Pump* or counterpulsation)).ab,ti.	3937
44	(Intraaortic and Balloon and (Pump* or counterpulsation)).ab,ti.	1667
45	IABP.ab,ti.	2199

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46	IABC.ab,ti.	120
47	Extracorporeal Membrane Oxygenation/	11447
48	(Extracorporeal and Membrane and Oxygenation$).ab,ti.	11804
49	(Extracorporeal and Life and Support$).ab,ti.	3112
50	ECMO.ab,ti.	8267
51	ECLS.ab,ti.	1293
52	Heart Lung Machine/	2095
53	(Heart and Lung and Machine$).ab,ti.	1060
54	Extracorporeal Circulation/	12973
55	(Extracorporeal and Circulat*).ab,ti.	11010
56	(extracorporeal and cardiopulmonary and resuscitation*).ab,ti.	1242
57	(extracorporeal and CPR).ab,ti.	427
58	ECPR.ab,ti.	380
59	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58	100032
60	22 and 59	12723
61	animals/ not humans/	4757950
62	exp Animals, Laboratory/	892881
63	exp Animal Experimentation/	9614
64	exp Models, Animal/	584650
65	exp rodentia/	3282798
66	(rat or rats or mouse or mice).ti.	1355142
67	61 or 62 or 63 or 64 or 65 or 66	5725191
68	60 not 67	11457
69	randomized controlled trial.pt. or randomized controlled trial/ or Randomized Controlled Trials as Topic/	658872
70	controlled clinical trial.pt.	94082
71	placebo.ab.	215311
72	randomly.ab.	351488
73	clinical trials as topic.sh.	194813
74	randomi?ed.ab.	609920
75	trial.ti.	234697
76	69 or 70 or 71 or 72 or 73 or 74 or 75	1439354
77	68 and 76	1339

Search strategy in Pubmed

#	Searches	Results
1	"Myocardial Infarction"[Mesh]	178446
2	(Myocard*[Title/Abstract]) AND Infarct*[Title/Abstract]	213501
3	(Cardiovascular[Title/Abstract]) AND Stroke*[Title/Abstract]	37057
4	(Heart[Title/Abstract]) AND Attack*[Title/Abstract]	11736

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5	(heart[Title/Abstract]) AND infarct*[Title/Abstract]	69148
6	STEMI[Title/Abstract]	11856
7	(((NSTEMI[Title/Abstract]) OR Non-STEMI[Title/Abstract])) OR ((Non[Title/Abstract]) AND STEMI[Title/Abstract])	4838
8	"Myocardial Ischemia"[Mesh]	437366
9	(Myocard*[Title/Abstract]) AND Ischemi*[Title/Abstract]	77593
10	(heart[Title/Abstract]) AND ischemi*[Title/Abstract]	74762
11	"Coronary Artery Disease"[Mesh]	64579
12	((Coronary[Title/Abstract]) AND Artery[Title/Abstract]) AND Disease*[Title/Abstract]	117973
13	(Coronary[Title/Abstract]) AND Arterioscleros*[Title/Abstract]	2525
14	"Acute Coronary Syndrome"[Mesh]	16526
15	((Acute[Title/Abstract]) AND Coronary[Title/Abstract]) AND Syndrom*[Title/Abstract]	37318
16	ACS[Title/Abstract]	23712
17	"Angina Pectoris"[Mesh]	43697
18	Stenocardia*[Title/Abstract]	952
19	(Angor[Title/Abstract]) AND Pectoris[Title/Abstract]	58
20	angina*[Title/Abstract]	56696
21	((Myocardial[Title/Abstract]) AND Preinfarction[Title/Abstract]) AND Syndrom*[Title/Abstract]	44
22	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21	617614
23	"Heart-Assist Devices"[Mesh]	14752
24	((Heart[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	10556
25	((Vascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1407
26	((cardiovascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1202
27	(Artificial[Title/Abstract]) AND Ventricle*[Title/Abstract]	1255
28	((Ventricle[Title/Abstract]) AND Assist*[Title/Abstract]) AND Device*[Title/Abstract]	1601
29	((Ventricular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	14282
30	((Artificial[Title/Abstract]) AND Heart[Title/Abstract]) AND Ventricle*[Title/Abstract]	610
31	(blood[Title/Abstract]) AND pump*[Title/Abstract]	17702
32	(cardiac[Title/Abstract]) AND resuscitat*[Title/Abstract]	21366
33	(counterpulsation[Title/Abstract]) AND device*[Title/Abstract]	451
34	((mechanical[Title/Abstract]) AND circulat*[Title/Abstract]) AND support*[Title/Abstract]	5748
35	((Cardiac[Title/Abstract]) AND Assist*[Title/Abstract]) AND	3326

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	System*[Title/Abstract]	
36	((Assist*[Title/Abstract])) AND Circulat*[Title/Abstract]	9223
37	LVAD[Title/Abstract]	5058
38	pVAD[Title/Abstract]	173
39	TandemHeart*[Title/Abstract]	204
40	HeartMate*[Title/Abstract]	1515
41	Impella*[Title/Abstract]	972
42	"Intra-Aortic Balloon Pumping"[Mesh]	4056
43	(((Intra[Title/Abstract]) AND Aortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	4026
44	((Intraaortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	5487
45	IABP[Title/Abstract]	2253
46	IABC[Title/Abstract]	127
47	"Extracorporeal Membrane Oxygenation"[Mesh]	11452
48	((Extracorporeal[Title/Abstract]) AND Membrane[Title/Abstract]) AND Oxygenation*[Title/Abstract]	12432
49	((Extracorporeal[Title/Abstract]) AND Life[Title/Abstract]) AND Support*[Title/Abstract]	3413
50	ECMO[Title/Abstract]	8706
51	ECLS[Title/Abstract]	1416
52	"Heart-Lung Machine"[Mesh]	2095
53	((Heart[Title/Abstract]) AND Lung[Title/Abstract]) AND Machine*[Title/Abstract]	1081
54	"Extracorporeal Circulation"[Mesh]	84949
55	(Extracorporeal[Title/Abstract]) AND Circulat*[Title/Abstract]	11650
56	((extracorporeal[Title/Abstract]) AND cardiopulmonary[Title/Abstract]) AND resuscitation*[Title/Abstract]	1349
57	(extracorporeal[Title/Abstract]) AND CPR[Title/Abstract]	477
58	ECPR[Title/Abstract]	482
59	#23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56 OR #57 OR #58	158913
60	#22 AND #59	20135
61	((animals[mh]) NOT humans[mh])	4791677
62	Animals, Laboratory[mh]	892913
63	Animal Experimentation[mh]	9615
64	Models, Animal[mh]	584678
65	rodentia[mh]	3282948
66	((rat[Title] OR rats[Title] OR mouse[Title] OR mice[Title]))	1355108

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67	#61 OR #62 OR #63 OR #64 OR #65 OR #66	5741699
68	#60 NOT #67	18053
69	((randomized controlled trial[Publication Type]) OR randomized controlled trial[mh]) OR Randomized Controlled Trials as Topic[mh]	661250
70	controlled clinical trial[Publication Type]	614085
71	((randomized[Title/Abstract]) OR randomised[Title/Abstract])	660183
72	placebo[Title/Abstract]	221488
73	randomly[Title/Abstract]	352402
74	trial[Title]	234625
75	#69 OR #70 OR #71 OR #72 OR #73 OR #74	1313992
76	#68 and #75	2105

Search strategy in Web of Science

#	Searches	Results
1	TS=(heart AND infarct*)	345636
2	TS=(myocard* AND infarct*)	452517
3	TS=(heart AND attack*)	44545
4	TS=(cardiovascular AND stroke*)	232512
5	TS=STEMI	20441
6	TS=(NSTEMI OR Non-STEMI OR (Non AND STEMI))	8228
7	TS=(Myocard* AND Ischemi*)	190331
8	TS=(heart AND ischemi*)	276064
9	TS=(Acute AND Coronary AND Syndrom*)	74970
10	TS=ACS	43248
11	TS=(Coronary AND Artery AND Disease*)	351532
12	TS=(Coronary AND Arterioscleros*)	48592
13	TS=Angina*	103224
14	TS=Stenocardia*	1163
15	TS=(Angor AND Pectoris)	472
16	TS=(Myocardial AND Preinfarction AND Syndrom*)	61
17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16	1094596
18	TS=(Heart AND Assist* AND (Device* OR pump*))	34749
19	TS=(Vascular AND Assist* AND (Device* OR pump*))	10298
20	TS=(cardiovascular AND Assist* AND (Device* OR pump*))	21034
21	TS=(Artificial AND Ventricle*)	10067
22	TS=((Ventricular OR Ventricle) AND Assist* AND (Device* OR pump*))	28212
23	TS=(Artificial AND Heart AND Ventricle*)	8861
24	TS=(blood AND pump*)	58329
25	TS=(cardiac AND resuscitat*)	40430

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26	TS=(counterpulsation AND device*)	1133
27	TS=(mechanical AND circulat* AND support*)	21978
28	TS=(Cardiac AND Assist* AND System*)	25077
29	TS=(Assist* AND Circulat*)	59084
30	TS=LVAD	9139
31	TS=pVAD	254
32	TS=TandemHeart*	276
33	TS=HeartMate*	2711
34	TS=Impella*	1654
35	TS=(Intra AND Aortic AND Balloon AND (Pump* OR counterpulsation))	7553
36	TS=(Intraaortic AND Balloon AND (Pump* OR counterpulsation))	2998
37	TS=IABP	2862
38	TS=IABC	283
39	TS=(Extracorporeal AND Membrane AND Oxygenation*)	21874
40	TS=(Extracorporeal AND Life AND Support*)	6398
41	TS=ECMO	12728
42	TS=ECLS	2140
43	TS=(Heart AND Lung AND Machine*)	3515
44	TS=(Extracorporeal AND Circulat*)	28102
45	TS=(extracorporeal AND cardiopulmonary AND resuscitation*)	2647
46	TS=(extracorporeal AND CPR)	764
47	TS=ECPR	664
48	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47	250465
49	#17 AND #48	47026
50	TS= clinical trial* OR TS=research design OR TS=comparative stud* OR TS=evaluation stud* OR TS=controlled trial* OR TS=follow-up stud* OR TS=prospective stud* OR TS=random* OR TS=placebo* OR TS=(single blind*) OR TS=(double blind*)	10462135
51	#49 AND #50	16870
52	#49 AND #50 limit clinical trials	2412

Search strategy in ClinicalTrials.gov

Basic search

104 Studies found for: Ventricular Assist Device OR Intra Aortic Balloon Pump OR Extracorporeal

Membrane Oxygenation | MI OR Myocardial Infarction OR Coronary Artery Disease

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Search strategy in Clinicaltrialsregister.eu

Basic search

78 Studies found for: "ventricular assist device" OR "intra aortic balloon pump" OR "Extracorporeal Membrane Oxygenation"

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Online supplementary file 3: The Endpoint Definitions of the Included Studies

Study	Endpoint	Definition
SEMPER FI[25]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding events according to the Global Use of Strategies to Open Coronary Arteries (GUSTO) trial.
	Arrhythmias	NA.
	Vascular complications	NA.
Zhou et al[29]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	Malignant arrhythmia (not detailed).
	Vascular complications	NA.
IABP-SHOCK II[8,9]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke identified by the presence of new neurologic symptoms in conjunction with signs of ischemia or bleeding on computed tomography.
	Bleeding	Severe or life-threatening bleeding and moderate bleeding as assessed according to the Global Use of Strategies to Open Occluded Coronary Arteries (GUSTO) criteria1.
	Arrhythmias	NA.
	Vascular complications	Peripheral ischemic vascular complications requiring surgical or interventional therapy.
CRISP AMI[28]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke (not detailed).
	Bleeding	Major bleeding events defined by theGlobal Use of Strategies To Open Coronary Arteries (GUSTO) trial and transfusions.
	Arrhythmias	NA.
	Vascular complications	Major vascular complications defined to include limb ischemia requiring intervention, distal embolization, major dissection, pseudoaneurysm or arteriovenousfistula, hematoma larger than 5 cm, and amputations.
Gu et al[30]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Vascular complications (not detailed).
Prondzinsky et al[31]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmia (not detailed).
	Vascular complications	NA.

Arias et al[33]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.
Vijayalakshmi et al[34]	Reinfarction	Reinfarction was defined as chest pain associated with ST-segment re-elevation and an increase in cardiac enzymes (3 times creatine kinase release).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmias including ventricular tachycardia, ventricular fibrillation or atrial fibrillation.
	Vascular complications	Vascular complications (not detailed).
IMPRESS[26]	Reinfarction	Reinfarction Defined by 3rd definition MI. Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following: 1. Symptoms of ischemia. 2. New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB). 3. Development of pathological Q waves in the ECG. 4. Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. 5. Identification of an intracoronary thrombus by angiography or autopsy.
	Repeat revascularization	Repeat PCI or CABG.
	Stroke-TIA	Any stroke confirmed by a neurologist and a concurring CT scan.
	Bleeding	1.Bleeding with associated serum hemoglobin level decrease of at least 5 g/dL (=3.1 mmol/L). The decrease in hemoglobin will be calculated as the last recorded Hb measurement preceding the onset of the bleeding, subtracted by the nadir Hb measurement (associated with the bleeding). 2. Bleeding necessitating a minimum of 2 packed cells of blood product transfusion (only blood transfusions that are explicitly related to the bleeding are taken into account). 3. The need for surgery to control the bleeding.
	Arrhythmias	Placement of implantable cardioverter-defibrillator.
	Vascular complications	1.A major bleed arising at the arterial access site (for major bleeding definition see above) requiring extraction of the device. 2. A thrombotic occlusion of the femoral artery. 3. Limb ischemia requiring extraction of either of the study devices. 4. The need for vascular surgery to correct a vascular complication.
Seyfarth et al[32]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Limb ischemia (not detailed).
Thiele et al[27]	Reinfarction	NA.

	Repeat revascularization	NA.
	Stroke-TIA	Cerebrovascular accidents with neurological dysfunction.
	Bleeding	Major bleedings requiring transfusion of blood components.
	Arrhythmias	NA.
	Vascular complications	Lower extremity ischaemia requiring surgical or interventional action.
Korbinian et al[35]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.

NA indicates not available.

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Online supplementary file 4: The Enrollment Information of the Included Studies

Study	Center of Enrollment	Time of Enrollment
SEMPER FI[25]	Catharina Hospital, Eindhoven, Netherlands	August 2014 to December 2018
Zhou et al[29]	Department of Cardiology of Liaocheng People's Hospital, Liaocheng, China	January 2014 to January 2016
IABP-SHOCK II[8,9]	University of Leipzig-Heart Center, Leipzig, Germany Heart Center Bad Krozingen, Bad Krozingen, Germany Asklepios Klinik Langen, Langen, Germany German Heart Center Munich, Munich, Germany Heart Center-Segeberger Kliniken, Bad Segeberg, Germany SLK Kliniken Heilbronn, Heilbronn, Germany University of Greifswald, Greifswald, Germany Martin-Luther-University Halle-Wittenberg, Halle (Saale), Germany Klinikum Links der Weser, Bremen, Germany Heart Center Ludwigshafen, Ludwigshafen, Germany Zentralklinik Bad Berka, Bad Berka, Germany University Clinic of Saarland, Homburg/Saar, Germany Friedrich-Schiller University Jena, Jena, Germany University of Lübeck, Lübeck, Germany Klinik Hennigsdorf, Hennigsdorf, Germany Otto-von-Guericke-University Magdeburg, Magdeburg, Germany Sana Heart Center Cottbus, Cottbus, Germany Heart Center Dresden, Dresden, Germany University of Würzburg, Würzburg, Germany Hospital München-Neuperlach, Germany Krankenhaus der Barmherzigen Brüder, Regensburg, Germany	June 2009 to March 2012

	Charité Berlin Campus Benjamin Franklin, Berlin, Germany Vivantes Klinikum am Urban, Berlin, Germany Carl-Thiem-Klinikum Cottbus, Cottbus, Germany University Rostock, Germany Kerckhoff Klinik Bad Nauheim, Bad Nauheim, Germany Helios Klinikum Erfurt, Erfurt, Germany Charité Berlin-Campus Mitte, Berlin, Germany Ludwigs-Maximilian University Munich-Hospital Großhadern, Germany Vivantes Auguste-Viktoria-Klinikum, Berlin, Germany Klinikum Weiden, Weiden, Germany Klinikum Fulda, Fulda, Germany Katharinenhospital Unna, Unna, Germany Philippusstift Essen, Essen, Germany Carl-von-Basedow-Klinikum Merseburg, Merseburg, Germany Klinikum Lippe-Detmold, Detmold, Germany Schwarzwald-Baar Klinikum VillingenSchwenningen, Villingen-Schwenningen, Germany	
CRISP AMI[28]	Flinders Medical Centre, Bedford Park, Australia Virga Jesse Hospital, Hasselt, Belgium Leeds General Infirmary, Leeds, England Saint Thomas Hospital, London, England Universitatsklinikum Leipzig, Leipzig, Germany University of Freiburg, Freiburg, Germany Klinik III Fur Innere Medizin der Universitat zu Koln, Dresden, Germany CARE Hospital, Hyderabad, India Baroda Heart Institute and Research Centre, Vadodara, India Sir Ganga Ram Hospital, New Delhi, India Gurunanak CARE Hospital, Hyderabad, India Medanta-The Medicity, Gurgaon, India	December 2008 to March 2011

	Tagore Hospital and Heart Care Centre, Jalandhar, India Indraprastha Apollo Hospital, New Delhi, India Apollo Gleneagles Hospital, Kolkata, India Saint John's Medical College and Hospital, Bangalore, India Mater Misercordiae Dublin, Dublin, Ireland Catherina Hospital, Eindhoven, Netherlands Royal Infirmary of Edinburgh, Edinburgh, Scotland Duke University Medical Center, Durham, United States University of Texas Health Science Center, Houston, United States Temple University Hospital, Philadelphia, United States Tufts Medical Center, Boston, United States Newark Beth Israel Medical Center, Newark, United States University of Rochester Medical Center-Strong Memorial Hospital, Rochester, United States Lancaster General Hospital, Lancaster, United States University of Alabama Medical Center, Birmingham, United States Washington Hospital Center, Washington, United States Allegheny General Hospital, Pittsburgh, United States University of Southern California University Hospital, Los Angeles, United States	
Gu et al[30]	Min Hang Hospital, Shanghai, China	January 2005 to February 2009
Prondzinsky et al[31]	Department of Medicine III, Department of Biometrics and Statistics (SU), Halle-Wittenberg, Germany	March 2003 to June 2004
Arias et al[33]	Departamento de Medicina Interna, Fundación Clínica Médica Sur, México	February 2001 to February 2003
Vijayalakshmi et al[34]	The James Cook University Hospital, Middlesbrough, United Kingdom	NA
IMPRESS[26]	Academic Medical Center, Amsterdam, Netherlands Haukeland University Hospital, Bergen, Norway	May 2012 to September 2015
Seyfarth et al[32]	Deutsches Herzzentrum München, Munich, Germany	August 2004 to March

	Medizinische Klinik rechts der Isar, Munich, Germany	2006
Thiele et al[27]	University of Leipzig-Heart Centre, Leipzig, Germany	August 2000 to December 2003
Korbinian et al[35]	University Hospital Munich, Munich, Germany	November 2015 to March 2018

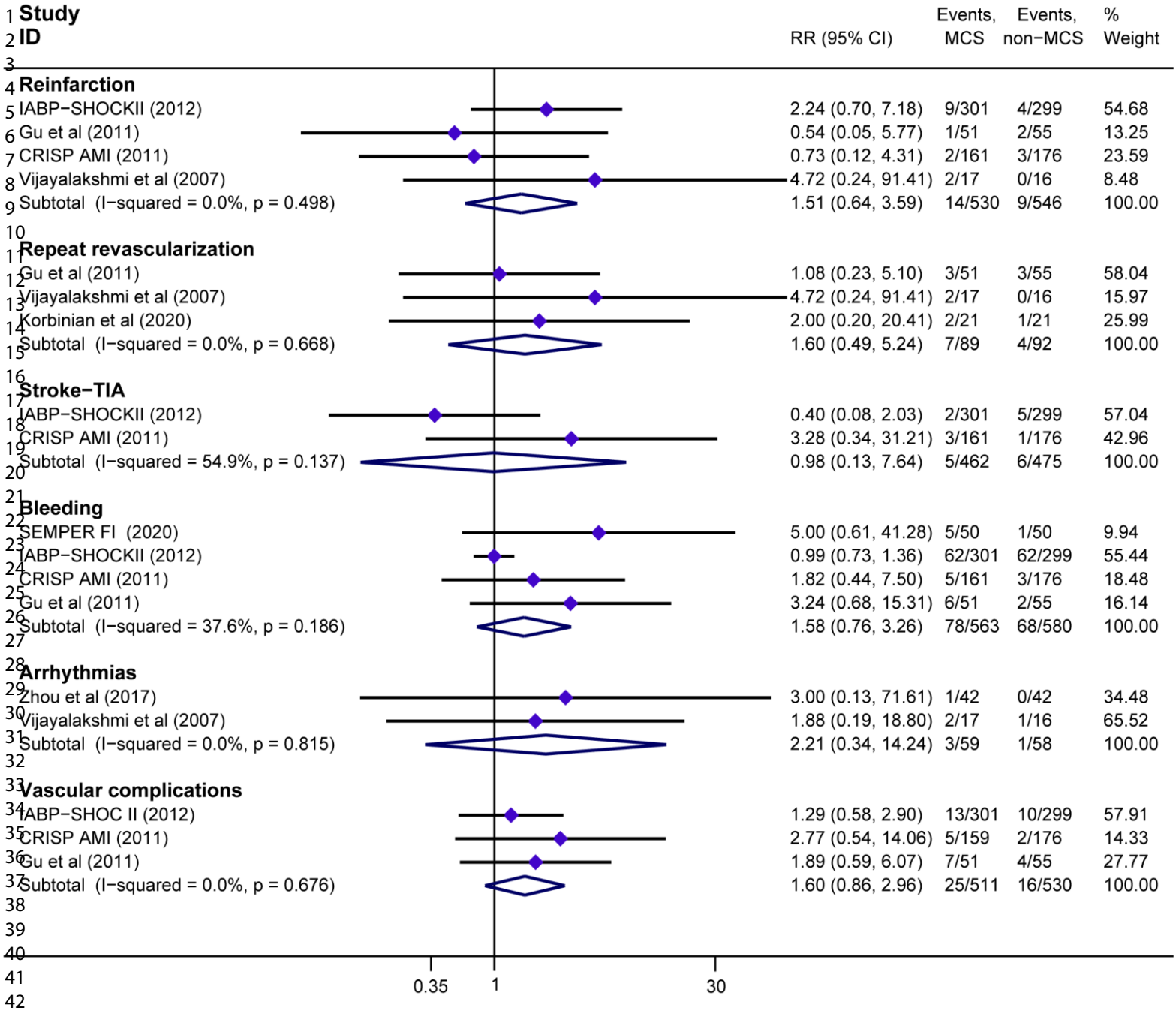
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Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

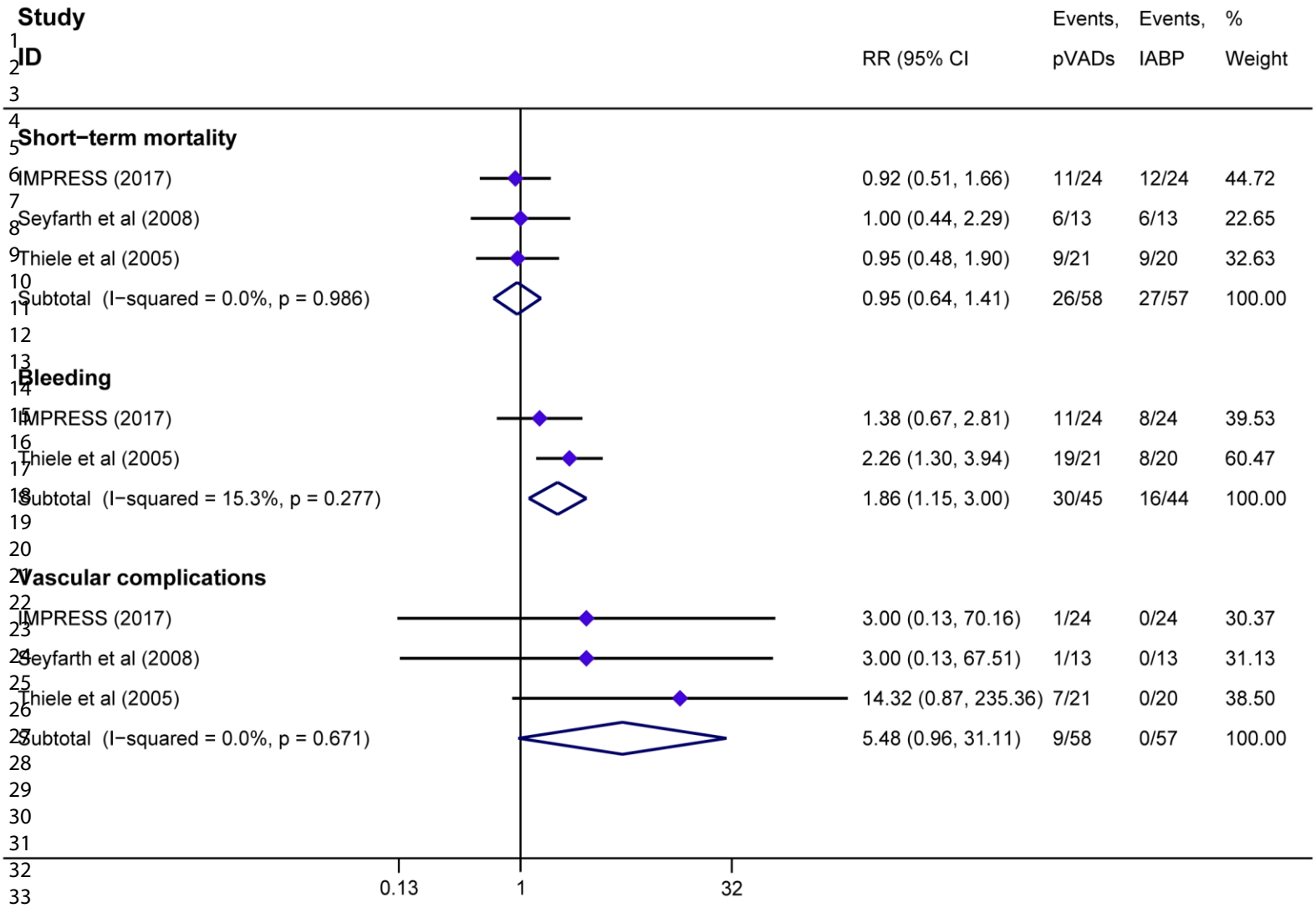
		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	
	In hospital	1.37 (0.50-3.78)
	30 days	0.70 (0.39-1.25)
	Different types of MCS use	
	IABP vs non-IABP	0.92 (0.57-1.47)
	ECLS vs non-ECLS	0.33 (0.01-1.47)
	Excluding high risk bias studies	
	Zhou et al[29], Arias et al[33] and Korbinian et al[35]	0.85 (0.51-1.45)
	Excluding each of the studies	
	SEMPER FI[25]	0.92 (0.57-1.48)
	Zhou et al[29]	0.80 (0.53-1.21)
	IABP-SHOCK II[8]	0.89 (0.45-1.74)
	CRISP AMI[28]	0.95 (0.59-1.55)
	Gu et al[30]	1.02 (0.66-1.58)
	Prondzinsky et al[31]	0.80 (0.48-1.34)
	Arias et al[33]	0.98 (0.58-1.66)
	Vijayalakshmi et al[34]	0.86 (0.55-1.34)
	Korbinian et al[35]	0.92 (0.57-1.47)
Long-term mortality		
	Inconsistent length of follow-up	
	6 months	0.72 (0.43-1.20)
	12 months	0.90 (0.35-2.31)
	Different types of MCS use	
	IABP vs non-IABP	0.87 (0.60-1.25)
	ECLS vs non-ECLS	0.50 (0.18-1.41)
	Excluding high risk bias studies	
	Zhou et al[29] and Korbinian et al[35]	0.72 (0.43-1.20)
	Excluding each of the studies	
	SEMPER FI[25]	0.82 (0.56-1.20)
	Zhou et al[29]	0.70 (0.45-1.09)
	IABP-SHOCK II[9]	0.68 (0.40-1.17)
	CRISP AMI[28]	0.89 (0.65-1.23)
	Gu et al[30]	0.92 (0.66-1.28)
	Korbinian et al[35]	0.87 (0.61-1.25)

CI indicates confidence interval.

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Online supplementary file 6: Forest plots for the secondary end points of AMI patients treated with MCS compared with non-MCS. TIA, transient ischaemic attack.



Online supplementary file 7: Forest plots for outcomes occurrence comparing pVADs use versus IABP use

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Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	
	IMPRESS[26]	0.97 (0.57-1.65)
	Seyfarth et al[32]	0.93 (0.59-1.46)
	Thiele et al[27]	0.94 (0.58-1.53)
Long-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	Not applicable

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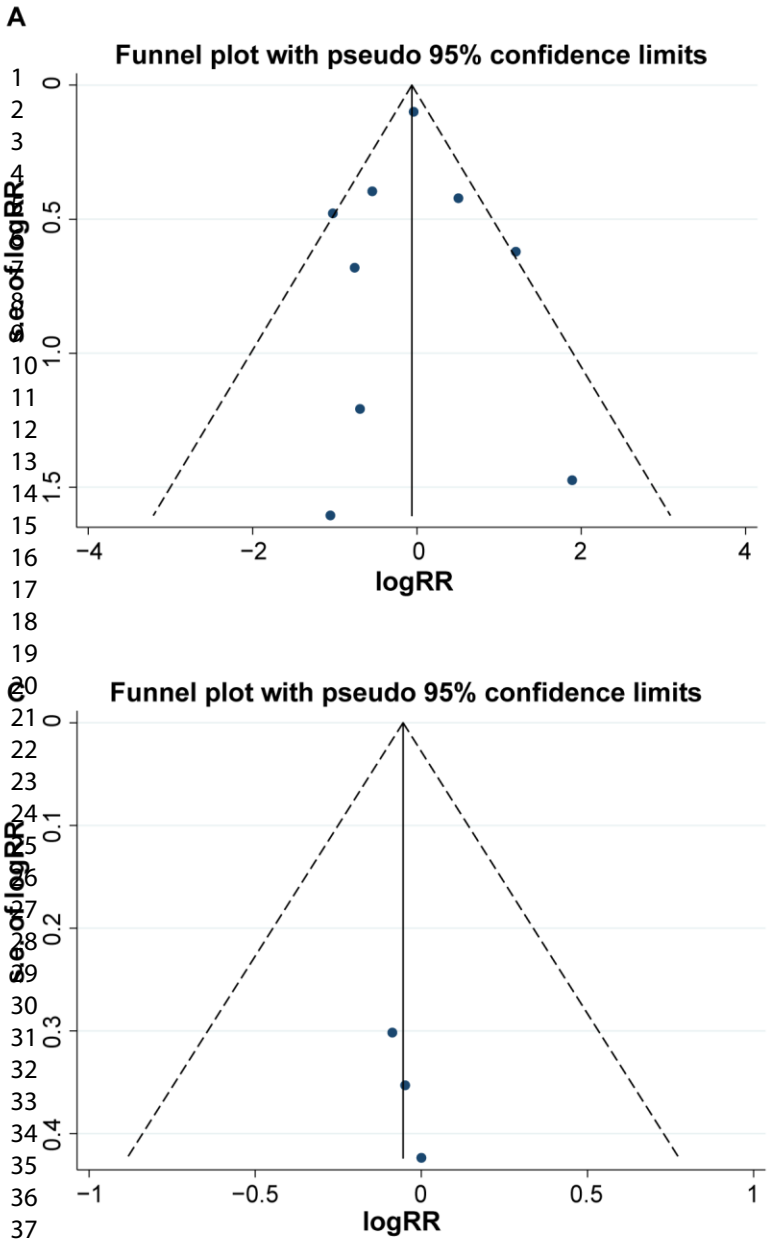
Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
SEMPER FI[25]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhou et al[29]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IABP-SHOCK II[8,9]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
CRISP AMI[28]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Gu et al[30]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Prondzinsky et al[31]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Arias et al[33]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Vijayalakshmi et al[34]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IMPRESS[26]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Seyfarth et al[32]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Thiele et al[27]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Korbinian et al[35]	Low risk	Low risk	Low risk	Low risk	Low risk	High risk	Low risk

Low risk

Unclear risk

High risk



Online supplementary file 10: Funnel plot of standard error for: A. MCS compared with non-MCS in short-term mortality; B. MCS compared with non-MCS in long-term mortality; C. pVADs compared with IABP in short-term mortality.

Online supplementary file 11: Publication bias on the association of primary outcome

		Egger's test (P)	Begg's test (P)
MCS vs non-MCS			
	Short-term mortality	0.886	0.917
	Long-term mortality	0.154	0.452
pVAD vs IABP			
	Short-term mortality	0.015*	0.296
	Long-term mortality	Not applicable	Not applicable

* Have statistical difference.

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PRISMA checklist for the meta-analysis

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6 and Online supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7

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Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7, and Online supplementary file 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9, Table 1, Table 2 and Online supplementary file 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 and Online supplementary file 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12, Online supplementary file 9, Online supplementary

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			file 10 and Online supplementary file 11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-11, Figure 3, Online supplementary file 5 and Online supplementary file 8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

NA indicates not available.

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